

FLUID DYNAMICS OF THE ATMOSPHERE AND OCEAN

Geoffrey K. Vallis
University of Exeter

ECMM719

Exeter, Spring 2016

CONTENTS

Preface	v
1 Equations with Rotation and Stratification	1
1.1 Review of Fluid Equations	1
1.1.1 Ideal Gas	2
1.2 Equations in a Rotating Frame	3
1.2.1 Rate of change of a vector	4
1.2.2 Velocity and acceleration in a rotating frame	5
1.2.3 Momentum equation in a rotating frame	6
1.2.4 Mass and tracer conservation in a rotating frame	7
1.3 Equations of Motion in Spherical Coordinates	7
1.3.1 Mass Conservation and Thermodynamic Equation	7
1.3.2 Momentum Equation	8
1.4 The primitive equations	9
1.4.1 The vector invariant form of the momentum equation	10
1.5 Cartesian Approximations: The Tangent Plane	10
1.5.1 The f -plane	10
1.5.2 The beta-plane approximation	11
1.6 The Boussinesq Approximation	12
1.6.1 Variation of density in the ocean	12
1.6.2 The Boussinesq equations	13
1.6.3 Energetics of the Boussinesq system	16
1.7 The Anelastic Approximation	17
1.8 Changing Vertical Coordinate	18
1.8.1 General relations	18
1.8.2 Pressure coordinates	19
1.8.3 Log-pressure coordinates	22
1.9 Scaling for Hydrostatic Balance	23

1.9.1	Preliminaries	23
1.9.2	Scaling and the aspect ratio	25
1.10	Geostrophic and Thermal Wind Balance	26
1.10.1	The Rossby number	26
1.10.2	Geostrophic balance	27
1.10.3	Taylor–Proudman effect	29
1.10.4	Thermal wind balance	30
1.10.5	Vertical velocity and hydrostatic balance	32
2	Shallow Water Systems	35
2.1	Dynamics of a Single, Shallow Layer	35
2.1.1	Momentum equations	35
2.1.2	Mass continuity equation	37
2.2	Reduced Gravity Equations	38
2.2.1	Pressure gradient in the active layer	39
2.3	Multi-Layer Shallow Water Equations	41
2.4	Geostrophic and Thermal-Wind Balance	43
2.5	Conservation Properties of Shallow Water Systems	45
2.5.1	Potential vorticity: a material invariant	45
2.5.2	Energy conservation: an integral invariant	48
2.6	Shallow Water Waves	49
2.6.1	Non-rotating shallow water waves	49
2.6.2	Rotating shallow water (Poincaré) waves	50
2.6.3	Kelvin waves	52
2.7	Geostrophic Adjustment	53
2.7.1	Non-rotating flow	54
2.7.2	Rotating flow	55
2.7.3	Energetics of adjustment	58
2.7.4	A variational perspective	59
3	Geostrophic Theory	61
3.1	Geostrophic Scaling	61
3.1.1	Scaling in the shallow water equations	61
3.2	The Planetary-Geostrophic Equations	64
3.2.1	Informal derivation	64
3.3	The Shallow Water Quasi-Geostrophic Equations	66
3.3.1	Single-layer shallow water quasi-geostrophic equations	67
3.3.2	Two-layer and multi-layer quasi-geostrophic systems	71
3.4	Geostrophic scaling in the stratified equations	73
3.4.1	Non-dimensional equations	74
3.5	Planetary-geostrophic equations for stratified flow	76
3.5.1	Simplifying the equations	76

3.5.2	Dimensional equations	77
3.5.3	Potential vorticity	77
3.5.4	Applicability to the ocean and atmosphere	78
3.6	The Continuously Stratified Quasi-Geostrophic System	78
3.6.1	Scaling and assumptions	78
3.6.2	Asymptotics	79
3.6.3	The potential vorticity equation	80
3.6.4	Dimensional equations	81
3.6.5	Buoyancy advection at the surface	82
3.6.6	Quasi-geostrophy in pressure coordinates	83
3.6.7	The two-level quasi-geostrophic system	84
3.7	Energetics of Quasi-Geostrophy	87
3.7.1	Conversion between APE and KE	88
3.7.2	Energetics of two-layer flows	88
3.7.3	Enstrophy conservation	89
3.8	Ekman Layers	90
3.8.1	Equations of motion and scaling	91
3.8.2	Integral properties of the Ekman layer	93
3.8.3	Sverdrup Balance	95
3.8.4	Frictional parameterization of the Ekman layer	96
4	Rossby Waves	99
4.1	Fundamentals and Formalities	99
4.1.1	Definitions and kinematics	99
4.1.2	Wave propagation and phase speed	100
4.1.3	The dispersion relation	101
4.2	Group Velocity	102
4.2.1	Superposition of two waves	102
4.3	Rossby Waves	104
4.3.1	The linear equation of motion	105
4.3.2	Waves in a single layer	105
4.3.3	The mechanism of Rossby waves	108
4.4	Rossby Waves in Stratified Quasi-Geostrophic Flow	108
4.4.1	Setting up the problem	108
4.4.2	Wave motion	109
4.5	Energy Propagation of Rossby Waves	111
4.6	Vertical Propagation of Rossby waves	114
4.6.1	Conditions for wave propagation	114
4.6.2	Dispersion relation and group velocity	117
4.7	The mechanism of jet production	117
4.7.1	I. The vorticity budget	117
4.7.2	II. Rossby waves and momentum flux	119

4.7.3	III. The pseudomomentum budget	120
5	Gravity Waves	127
5.1	Internal Waves in a Continuously Stratified Fluid	127
5.1.1	Hydrostatic internal waves	129
5.2	Internal Waves in a Rotating Frame of Reference	129
5.2.1	Equations of motion	130
5.2.2	Dispersion Relation	131
5.3	Parcel arguments for the dispersion relation	131
5.3.1	Non-rotating case	131
5.3.2	Rotating Case	132
5.4	More Properties of Internal Waves	134
5.4.1	Group velocity and phase speed	135
5.4.2	Energetics of internal waves	139

PREFACE

December 11, 2015

These are a set of lecture notes for ECMM917, Fluid Dynamics of the Atmosphere and Ocean, given at the University of Exeter. The notes are not necessarily self-contained – you may need to look in books for a full understanding. Sections marked with a black diamond are, like the ski slopes, more advanced.

CHAPTER 1

EQUATIONS WITH ROTATION AND STRATIFICATION

WEEKS 1 TO 3

1.1 REVIEW OF FLUID EQUATIONS

First we just write down the equations without derivation. For dry air, or for a salt-free liquid, the equations of motion may be written as follows:

The *mass continuity equation*:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.1)$$

If density is constant this reduces to $\nabla \cdot \mathbf{v} = 0$.

The *momentum equation*:

$$\frac{D\mathbf{v}}{Dt} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \mathbf{v} + \mathbf{F}, \quad (1.2)$$

where $\mathbf{F}$ represents the effects of body forces such as gravity and ν is the kinematic viscosity. If density is constant, or pressure, p , is given as a function of density alone (e.g., $p = C\rho^\gamma$ where γ is a constant), then (1.1) and (1.2) form a complete system.

The *thermodynamic equation*:

$$\frac{DI}{Dt} + \frac{p}{\rho} \nabla \cdot \mathbf{v} = \dot{Q}, \quad (1.3)$$

where $\dot{Q}$ represents diabatic sources such as heating and diffusion, I is internal energy. In the ideal gas case the internal energy is given by $I = c_v T$ where T is temperature.

An equation of state:

$$p = f(I, \rho), \quad (1.4)$$

where f is some known function. For example, for an ideal gas, $p = \rho R I / c_v$ or, more simply, $p = \rho R T$, where R is the ideal gas constant for the gas at hand and T is temperature.

The above four equations have four unknowns: velocity (a vector), temperature, pressure and density. The equations are called the *Euler equations* if the viscous term is omitted, and the *Navier–Stokes equations* if viscosity is included.¹

1.1.1 Ideal Gas

Let us look at the ideal gas case in a little more detail. For fluid dynamical purposes the ideal gas equation of state is usually written in the form

$$p = \rho R T \quad (1.5)$$

where R is the gas constant of the gas in question, related to the universal gas constant R_u by $R = R_u / m$, where m is the molecular weight of the gas.

The internal energy of an ideal gas is given by $I = c_v T$ where c_v is the heat capacity at constant volume. It is a function of temperature alone, and in fact is almost a constant. For an ideal gas we also have $c_p - c_v = R$, where c_p is the heat capacity at constant pressure.

For an ideal gas the first law of thermodynamics may be written in either of the two equivalent forms

$$dQ = c_v dT + p d\alpha \quad \text{or} \quad dQ = c_p dT - \alpha dp, \quad (1.6a,b)$$

where the second expression is derived using $\alpha = RT/p$. Forming the material derivative of the above gives two forms of the internal energy equation:

$$c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q} \quad \text{or} \quad c_p \frac{DT}{Dt} - \frac{RT}{p} \frac{Dp}{Dt} = \dot{Q}. \quad (1.7a,b)$$

Using the mass continuity equation, (1.7a) is equivalent to

$$\frac{DT}{Dt} + \frac{p}{c_v \rho} \nabla \cdot \mathbf{v} = \frac{\dot{Q}}{c_v}. \quad (1.8)$$

Alternatively, again using the ideal gas equation, we may eliminate T in favour of p and α and obtain

$$\frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{v} = \dot{Q} \frac{\rho R}{c_v}. \quad (1.9)$$

where $\gamma = c_p / c_v$.

Potential temperature

Using the ideal gas equation we can write (1.6b) as

$$d\eta = \frac{dQ}{T} = c_p d \ln T - R d \ln p. \quad (1.10)$$

where η is the specific entropy, which is a function of state. Now, let us *define* the *potential temperature*, θ , by the expression

$$\theta \equiv T \left(\frac{p_0}{p} \right)^\kappa, \quad (1.11)$$

where $\kappa = R/c_p$. It straightforwardly follows that

$$c_p d \ln \theta = c_p d \ln T - R d \ln p, \quad (1.12)$$

and therefore the first law of thermodynamics can be written at

$$dQ = c_p \left(\frac{T}{\theta} \right) d\theta. \quad (1.13)$$

Taking the material derivative we have

$$c_p \frac{D\theta}{Dt} = \frac{\theta}{T} \dot{Q}. \quad (1.14)$$

This is a useful form because it just involves the material derivative of one quantity. The potential temperature is, in the absence of diabatic terms, a materially conserved quantity, unlike temperature. It is closely related to entropy, and in particular

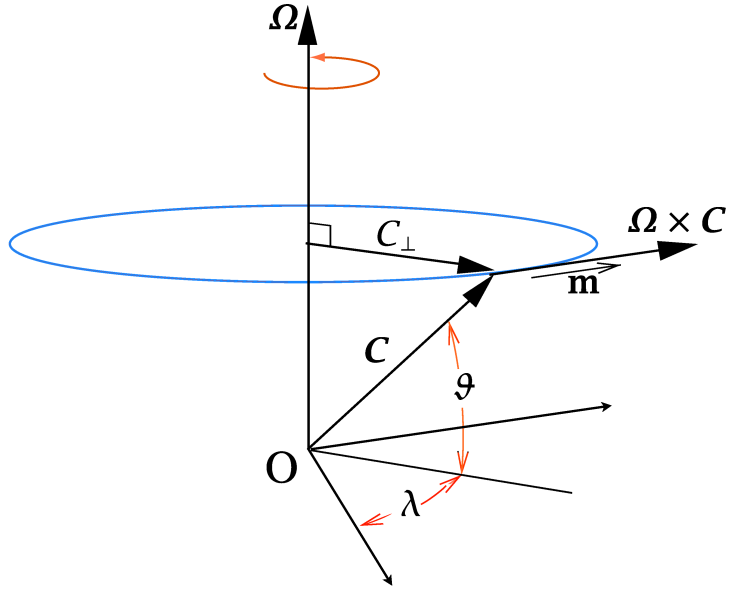
$$d\eta = c_p d \ln \theta. \quad (1.15)$$

The potential temperature is the temperature that a fluid would have if moved adiabatically to the reference pressure p_0 , but the explicit demonstration of this is left to the reader. Indeed, potential temperature may be defined this way, and for an ideal gas this is equivalent to (1.11).

1.2 THE EQUATIONS OF MOTION IN A ROTATING FRAME OF REFERENCE

Newton's second law of motion, that the acceleration on a body is proportional to the imposed force divided by the body's mass, applies in so-called inertial frames of reference; that is, frames that are stationary or moving only with a constant rectilinear velocity relative to the distant galaxies. Now the Earth spins round its own axis with a period of almost 24 hours (23h 56m) and so the surface of the Earth manifestly is not an inertial frame. Nevertheless, it is very convenient to describe the flow relative to the Earth's surface (which in fact is moving at speeds of up to a few hundreds of metres per second), rather than in some inertial frame. This necessitates recasting the equations into a form that is appropriate for a rotating frame of reference, and that is the subject of this section.

Figure 1.1 A vector $\mathbf{C}$ rotating at an angular velocity $\mathbf{\Omega}$. It appears to be a constant vector in the rotating frame, whereas in the inertial frame it evolves according to $(d\mathbf{C}/dt)_I = \mathbf{\Omega} \times \mathbf{C}$.



1.2.1 Rate of change of a vector

Consider first a vector $\mathbf{C}$ of constant length rotating relative to an inertial frame at a constant angular velocity $\mathbf{\Omega}$. Then, in a frame rotating with that same angular velocity it appears stationary and constant. If in a small interval of time δt the vector $\mathbf{C}$ rotates through a small angle $\delta \lambda$ then the change in $\mathbf{C}$, as perceived in the inertial frame, is given by (see Fig. 1.1)

$$\delta \mathbf{C} = |\mathbf{C}| \cos \vartheta \delta \lambda \mathbf{m}, \quad (1.16)$$

where the vector $\mathbf{m}$ is the unit vector in the direction of change of $\mathbf{C}$, which is perpendicular to both $\mathbf{C}$ and $\mathbf{\Omega}$. But the rate of change of the angle λ is just, by definition, the angular velocity so that $\delta \lambda = |\mathbf{\Omega}| \delta t$ and

$$\delta \mathbf{C} = |\mathbf{C}| |\mathbf{\Omega}| \sin \hat{\vartheta} \mathbf{m} \delta t = \mathbf{\Omega} \times \mathbf{C} \delta t. \quad (1.17)$$

using the definition of the vector cross product, where $\hat{\vartheta} = (\pi/2 - \vartheta)$ is the angle between $\mathbf{\Omega}$ and $\mathbf{C}$. Thus

$$\left(\frac{d\mathbf{C}}{dt} \right)_I = \mathbf{\Omega} \times \mathbf{C}, \quad (1.18)$$

where the left-hand side is the rate of change of $\mathbf{C}$ as perceived in the inertial frame.

Now consider a vector $\mathbf{B}$ that changes in the inertial frame. In a small time δt the change in $\mathbf{B}$ as seen in the rotating frame is related to the change seen in the inertial frame by

$$(\delta \mathbf{B})_I = (\delta \mathbf{B})_R + (\delta \mathbf{B})_{rot}, \quad (1.19)$$

where the terms are, respectively, the change seen in the inertial frame, the change due to the vector itself changing as measured in the rotating frame, and the change due to the rotation.

Using (1.17) $(\delta \mathbf{B})_{rot} = \boldsymbol{\Omega} \times \mathbf{B} \delta t$, and so the rates of change of the vector $\mathbf{B}$ in the inertial and rotating frames are related by

$$\left(\frac{d\mathbf{B}}{dt} \right)_I = \left(\frac{d\mathbf{B}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{B}. \quad (1.20)$$

This relation applies to a vector $\mathbf{B}$ that, as measured at any one time, is the same in both inertial and rotating frames.

1.2.2 Velocity and acceleration in a rotating frame

The velocity of a body is not measured to be the same in the inertial and rotating frames, so care must be taken when applying (1.20) to velocity. First apply (1.20) to $\mathbf{r}$, the position of a particle to obtain

$$\left(\frac{d\mathbf{r}}{dt} \right)_I = \left(\frac{d\mathbf{r}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{r} \quad (1.21)$$

or

$$\mathbf{v}_I = \mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}. \quad (1.22)$$

We refer to $\mathbf{v}_R$ and $\mathbf{v}_I$ as the relative and inertial velocity, respectively, and (1.22) relates the two. Apply (1.20) again, this time to the velocity $\mathbf{v}_R$ to give

$$\left(\frac{d\mathbf{v}_R}{dt} \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R, \quad (1.23)$$

or, using (1.22)

$$\left(\frac{d}{dt} (\mathbf{v}_I - \boldsymbol{\Omega} \times \mathbf{r}) \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R, \quad (1.24)$$

or

$$\left(\frac{d\mathbf{v}_I}{dt} \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R + \frac{d\boldsymbol{\Omega}}{dt} \times \mathbf{r} + \boldsymbol{\Omega} \times \left(\frac{d\mathbf{r}}{dt} \right)_I. \quad (1.25)$$

Then, noting that

$$\left(\frac{d\mathbf{r}}{dt} \right)_I = \left(\frac{d\mathbf{r}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{r} = (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}), \quad (1.26)$$

and assuming that the rate of rotation is constant, (1.25) becomes

$$\left(\frac{d\mathbf{v}_R}{dt} \right)_R = \left(\frac{d\mathbf{v}_I}{dt} \right)_I - 2\boldsymbol{\Omega} \times \mathbf{v}_R - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}). \quad (1.27)$$

This equation may be interpreted as follows. The term on the left-hand side is the rate of change of the relative velocity as measured in the rotating frame. The first term on the right-hand side is the rate of change of the inertial velocity as measured in the inertial frame (the inertial acceleration, which is, by Newton's second law, equal to the force on a fluid parcel divided by its

mass). The second and third terms on the right-hand side (including the minus signs) are the *Coriolis force* and the *centrifugal force* per unit mass. Neither of these are true forces — they may be thought of as quasi-forces (i.e., ‘as if’ forces); that is, when a body is observed from a rotating frame it seems to behave as if unseen forces are present that affect its motion. If (1.27) is written, as is common, with the terms $+2\boldsymbol{\Omega} \times \mathbf{v}_r$ and $+\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ on the left-hand side then these terms should be referred to as the Coriolis and centrifugal *accelerations*.

Centrifugal force

If $\mathbf{r}_\perp$ is the perpendicular distance from the axis of rotation (see Fig. 1.1 and substitute $\mathbf{r}$ for $\mathbf{C}$), then, because $\boldsymbol{\Omega}$ is perpendicular to $\mathbf{r}_\perp$, $\boldsymbol{\Omega} \times \mathbf{r} = \boldsymbol{\Omega} \times \mathbf{r}_\perp$. Then, using the vector identity $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_\perp) = (\boldsymbol{\Omega} \cdot \mathbf{r}_\perp)\boldsymbol{\Omega} - (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega})\mathbf{r}_\perp$ and noting that the first term is zero, we see that the centrifugal force per unit mass is just given by

$$\mathbf{F}_{ce} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \Omega^2 \mathbf{r}_\perp. \quad (1.28)$$

This may usefully be written as the gradient of a scalar potential,

$$\mathbf{F}_{ce} = -\nabla \Phi_{ce}. \quad (1.29)$$

where $\Phi_{ce} = -(\Omega^2 r_\perp^2)/2 = -(\boldsymbol{\Omega} \times \mathbf{r}_\perp)^2/2$.

Coriolis force

The Coriolis force per unit mass is:

$$\mathbf{F}_{Co} = -2\boldsymbol{\Omega} \times \mathbf{v}_R. \quad (1.30)$$

It plays a central role in much of geophysical fluid dynamics and will be considered extensively later on. For now, we just note three basic properties.

- (i) There is no Coriolis force on bodies that are stationary in the rotating frame.
- (ii) The Coriolis force acts to deflect moving bodies at right angles to their direction of travel.
- (iii) The Coriolis force does no work on a body because it is perpendicular to the velocity, and so $\mathbf{v}_R \cdot (\boldsymbol{\Omega} \times \mathbf{v}_R) = 0$.

1.2.3 Momentum equation in a rotating frame

Since (1.27) simply relates the accelerations of a particle in the inertial and rotating frames, then in the rotating frame of reference the momentum equation may be written

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \nabla \Phi, \quad (1.31)$$

incorporating the centrifugal term into the potential, Φ . We have dropped the subscript R ; henceforth, unless we need to be explicit, all velocities without a subscript will be considered to be relative to the rotating frame.

1.2.4 Mass and tracer conservation in a rotating frame

Let ϕ be a scalar field that, in the inertial frame, obeys

$$\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v}_I = 0. \quad (1.32)$$

Now, observers in both the rotating and inertial frame measure the same value of ϕ . Further, $D\phi/Dt$ is simply the rate of change of ϕ associated with a material parcel, and therefore is reference frame invariant. Thus,

$$\left(\frac{D\phi}{Dt} \right)_R = \left(\frac{D\phi}{Dt} \right)_I, \quad (1.33)$$

where $(D\phi/Dt)_R = (\partial\phi/\partial t)_R + \mathbf{v}_R \cdot \nabla\phi$ and $(D\phi/Dt)_I = (\partial\phi/\partial t)_I + \mathbf{v}_I \cdot \nabla\phi$ and the local temporal derivatives $(\partial\phi/\partial t)_R$ and $(\partial\phi/\partial t)_I$ are evaluated at fixed locations in the rotating and inertial frames, respectively.

Further, using (1.22), we have that we have that

$$\nabla \cdot \mathbf{v}_I = \nabla \cdot (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}) = \nabla \cdot \mathbf{v}_R \quad (1.34)$$

since $\nabla \cdot (\boldsymbol{\Omega} \times \mathbf{r}) = 0$. Thus, using (1.33) and (1.34), (1.32) is equivalent to

$$\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v}_R = 0, \quad (1.35)$$

where all observables are measured in the *rotating* frame. Thus, the equation for the evolution of a scalar whose measured value is the same in rotating and inertial frames is unaltered by the presence of rotation. In particular, the mass conservation equation is unaltered by the presence of rotation.

The individual components of the material derivative differ in the rotating and inertial frames. In particular

$$\left(\frac{\partial\phi}{\partial t} \right)_I = \left(\frac{\partial\phi}{\partial t} \right)_R - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla\phi \quad (1.36)$$

because $\boldsymbol{\Omega} \times \mathbf{r}$ is the velocity, in the inertial frame, of a uniformly rotating body. Similarly,

$$\mathbf{v}_I \cdot \nabla\phi = (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla\phi. \quad (1.37)$$

Adding the last two equations reprises (1.33).

1.3 ♦ EQUATIONS OF MOTION IN SPHERICAL COORDINATES

1.3.1 Mass Conservation and Thermodynamic Equation

The mass conservation equation expanded in spherical co-ordinates, is

$$\begin{aligned} \frac{\partial\rho}{\partial t} + \frac{u}{r \cos\vartheta} \frac{\partial\rho}{\partial\lambda} + \frac{v}{r} \frac{\partial\rho}{\partial\vartheta} + w \frac{\partial\rho}{\partial r} \\ + \frac{\rho}{r \cos\vartheta} \left[\frac{\partial u}{\partial\lambda} + \frac{\partial}{\partial\vartheta} (v \cos\vartheta) + \frac{1}{r} \frac{\partial}{\partial r} (wr^2 \cos\vartheta) \right] = 0. \end{aligned} \quad (1.38)$$

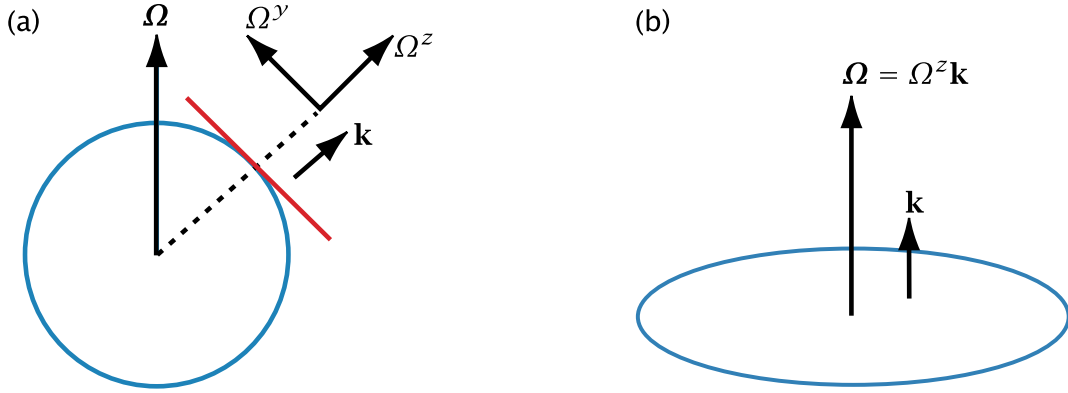


Figure 1.2 (a) On the sphere the rotation vector Ω can be decomposed into two components, one in the local vertical and one in the local horizontal, pointing toward the pole. That is, $\Omega = \Omega_y \mathbf{j} + \Omega_z \mathbf{k}$ where $\Omega_y = \Omega \cos \vartheta$ and $\Omega_z = \Omega \sin \vartheta$. In geophysical fluid dynamics, the rotation vector in the local vertical is often the more important component in the horizontal momentum equations. On a rotating disk, (b), the rotation vector Ω is parallel to the local vertical $\mathbf{k}$.

Equivalently this is the same as

$$\frac{\partial \rho}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial(u\rho)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta}(v\rho \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 w \rho) = 0. \quad (1.39)$$

The thermodynamic equation is a tracer advection equation. The (adiabatic) potential temperature in spherical coordinate form is

$$\frac{D\theta}{Dt} = \frac{\partial \theta}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \theta}{\partial \lambda} + \frac{v}{r} \frac{\partial \theta}{\partial \vartheta} + w \frac{\partial \theta}{\partial r} = 0, \quad (1.40)$$

and similarly for tracers such as water vapour or salt.

1.3.2 Momentum Equation

The momentum equation is:

$$\frac{Du}{Dt} - \left(2\Omega + \frac{u}{r \cos \vartheta} \right) (v \sin \vartheta - w \cos \vartheta) = -\frac{1}{\rho r \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (1.41a)$$

$$\frac{Dv}{Dt} + \frac{wv}{r} + \left(2\Omega + \frac{u}{r \cos \vartheta} \right) u \sin \vartheta = -\frac{1}{\rho r} \frac{\partial p}{\partial \vartheta}, \quad (1.41b)$$

$$\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} - 2\Omega u \cos \vartheta = -\frac{1}{\rho} \frac{\partial p}{\partial r} - g. \quad (1.41c)$$

The terms involving Ω are called Coriolis terms, and the quadratic terms on the left-hand sides involving $1/r$ are often called metric terms.

1.4 ♦ THE PRIMITIVE EQUATIONS

The so-called *primitive equations* of motion are simplifications of the equations that make three related approximations:

- (i) *The hydrostatic approximation.* In the vertical momentum equation the gravitational term is assumed to be balanced by the pressure gradient term, so that

$$\frac{\partial p}{\partial z} = -\rho g. \quad (1.42)$$

The advection of vertical velocity, the Coriolis terms, and the metric term $(u^2 + v^2)/r$ are all neglected.

- (ii) *The shallow-fluid approximation.* We write $r = a + z$ where the constant a is the radius of the Earth and z increases in the radial direction. The coordinate r is then replaced by a except where it is used as the differentiating argument. Thus, for example,

$$\frac{1}{r^2} \frac{\partial(r^2 w)}{\partial r} \rightarrow \frac{\partial w}{\partial z}. \quad (1.43)$$

- (iii) *The traditional approximation.* Coriolis terms in the horizontal momentum equations involving the vertical velocity, and the still smaller metric terms uw/r and vw/r , are neglected.

The second and third of these approximations should be taken, or not, together, the underlying reason being that they both relate to the presumed small aspect ratio of the motion, so the approximations succeed or fail together.

Making these approximations, the momentum equations become

$$\frac{Du}{Dt} - 2\Omega \sin \vartheta v - \frac{uv}{a} \tan \vartheta = -\frac{1}{a\rho \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (1.44a)$$

$$\frac{Dv}{Dt} + 2\Omega \sin \vartheta u + \frac{u^2 \tan \vartheta}{a} = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad (1.44b)$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (1.44c)$$

where

$$\frac{D}{Dt} = \left(\frac{\partial}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{a} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial z} \right). \quad (1.45)$$

We note the ubiquity of the factor $2\Omega \sin \vartheta$, and take the opportunity to define the *Coriolis parameter*, $f \equiv 2\Omega \sin \vartheta$.

The corresponding mass conservation equation for a shallow fluid layer is:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial \rho}{\partial \lambda} + \frac{v}{a} \frac{\partial \rho}{\partial \vartheta} + w \frac{\partial \rho}{\partial z} \\ + \rho \left[\frac{1}{a \cos \vartheta} \frac{\partial u}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{\partial w}{\partial z} \right] = 0, \end{aligned} \quad (1.46)$$

or equivalently,

$$\frac{\partial \rho}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial(u\rho)}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(v\rho \cos \vartheta) + \frac{\partial(w\rho)}{\partial z} = 0. \quad (1.47)$$

1.4.1 The vector invariant form of the momentum equation

The ‘vector invariant’ form of the momentum equation is so-called because it appears to take the same form in all coordinate systems — there is no advective derivative of the coordinate system to worry about. With the aid of the identity $(\mathbf{v} \cdot \nabla)\mathbf{v} = -\mathbf{v} \times \boldsymbol{\omega} + \nabla(\mathbf{v}^2/2)$, where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$ is the relative vorticity, the three-dimensional momentum equation, (1.31), may be written:

$$\frac{\partial \mathbf{v}}{\partial t} + (2\boldsymbol{\Omega} + \boldsymbol{\omega}) \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \frac{1}{2} \nabla \mathbf{v}^2 + \mathbf{g}, \quad (1.48)$$

and this is the vector invariant momentum equation.

We can write the horizontal momentum equations of the primitive equations in a similar way. Making the traditional and shallow fluid approximations, the horizontal components of (1.48) become

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{f} + \mathbf{k}\zeta) \times \mathbf{u} + w \frac{\partial \mathbf{u}}{\partial z} = -\frac{1}{\rho} \nabla_z p - \frac{1}{2} \nabla \mathbf{u}^2, \quad (1.49)$$

where $\mathbf{u} = (u, v, 0)$, $\mathbf{f} = \mathbf{k} 2\Omega \sin \vartheta$, ∇_z is the horizontal gradient operator (the gradient at a constant value of z). For reference, in spherical coordinates ζ is given by

$$\zeta = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(u \cos \vartheta) = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a} \frac{\partial u}{\partial \vartheta} + \frac{u}{a} \tan \vartheta. \quad (1.50)$$

The separate components of the momentum equation are given by:

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w \frac{\partial u}{\partial z} = -\frac{1}{a \cos \vartheta} \left(\frac{1}{\rho} \frac{\partial p}{\partial \lambda} + \frac{1}{2} \frac{\partial \mathbf{u}^2}{\partial \lambda} \right), \quad (1.51)$$

and

$$\frac{\partial v}{\partial t} + (f + \zeta)u + w \frac{\partial v}{\partial z} = -\frac{1}{a} \left(\frac{1}{\rho} \frac{\partial p}{\partial \vartheta} + \frac{1}{2} \frac{\partial \mathbf{u}^2}{\partial \vartheta} \right). \quad (1.52)$$

1.5 CARTESIAN APPROXIMATIONS: THE TANGENT PLANE

1.5.1 The f -plane

Although the rotation of the Earth is central for many dynamical phenomena, the sphericity of the Earth is not always so. This is especially true for phenomena on a scale somewhat smaller than global where the use of spherical coordinates becomes awkward, and it is more convenient to use a locally Cartesian representation of the equations. Referring to Fig. 1.2 we will define a plane tangent to the surface of the Earth at a latitude ϑ_0 , and then use a Cartesian

coordinate system (x, y, z) to describe motion on that plane. For small excursions on the plane, $(x, y, z) \approx (a\lambda \cos \vartheta_0, a(\vartheta - \vartheta_0), z)$. Consistently, the velocity is $\mathbf{v} = (u, v, w)$, so that u, v and w are the components of the velocity *in the tangent plane*, in approximately in the east–west, north–south and vertical directions, respectively.

The momentum equations for flow in this plane are then

$$\frac{\partial u}{\partial t} + (\mathbf{v} \cdot \nabla)u + 2(\Omega^y w - \Omega^z v) = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (1.53a)$$

$$\frac{\partial v}{\partial t} + (\mathbf{v} \cdot \nabla)v + 2(\Omega^z u - \Omega^x w) = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (1.53b)$$

$$\frac{\partial w}{\partial t} + (\mathbf{v} \cdot \nabla)w + 2(\Omega^x v - \Omega^y u) = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (1.53c)$$

where the rotation vector $\boldsymbol{\Omega} = \Omega^x \mathbf{i} + \Omega^y \mathbf{j} + \Omega^z \mathbf{k}$ and $\Omega^x = 0$, $\Omega^y = \Omega \cos \vartheta_0$ and $\Omega^z = \Omega \sin \vartheta_0$. If we make the traditional approximation, and so ignore the components of $\boldsymbol{\Omega}$ not in the direction of the local vertical, then

$$\frac{Du}{Dt} - f_0 v = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (1.54a)$$

$$\frac{Dv}{Dt} + f_0 u = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (1.54b)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g. \quad (1.54c)$$

where $f_0 = 2\Omega^z = 2\Omega \sin \vartheta_0$. Defining the horizontal velocity vector $\mathbf{u} = (u, v, 0)$, the first two equations may also be written as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f}_0 \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (1.55)$$

where $D\mathbf{u}/Dt = \partial\mathbf{u}/\partial t + \mathbf{v} \cdot \nabla \mathbf{u}$, $\mathbf{f}_0 = 2\Omega \sin \vartheta_0 \mathbf{k} = f_0 \mathbf{k}$, and $\mathbf{k}$ is the direction perpendicular to the plane (it does not change its orientation with latitude). These equations are, evidently, exactly the same as the momentum equations in a system in which the rotation vector is aligned with the local vertical, as illustrated in the right-hand panel in Fig. 1.2 (on page 8). They will describe flow on the surface of a rotating sphere to a good approximation provided the flow is of limited latitudinal extent so that the effects of sphericity are unimportant; we have made what is known as the *f-plane* approximation since the Coriolis parameter is a constant. We may in addition make the hydrostatic approximation, in which case (1.54c) becomes the familiar $\partial p / \partial z = -\rho g$.

1.5.2 The beta-plane approximation

The magnitude of the vertical component of rotation varies with latitude, and this has important dynamical consequences. We can approximate this effect by allowing the effective rotation vector

to vary. Thus, noting that, for small variations in latitude,

$$f = 2\Omega \sin \vartheta \approx 2\Omega \sin \vartheta_0 + 2\Omega(\vartheta - \vartheta_0) \cos \vartheta_0, \quad (1.56)$$

then on the tangent plane we may mimic this by allowing the Coriolis parameter to vary as

$$f = f_0 + \beta y, \quad (1.57)$$

where $f_0 = 2\Omega \sin \vartheta_0$ and $\beta = \partial f / \partial y = (2\Omega \cos \vartheta_0)/a$. This important approximation is known as the *beta-plane*, or *β -plane*, approximation; it captures the the most important *dynamical* effects of sphericity, without the complicating *geometric* effects, which are not essential to describe many phenomena. The momentum equations (1.54) are unaltered except that f_0 is replaced by $f_0 + \beta y$ to represent a varying Coriolis parameter. Thus, sphericity combined with rotation is dynamically equivalent to a *differentially rotating* system. For future reference, we write down the β -plane horizontal momentum equations:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (1.58)$$

where $\mathbf{f} = (f_0 + \beta y)\hat{\mathbf{k}}$. In component form this equation becomes

$$\frac{Du}{Dt} - fv = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad \frac{Dv}{Dt} + fu = -\frac{1}{\rho} \frac{\partial p}{\partial y}. \quad (1.59a,b)$$

The mass conservation, thermodynamic and hydrostatic equations in the β -plane approximation are the same as the usual Cartesian, f -plane, forms of those equations.

1.6 THE BOUSSINESQ APPROXIMATION

The density variations in the ocean are quite small compared to the mean density, and we may exploit this to derive somewhat simpler but still quite accurate equations of motion. Let us first examine how much density does vary in the ocean.

1.6.1 Variation of density in the ocean

The variations of density in the ocean are due to three effects: the compression of water by pressure (which we denote as $\Delta_p \rho$), the thermal expansion of water if its temperature changes ($\Delta_T \rho$), and the haline contraction if its salinity changes ($\Delta_S \rho$). How big are these? An appropriate equation of state to approximately evaluate these effects is the linear one

$$\rho = \rho_0 \left[1 - \beta_T (T - T_0) + \beta_S (S - S_0) + \frac{p}{\rho_0 c_s^2} \right], \quad (1.60)$$

where $\beta_T \approx 2 \times 10^{-4} \text{ K}^{-1}$, $\beta_S \approx 10^{-3} \text{ psu}^{-1}$ and $c_s \approx 1500 \text{ m s}^{-1}$. The three effects may then be evaluated as follows.

Pressure compressibility. We have $\Delta_p \rho \approx \Delta p / c_s^2 \approx \rho_0 g H / c_s^2$, where H is the depth and the pressure change is quite accurately evaluated using the hydrostatic approximation. Thus,

$$\frac{|\Delta_p \rho|}{\rho_0} \ll 1 \quad \text{if} \quad \frac{gH}{c_s^2} \ll 1, \quad (1.61)$$

or if $H \ll c_s^2 / g$. The quantity $c_s^2 / g \approx 200$ km is the density scale height of the ocean. Thus, the pressure at the bottom of the ocean (say $H = 10$ km in the deep trenches), enormous as it is, is insufficient to compress the water enough to make a significant change in its density. Changes in density due to dynamical variations of pressure are small if the Mach number is small, and this is also the case.

Thermal expansion. We have $\Delta_T \rho \approx -\beta_T \rho_0 \Delta T$ and therefore

$$\frac{|\Delta_T \rho|}{\rho_0} \ll 1 \quad \text{if} \quad \beta_T \Delta T \ll 1. \quad (1.62)$$

For $\Delta T = 20$ K, $\beta_T \Delta T \approx 4 \times 10^{-3}$, and evidently we would require temperature differences of order β_T^{-1} , or 5000 K to obtain order one variations in density.

Saline contraction. We have $\Delta_S \rho \approx \beta_S \rho_0 \Delta S$ and therefore

$$\frac{|\Delta_S \rho|}{\rho_0} \ll 1 \quad \text{if} \quad \beta_S \Delta S \ll 1. \quad (1.63)$$

As changes in salinity in the ocean rarely exceed 5 psu, for which $\beta_S \Delta S = 5 \times 10^{-3}$, the fractional change in the density of seawater is correspondingly very small.

Evidently, fractional density changes in the ocean are very small.

1.6.2 The Boussinesq equations

The *Boussinesq equations* are a set of equations that exploit the smallness of density variations in many liquids. We write

$$\rho = \rho_0 + \delta \rho(x, y, z, t) \quad (1.64a)$$

where ρ_0 is a constant and we assume that

$$|\delta \rho| \ll \rho_0. \quad (1.65)$$

Associated with the reference density is a reference pressure that is defined to be in hydrostatic balance with it. That is,

$$p = p_0(z) + \delta p(x, y, z, t) \quad (1.66a)$$

where

$$\frac{dp_0}{dz} \equiv -g\rho_0 \quad (1.67)$$

Note that $\nabla_z p = \nabla_z \delta p$.

Momentum equations

To obtain the Boussinesq equations we use $\rho = \rho_0 + \delta\rho$, and assume $\delta\rho/\rho_0$ is small. Without approximation, the momentum equation can be written as

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla\delta p - \frac{\partial p_0}{\partial z}\mathbf{k} - g(\rho_0 + \delta\rho)\mathbf{k}, \quad (1.68)$$

and using (1.67a) this becomes, again without approximation,

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla\delta p - g\delta\rho\mathbf{k}. \quad (1.69)$$

If density variations are small this becomes

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\nabla\phi + b\mathbf{k}, \quad (1.70)$$

where $\phi = \delta p/\rho_0$ and $b = -g\delta\rho/\rho_0$ is the *buoyancy*. Note that we should not and do not neglect the term $g\delta\rho$, for there is no reason to believe it to be small ($\delta\rho$ may be small, but g is big). Equation (1.70) is the momentum equation in the Boussinesq approximation, and it is common to say that the Boussinesq approximation ignores all variations of density of a fluid in the momentum equation, except when associated with the gravitational term.

For most large-scale motions in the ocean the *deviation* pressure and density fields are also approximately in hydrostatic balance, and in that case the vertical component of (1.70) becomes

$$\frac{\partial\phi}{\partial z} = b. \quad (1.71)$$

A condition for (1.71) to hold is that vertical accelerations are small *compared to* $g\delta\rho/\rho_0$, *and not compared to the acceleration due to gravity itself*. For more discussion of this point, see section 1.9.

Mass Conservation

The unapproximated mass conservation equation is

$$\frac{D\delta\rho}{Dt} + (\rho_0 + \delta\rho)\nabla \cdot \mathbf{v} = 0. \quad (1.72)$$

Provided that time scales advectively — that is to say that D/Dt scales in the same way as $\mathbf{v} \cdot \nabla$ — then we may approximate this equation by

$$\nabla \cdot \mathbf{v} = 0, \quad (1.73)$$

which is the same as that for a constant density fluid. This *absolutely does not* allow one to go back and use (1.72) to say that $D\delta\rho/Dt = 0$; the evolution of density is given by the thermodynamic equation in conjunction with an equation of state, and this should not be confused with the mass conservation equation. Note also that in eliminating the time-derivative of density we eliminate the possibility of sound waves.

Summary of Boussinesq Equations

The simple Boussinesq equations are, for an inviscid fluid:

$$\text{momentum equations:} \quad \frac{D\mathbf{v}}{Dt} + \mathbf{f} \times \mathbf{v} = -\nabla\phi + b\mathbf{k}, \quad (\text{B.1})$$

$$\text{mass conservation:} \quad \nabla \cdot \mathbf{v} = 0, \quad (\text{B.2})$$

$$\text{buoyancy equation:} \quad \frac{Db}{Dt} = \dot{b}. \quad (\text{B.3})$$

A more general form replaces the buoyancy equation by:

$$\text{thermodynamic equation:} \quad \frac{D\theta}{Dt} = \dot{\theta}, \quad (\text{B.4})$$

$$\text{salinity equation:} \quad \frac{DS}{Dt} = \dot{S}, \quad (\text{B.5})$$

$$\text{equation of state:} \quad b = b(\theta, S, z). \quad (\text{B.6})$$

Thermodynamic equation and equation of state

[This section is even more informal and non-rigorous than other sections.] We write the thermodynamic equation as

$$\frac{DI}{Dt} + \frac{p}{\rho} \nabla \cdot \mathbf{v} = \dot{Q} \quad (1.74)$$

We neglect the second term on the left-hand side (because the fluid is incompressible), and write the first term in terms of temperature

$$C \frac{DT}{Dt} = \dot{Q} \quad (1.75)$$

where c is the heat capacity of the fluid. We further suppose that the temperature is linearly related to the buoyancy, b . That is, we assume a linear equation of state, $b = b_0 (1 + A(T - T_0))$ where A is a constant coefficient of thermal expansion. The thermodynamic equation becomes

$$\frac{Db}{Dt} = Q_b, \quad (1.76)$$

where $Q_b = A\dot{Q}/C$. The momentum equation (1.70), mass continuity equation (1.73) and thermodynamic equation (1.76) then form a closed set, called the *Boussinesq equations*.

♦ *Mean stratification and the buoyancy frequency*

The processes that cause density to vary in the vertical often differ from those that cause it to vary in the horizontal. For this reason it is sometimes useful to write $\rho = \rho_0 + \tilde{\rho}(z) + \rho'(x, y, z, t)$ and define $\tilde{b}(z) \equiv -g\tilde{\rho}/\rho_0$ and $b' \equiv -g\rho'/\rho_0$. The thermodynamic equation (1.74) becomes

$$\frac{Db'}{Dt} + N^2 w = 0, \quad (1.77)$$

where D/Dt remains a three-dimensional operator and

$$N^2(z) = \frac{d\tilde{b}_\sigma}{dz}, \quad (1.78)$$

The quantity N^2 is a measure of the *mean stratification* of the fluid. N is known as the buoyancy frequency, something we return to later on.

1.6.3 Energetics of the Boussinesq system

In a uniform gravitational field but with no other forcing or dissipation, we write the simple Boussinesq equations as

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = b\mathbf{k} - \nabla\phi, \quad \nabla \cdot \mathbf{v} = 0, \quad \frac{Db}{Dt} = 0. \quad (1.79a,b,c)$$

From (1.79a) and (1.79b) the kinetic energy density evolution is given by

$$\frac{1}{2} \frac{Dv^2}{Dt} = bw - \nabla \cdot (\phi \mathbf{v}), \quad (1.80)$$

where the constant reference density ρ_0 is omitted. Let us now define the potential $\Phi \equiv -z$, so that $\nabla\Phi = -\mathbf{k}$ and

$$\frac{D\Phi}{Dt} = \nabla \cdot (\mathbf{v}\Phi) = -w, \quad (1.81)$$

and using this and (1.79c) gives

$$\frac{D}{Dt}(b\Phi) = -wb. \quad (1.82)$$

Adding (1.82) to (1.80) and expanding the material derivative gives

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \mathbf{v}^2 + b\Phi \right) + \nabla \cdot \left[\mathbf{v} \left(\frac{1}{2} \mathbf{v}^2 + b\Phi + \phi \right) \right] = 0. \quad (1.83)$$

This constitutes an energy equation for the Boussinesq system. The energy density (divided by ρ_0) is just $\mathbf{v}^2/2 + b\Phi$. What does the term $b\Phi$ represent? Its integral, multiplied by ρ_0 , is the potential energy of the flow minus that of the basic state, or $\int g(\rho - \rho_0)z \, dz$. If there were a heating term on the right-hand side of (1.79c) this would directly provide a source of potential energy, rather than internal energy as in the compressible system. Because the fluid is incompressible, there is no conversion from kinetic and potential energy into internal energy.

1.7 ♦ THE ANELASTIC APPROXIMATION

We don't derive the equations here — we just write them down.

In the atmosphere the density varies significantly, especially in the vertical. However deviations of both ρ and p from a statically balanced state are often quite small, and the relative vertical variation of potential temperature is also small. We can usefully exploit these observations to give a somewhat simplified set of equations, useful both for theoretical and numerical analyses because sound waves are eliminated by way of an 'anelastic' approximation. We set

$$\rho = \bar{\rho}(z) + \delta\rho(x, y, z, t), \quad p = \bar{p}(z) + \delta p(x, y, z, t), \quad (1.84a,b)$$

where we assume that $|\delta\rho| \ll |\bar{\rho}|$ and we define $\bar{p}$ such that

$$\frac{\partial \bar{p}}{\partial z} \equiv -g\bar{\rho}(z). \quad (1.85)$$

The notation is similar to that for the Boussinesq case except that, importantly, the density basic state is now a (given) function of vertical coordinate. As with the Boussinesq case, the idea is to ignore dynamic variations of density (i.e., of $\delta\rho$) except where associated with gravity. We also assume an ideal gas equation of state. After some algebra we obtain the following.

The horizontal momentum equation:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad (1.86)$$

where $\phi = \delta p / \bar{\rho}$, and this is similar to the corresponding equation in the Boussinesq approximation.

The vertical component of the momentum equation:

$$\frac{Dw}{Dt} = g\delta s - \frac{\partial \phi}{\partial z}, \quad (1.87)$$

$\delta s = \delta\theta/\theta_0$, where θ_0 is a constant. If we define a buoyancy by $b_a \equiv g\delta s = g\delta\theta/\theta_0$, then (1.86) and (1.87) have the same form as the Boussinesq momentum equations, but with a slightly different definition of buoyancy.

The mass continuity equation:

$$\nabla \cdot \mathbf{u} + \frac{1}{\bar{\rho}} \frac{\partial}{\partial z} (\bar{\rho} w) = 0. \quad (1.88)$$

It is here that the eponymous anelastic approximation arises: the elastic compressibility of the fluid is neglected, and this serves to eliminate sound waves.

The thermodynamic equation:

$$\frac{Db_a}{Dt} = Q_s. \quad (1.89)$$

Summarizing, the complete set of anelastic equations, with rotation but with no dissipation or diabatic terms, is

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = \mathbf{k}b_a - \nabla\phi \quad (1.90a)$$

$$\frac{Db_a}{Dt} = 0 \quad (1.90b)$$

$$\nabla \cdot (\tilde{\rho}\mathbf{v}) = 0 \quad (1.90c)$$

where $b_a = g\delta s = g\delta\theta/\theta_0$. The main difference between the anelastic and Boussinesq sets of equations is in the mass continuity equation, and when $\tilde{\rho} = \rho_0 = \text{constant}$ the two equation sets are identical. However, whereas the Boussinesq approximation is a very good one for ocean dynamics, the anelastic approximation is much less so for large-scale atmosphere flow. The equations also provide a good jumping-off point for theoretical studies. Because of their similarity, the Boussinesq and anelastic approximations are sometimes referred to as the strong and weak Boussinesq approximations, respectively.

1.8 ♦ CHANGING VERTICAL COORDINATE

Although using z as a vertical coordinate is a natural choice given our Cartesian worldview, it is not the only option, nor is it always the most useful one. Any variable that has a one-to-one correspondence with z in the vertical, so any variable that varies monotonically with z , could be used; pressure and, perhaps surprisingly, entropy, are common choices. In the atmosphere pressure almost always falls monotonically with height, and using it instead of z provides a useful simplification of the mass conservation and geostrophic relations, as well as a more direct connection with observations, which are often taken at fixed values of pressure. (In the ocean pressure coordinates are essentially almost the same as height coordinates because density is almost constant.) Entropy seems an exotic vertical coordinate, but it is very useful in adiabatic flow.

1.8.1 General relations

First consider a general vertical coordinate, ξ . Any variable Ψ that is a function of the coordinates (x, y, z, t) may be expressed instead in terms of (x, y, ξ, t) by considering z to be function of the independent variables (x, y, ξ, t) ; that is, we let $\Psi(x, y, \xi, t) = \Psi(x, y, z(x, y, \xi, t), t)$. Derivatives with respect to z and ξ are related by

$$\frac{\partial\Psi}{\partial\xi} = \frac{\partial\Psi}{\partial z} \frac{\partial z}{\partial\xi} \quad \text{and} \quad \frac{\partial\Psi}{\partial z} = \frac{\partial\Psi}{\partial\xi} \frac{\partial\xi}{\partial z}. \quad (1.91a,b)$$

Horizontal derivatives in the two coordinate systems are related by the chain rule,

$$\left(\frac{\partial\Psi}{\partial x}\right)_\xi = \left(\frac{\partial\Psi}{\partial x}\right)_z + \left(\frac{\partial z}{\partial x}\right)_\xi \frac{\partial\Psi}{\partial z}, \quad (1.92)$$

and similarly for time.

The material derivative in ξ coordinates may be derived by transforming the original expression in z coordinates using the chain rule, but because (x, y, t, ξ) are independent coordinates, and noting that the ‘vertical velocity’ in ξ coordinates is just $\dot{\xi}$ (i.e., $D\xi/Dt$, just as the vertical velocity in z coordinates is $w = Dz/Dt$), we can write down

$$\frac{D\Psi}{Dt} = \frac{\partial\Psi}{\partial t} \Big|_{x,y,\xi} + \mathbf{u} \cdot \nabla_{\xi} \Psi + \dot{\xi} \frac{\partial\Psi}{\partial \xi}, \quad (1.93)$$

where ∇_{ξ} is the gradient operator at constant ξ . The operator D/Dt is physically the same in z or ξ coordinates because it is the total derivative of some property of a fluid parcel, and this is independent of the coordinate system. However, the individual terms within it will differ between coordinate systems.

1.8.2 Pressure coordinates

In pressure coordinates the analogue of the vertical velocity is $\omega \equiv Dp/Dt$, and the advective derivative itself is given by

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + \omega \frac{\partial}{\partial p}. \quad (1.94)$$

Note, though, that advective derivative is the same operator as it is in height coordinates, since it is just the total derivative of a given fluid parcel; it is just written with different coordinates.

To obtain an expression for the pressure force, now let $\xi = p$ in (1.92) and apply the relationship to p itself to give

$$0 = \left(\frac{\partial p}{\partial x} \right)_z + \left(\frac{\partial z}{\partial x} \right)_p \frac{\partial p}{\partial z}, \quad (1.95)$$

which, using the hydrostatic relationship, gives

$$\left(\frac{\partial p}{\partial x} \right)_z = \rho \left(\frac{\partial \Phi}{\partial x} \right)_p, \quad (1.96)$$

where $\Phi = gz$ is the *geopotential*. Thus, the horizontal pressure force in the momentum equations is

$$\frac{1}{\rho} \nabla_z p = \nabla_p \Phi, \quad (1.97)$$

where the subscripts on the gradient operator indicate that the horizontal derivatives are taken at constant z or constant p . The horizontal momentum equation thus becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_p \Phi, \quad (1.98)$$

where D/Dt is given by (1.94). The hydrostatic equation in height coordinates is $\partial p / \partial z = -\rho g$ and in pressure coordinates this becomes

$$\frac{\partial \Phi}{\partial p} = -\frac{1}{\rho} \quad \text{or} \quad \frac{\partial \Phi}{\partial p} = -\frac{RT}{p}, \quad (1.99a,b)$$

using the ideal gas relation

The mass conservation equation simplifies attractively in pressure coordinates, if the hydrostatic approximation is used. Recall that the mass conservation equation can be derived from the material form

$$\frac{D}{Dt}(\rho \delta V) = 0, \quad (1.100)$$

where $\delta V = \delta x \delta y \delta z$ is a volume element. But by the hydrostatic relationship $\rho \delta z = -(1/g)\delta p$ and thus

$$\frac{D}{Dt}(\delta x \delta y \delta p) = 0. \quad (1.101)$$

This is completely analogous to the expression for the material conservation of volume in an incompressible fluid. Thus, without further ado, we write the mass conservation in pressure coordinates as

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0, \quad (1.102)$$

where the horizontal derivative is taken at constant pressure.

The (adiabatic) thermodynamic equation is, of course, still $D\theta/Dt = 0$, and θ may be related to pressure and temperature using its definition and the ideal gas equation to complete the equation set. However, because the hydrostatic equation is written in terms of temperature and not potential temperature it is convenient to write the thermodynamic equation accordingly. To do this we begin with the thermodynamic equation in the form of (1.7b), namely $c_p DT/Dt - \alpha Dp/Dt = 0$. Since $\omega \equiv Dp/Dt$ this equation is simply

$$c_p \frac{DT}{Dt} - \frac{RT}{p} \omega = 0, \quad (1.103)$$

which is an appropriate thermodynamic equation in pressure coordinates. It is sometimes useful to write this as

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} - \omega S_p = 0, \quad (1.104)$$

where

$$S_p = \frac{\kappa T}{p} - \frac{\partial T}{\partial p} = -\frac{T}{\theta} \frac{\partial \theta}{\partial p} \quad (1.105)$$

using the ideal gas equation and the definition of potential temperature, with $\kappa = R/c_p$. Evidently, S_p is an appropriate measure of static stability in pressure coordinates and it is closely related to the buoyancy frequency N , as we see in the next subsection.

The main practical difficulty with the pressure-coordinate equations is the lower boundary condition. Using

$$w \equiv \frac{Dz}{Dt} = \frac{\partial z}{\partial t} + \mathbf{u} \cdot \nabla_p z + \omega \frac{\partial z}{\partial p}, \quad (1.106)$$

and (1.99), the boundary condition of $w = 0$ at $z = z_s$ becomes

$$\frac{\partial \Phi}{\partial t} + \mathbf{u} \cdot \nabla_p \Phi - \alpha \omega = 0 \quad (1.107)$$

Equations of Motion in Pressure and Log-pressure Coordinates

The adiabatic, inviscid primitive equations in pressure coordinates are:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_p \Phi, \quad (\text{P.1})$$

$$\frac{\partial \Phi}{\partial p} = \frac{-RT}{p}, \quad (\text{P.2})$$

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0, \quad (\text{P.3})$$

$$c_p \frac{DT}{Dt} - \frac{RT}{p} \omega = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} - \omega S_p = 0. \quad (\text{P.4})$$

where $S_p = \kappa T/p - \partial T/\partial p$. The above equations are, respectively, the horizontal momentum equation, the hydrostatic equation, the mass continuity equation and the thermodynamic equation.

The corresponding equations in log-pressure coordinates are

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_Z \Phi, \quad (\text{P.5})$$

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H}, \quad (\text{P.6})$$

$$\nabla_Z \cdot \mathbf{u} + \frac{1}{\rho_R} \frac{\partial \rho_R W}{\partial z} = 0, \quad (\text{P.7})$$

$$c_p \frac{DT}{Dt} + W \frac{RT}{H} = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + WS_Z = 0. \quad (\text{P.8})$$

where $\rho_R = \rho_0 \exp(-Z/H)$ and $S_Z = \kappa T/H + \partial T/\partial Z$ where $\kappa = R/c_p$. The thermodynamic equation may also be written as

$$\frac{\partial}{\partial t} \frac{\partial \Phi}{\partial Z} + u \frac{\partial}{\partial x} \frac{\partial \Phi}{\partial Z} + v \frac{\partial}{\partial y} \frac{\partial \Phi}{\partial Z} + WN_*^2 = 0, \quad (\text{P.9})$$

where $N_*^2 = (R/H)S_Z$.

at $p(x, y, z_s, t)$. In theoretical studies, it is common to assume that the lower boundary is in fact a constant pressure surface and simply assume that $\omega = 0$, or sometimes the condition $\omega = -\alpha^{-1}\partial\Phi/\partial t$ is used. For realistic studies (with general circulation models, say) the fact that the level $z = 0$ is not a coordinate surface must be properly accounted for. For this reason, and especially if the lower boundary is uneven because of the presence of topography, so-called *sigma coordinates* are sometimes used, in which the vertical coordinate is chosen so that the lower boundary is a coordinate surface. Sigma coordinates may use height itself as a vertical measure (typical in oceanic applications) or use pressure (typical in atmospheric applications). In the latter case the vertical coordinate is $\sigma = p/p_s$ where $p_s(x, y, t)$ is the surface pressure. The difficulty of applying (1.107) is replaced by a prognostic equation for the surface pressure, derived from the mass conservation equation.

Finally, we note that the pressure coordinate equations (collected together in the shaded box on the preceding page) are isomorphic to the hydrostatic general Boussinesq equations (see the shaded box on page 15) with $z \Leftrightarrow -p$, $w \Leftrightarrow -\omega$, $\phi \Leftrightarrow \Phi$, $b \Leftrightarrow \alpha$, and an equation of state $b = b(\theta, z) \Leftrightarrow \alpha = \alpha(\theta, p)$. In an ideal gas, for example, $\alpha = (\theta R/p_R)(p_R/p)^{1/\gamma}$. Any conservation laws that hold in one system can therefore be expected to have an analog in the other.

1.8.3 Log-pressure coordinates

A variant of pressure coordinates is *log-pressure* coordinates, in which the vertical coordinate is $Z = -H \ln(p/p_R)$ where p_R is a reference pressure (say 1000 mb) and H is a constant (for example a scale height RT_0/g where T_0 is a constant) so that Z has units of length. (Uppercase letters are conventionally used for some variables in log-pressure coordinates, and these are not to be confused with scaling parameters.) The ‘vertical velocity’ for the system is now

$$W \equiv \frac{DZ}{Dt}, \quad (1.108)$$

and the advective derivative is

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + W \frac{\partial}{\partial Z}. \quad (1.109)$$

The horizontal momentum equation is unaltered from (1.98), although we use (1.109) to evaluate the advective derivative. The hydrostatic equation, (1.99b), straightforwardly becomes

$$\frac{\partial\Phi}{\partial Z} = \frac{RT}{H} \quad (1.110)$$

The mass continuity equation (1.102) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial W}{\partial Z} - \frac{W}{H} = 0. \quad (1.111)$$

which may be written as

$$\nabla_Z \cdot \mathbf{u} + \frac{1}{\rho_R} \frac{\partial(\rho_R W)}{\partial Z} = 0 \quad (1.112)$$

where $\rho_R = \rho_0 \exp(-Z/H)$, so giving a form similar to the mass conservation equation in the anelastic equations. (The value of the constant ρ_0 may be set to one.)

As with pressure coordinates, it is convenient to write the thermodynamic equation in terms of temperature and not potential temperature, and in an analogous procedure to that leading to (1.103) we straightforwardly obtain

$$c_p \frac{DT}{Dt} + W \frac{RT}{H} = 0. \quad (1.113)$$

This may be written as

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + WS_Z = 0 \quad (1.114)$$

where

$$S_Z = \frac{\kappa T}{H} + \frac{\partial T}{\partial Z}. \quad (1.115)$$

and we may note that $S_Z = S_p p/H$. Using the hydrostatic equation we may write (1.114) as

$$\frac{\partial}{\partial t} \frac{\partial \Phi}{\partial Z} + u \frac{\partial}{\partial x} \frac{\partial \Phi}{\partial Z} + v \frac{\partial}{\partial y} \frac{\partial \Phi}{\partial Z} + WN_*^2 = 0, \quad (1.116)$$

where $N_*^2 = (R/H)S_Z$. The quantity N_* is not exactly equal to the square of the buoyancy frequency as normally defined (for an ideal gas $N^2 = (g/\theta)\partial\theta/\partial z$), but the two can be shown to be related by $N_*/N = p/(\rho gH) = RT/gH$, and are equal for an isothermal atmosphere.

Finally, we note that integrating the hydrostatic equation between two pressure levels gives, with $\Phi = gz$,

$$z(p_2) - z(p_1) = -\frac{R}{g} \int_{p_1}^{p_2} T \, d \ln p. \quad (1.117)$$

Thus, the thickness of the layer is proportional to the average temperature of the layer.

1.9 SCALING FOR HYDROSTATIC BALANCE

We now look in more detail at the conditions required for hydrostatic balance to hold. Along with geostrophic balance, considered in the next section, it is one of the most fundamental balances in geophysical fluid dynamics. The corresponding states, hydrostasy and geostrophy, are not exactly realized, but their approximate satisfaction has profound consequences on the behaviour of the atmosphere and ocean.

1.9.1 Preliminaries

Consider the relative sizes of terms in (1.53c):

$$\frac{W}{T} + \frac{UW}{L} + \frac{W^2}{H} + \Omega U \sim \left| \frac{1}{\rho} \frac{\partial p}{\partial z} \right| + g. \quad (1.118)$$

For most large-scale motion in the atmosphere and ocean the terms on the right-hand side are orders of magnitude larger than those on the left, and therefore must be approximately equal. Explicitly, suppose $W \sim 1 \text{ cm s}^{-1}$, $L \sim 10^5 \text{ m}$, $H \sim 10^3 \text{ m}$, $U \sim 10 \text{ m s}^{-1}$, $T = L/U$. Then by substituting into (1.118) it seems that the pressure term is the only one which could balance the gravitational term, and we are led to approximate (1.53c) by,

$$\frac{\partial p}{\partial z} = -\rho g. \quad (1.119)$$

This equation, which is a vertical momentum equation, is known as *hydrostatic balance*.

However, (1.119) is not always a useful equation! Let us suppose that the density is a constant, ρ_0 . We can then write the pressure as

$$p(x, y, z, t) = p_0(z) + p'(x, y, z, t), \quad (1.120)$$

where

$$\frac{\partial p_0}{\partial z} = -\rho_0 g. \quad (1.121)$$

That is, p_0 and ρ_0 are in hydrostatic balance. On the f -plane, the inviscid vertical momentum equation becomes, without approximation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial z}. \quad (1.122)$$

Thus, *for constant density fluids the gravitational term has no dynamical effect*: there is no buoyancy force, and the pressure term in the horizontal momentum equations can be replaced by p' . Hydrostatic balance, and in particular (1.121), is certainly not an appropriate vertical momentum equation in this case. If the fluid is stratified, we should therefore subtract off the hydrostatic pressure associated with the mean density before we can determine whether hydrostasy is a useful *dynamical* approximation, accurate enough to determine the horizontal pressure gradients. This is automatic in the Boussinesq equations, where the vertical momentum equation is

$$\frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b, \quad (1.123)$$

and the hydrostatic balance of the basic state is already subtracted out. In the more general equation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (1.124)$$

we need to compare the advective term on the left-hand side with the pressure variations arising from horizontal flow in order to determine whether hydrostasy is an appropriate vertical momentum equation. Nevertheless, if we only need to determine the pressure for use in an equation of state then we simply need to compare the sizes of the dynamical terms in (1.53c) with g itself, in order to determine whether a hydrostatic approximation will suffice.

1.9.2 Scaling and the aspect ratio

In a Boussinesq fluid we write the horizontal and vertical momentum equations as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad \frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b. \quad (1.125a,b)$$

With $\mathbf{f} = 0$, (1.125a) implies the scaling

$$\phi \sim U^2. \quad (1.126)$$

If we use mass conservation, $\nabla_z \cdot \mathbf{u} + \partial w / \partial z = 0$, to scale vertical velocity then

$$w \sim W = \frac{H}{L} U = \alpha U, \quad (1.127)$$

where $\alpha \equiv H/L$ is the aspect ratio. The advective terms in the vertical momentum equation all scale as

$$\frac{Dw}{Dt} \sim \frac{UW}{L} = \frac{U^2 H}{L^2}. \quad (1.128)$$

Using (1.126) and (1.128) the ratio of the advective term to the pressure gradient term in the vertical momentum equations then scales as

$$\frac{|Dw/Dt|}{|\partial \phi / \partial z|} \sim \frac{U^2 H / L^2}{U^2 / H} \sim \left(\frac{H}{L} \right)^2. \quad (1.129)$$

Thus, the condition for hydrostasy, that $|Dw/Dt|/|\partial \phi / \partial z| \ll 1$, is:

$$\alpha^2 \equiv \left(\frac{H}{L} \right)^2 \ll 1. \quad (1.130)$$

The advective term in the vertical momentum may then be neglected. Thus, hydrostatic balance is a *small aspect ratio approximation*.

We can obtain the same result more formally by non-dimensionalizing the momentum equations. Using uppercase symbols to denote scaling values we write

$$\begin{aligned} (x, y) &= L(\hat{x}, \hat{y}), & z &= H\hat{z}, & \mathbf{u} &= U\hat{\mathbf{u}}, & w &= W\hat{w} = \frac{HU}{L}\hat{w}, \\ t &= T\hat{t} = \frac{L}{U}\hat{t}, & \phi &= \Phi\hat{\phi} = U^2\hat{\phi}, & b &= B\hat{b} = \frac{U^2}{H}\hat{b}, \end{aligned} \quad (1.131)$$

where the hatted variables are non-dimensional and the scaling for w is suggested by the mass conservation equation, $\nabla_z \cdot \mathbf{u} + \partial w / \partial z = 0$. Substituting (1.131) into (1.125) (with $\mathbf{f} = 0$) gives us the non-dimensional equations

$$\frac{D\hat{\mathbf{u}}}{D\hat{t}} = -\nabla\hat{\phi}, \quad \alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} + \hat{b}, \quad (1.132a,b)$$

where $D/D\hat{t} = \partial/\partial\hat{t} + \hat{u}\partial/\partial\hat{x} + \hat{v}\partial/\partial\hat{y} + \hat{w}\partial/\partial\hat{z}$ and we use the convention that when ∇ operates on non-dimensional quantities the operator itself is non-dimensional. From (1.132b) it is clear that hydrostatic balance pertains when $\alpha^2 \ll 1$.

Variable	Scaling symbol	Meaning	Atmos. value	Ocean value
(x, y)	L	Horizontal length scale	10^6 m	10^5 m
t	T	Time scale	1 day (10^5 s)	10 days (10^6 s)
(u, v)	U	Horizontal velocity	10 m s $^{-1}$	0.1 m s $^{-1}$
	Ro	Rossby number, U/fL	0.1	0.01

Table 1.1 Scales of large-scale flow in atmosphere and ocean. The choices given are representative of large-scale mid-latitude eddying motion in both systems.

1.10 GEOSTROPHIC AND THERMAL WIND BALANCE

We now consider the dominant dynamical balance in the horizontal components of the momentum equation. In the horizontal plane (meaning along geopotential surfaces) we find that the Coriolis term is much larger than the advective terms and the dominant balance is between it and the horizontal pressure force. This balance is called *geostrophic balance*, and it occurs when the Rossby number is small, as we now investigate.

1.10.1 The Rossby number

The *Rossby number* characterizes the importance of rotation in a fluid. It is, essentially, the ratio of the magnitude of the relative acceleration to the Coriolis acceleration, and it is of fundamental importance in geophysical fluid dynamics. It arises from a simple scaling of the horizontal momentum equation, namely

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (1.133a)$$

$$\frac{U^2/L}{fU} \quad (1.133b)$$

where U is the approximate magnitude of the horizontal velocity and L is a typical length scale over which that velocity varies. (We assume that $W/H \lesssim U/L$, so that vertical advection does not dominate the advection.) The ratio of the sizes of the advective and Coriolis terms is defined to be the Rossby number,

$$Ro \equiv \frac{U}{fL}. \quad (1.134)$$

If the Rossby number is small then rotation effects are important, and as the values in Table 1.1 indicate this is the case for large-scale flow in both ocean and atmosphere.

Another intuitive way to think about the Rossby number is in terms of time scales. The Rossby number based on a time scale is

$$Ro_T \equiv \frac{1}{fT}, \quad (1.135)$$

where T is a time scale associated with the dynamics at hand. If the time scale is an advective one, meaning that $T \sim L/U$, then this definition is equivalent to (1.134). Now, $f = 2\Omega \sin \vartheta$, where Ω is the angular velocity of the rotating frame and equal to $2\pi/T_p$ where T_p is the period of rotation (24 hours). Thus,

$$Ro_T = \frac{T_p}{4\pi T \sin \vartheta} = \frac{T_i}{T}, \quad (1.136)$$

where $T_i = 1/f$ is the ‘inertial time scale’, about three hours in mid-latitudes. Thus, for phenomena with time scales much longer than this, such as the motion of the Gulf Stream or a mid-latitude atmospheric weather system, the effects of the Earth’s rotation can be expected to be important, whereas a short-lived phenomena, such as a cumulus cloud or tornado, may be oblivious to such rotation. The expressions (1.134) and (1.135) are, of course, just approximate measures of the importance of rotation.

1.10.2 Geostrophic balance

If the Rossby number is sufficiently small in (1.133a) then the rotation term will dominate the nonlinear advection term, and if the time period of the motion scales advectively then the rotation term also dominates the local time derivative. The only term that can then balance the rotation term is the pressure term, and therefore we must have

$$\mathbf{f} \times \mathbf{u} \approx -\frac{1}{\rho} \nabla_z p, \quad (1.137)$$

or, in Cartesian component form

$$fu \approx -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv \approx \frac{1}{\rho} \frac{\partial p}{\partial x}. \quad (1.138)$$

This balance is known as *geostrophic balance*, and its consequences are profound, giving geophysical fluid dynamics a special place in the broader field of fluid dynamics. We *define* the geostrophic velocity by

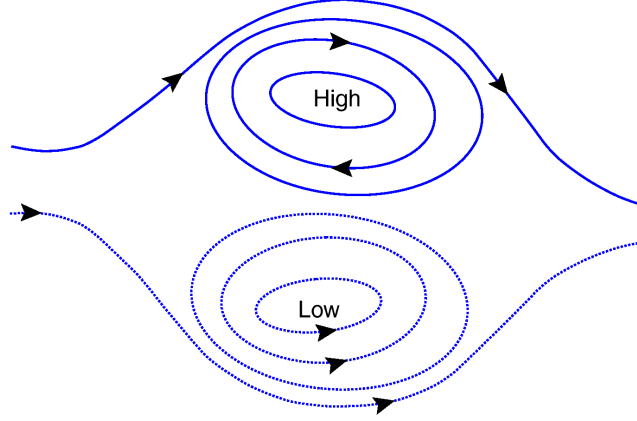
$$fu_g \equiv -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv_g \equiv \frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (1.139)$$

and for low Rossby number flow $u \approx u_g$ and $v \approx v_g$. In spherical coordinates the geostrophic velocity is

$$fu_g = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad fv_g = \frac{1}{\rho a \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (1.140)$$

where $f = 2\Omega \sin \vartheta$. Geostrophic balance has a number of immediate ramifications:

Figure 1.3 Schematic of geostrophic flow with a positive value of the Coriolis parameter f . Flow is parallel to the lines of constant pressure (isobars). Cyclonic flow is anticlockwise around a low pressure region and anticyclonic flow is clockwise around a high. If f were negative, as in the Southern Hemisphere, (anti)cyclonic flow would be (anti)clockwise.



- Geostrophic flow is parallel to lines of constant pressure (isobars). If $f > 0$ the flow is anticlockwise round a region of low pressure and clockwise around a region of high pressure (see Fig. 1.3).
- If the Coriolis force is constant and if the density does not vary in the horizontal the geostrophic flow is horizontally non-divergent and

$$\nabla_z \cdot \mathbf{u}_g = \frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = 0. \quad (1.141)$$

We may define the *geostrophic streamfunction*, ψ , by

$$\psi \equiv \frac{p}{f_0 \rho_0}, \quad (1.142)$$

whence

$$u_g = -\frac{\partial \psi}{\partial y}, \quad v_g = \frac{\partial \psi}{\partial x}. \quad (1.143)$$

The vertical component of vorticity, ζ , is then given by

$$\zeta = \mathbf{k} \cdot \nabla \times \mathbf{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \nabla_z^2 \psi. \quad (1.144)$$

- If the Coriolis parameter is not constant, then cross-differentiating (1.139) gives, for constant density geostrophic flow,

$$v_g \frac{\partial f}{\partial y} + f \nabla_z \cdot \mathbf{u}_g = 0, \quad (1.145)$$

which, using the mass continuity equation $\nabla_z \cdot \mathbf{u}_g = -\partial w / \partial z$,

$$\beta v_g = f \frac{\partial w}{\partial z}. \quad (1.146)$$

where $\beta \equiv \partial f / \partial y = 2\Omega \cos \vartheta / a$. This geostrophic vorticity balance is sometimes known as ‘Sverdrup balance’, although that expression is better restricted to the case when the vertical velocity from a wind stress.

1.10.3 Taylor–Proudman effect

If $\beta = 0$, then (1.146) implies that the vertical velocity is not a function of height. In fact, in that case none of the components of velocity vary with height if density is also constant. To show this, in the limit of zero Rossby number we first write the three-dimensional momentum equation as

$$\mathbf{f}_0 \times \mathbf{v} = -\nabla \phi - \nabla \chi, \quad (1.147)$$

where $\mathbf{f}_0 = 2\boldsymbol{\Omega} = 2\Omega \mathbf{k}$, $\phi = p / \rho_0$, and $\nabla \chi$ represents other potential forces. If $\chi = gz$ then the vertical component of this equation represents hydrostatic balance, and the horizontal components represent geostrophic balance. On taking the curl of this equation, the terms on the right-hand side vanish and the left-hand side becomes

$$(\mathbf{f}_0 \cdot \nabla) \mathbf{v} - \mathbf{f}_0 \nabla \cdot \mathbf{v} - (\mathbf{v} \cdot \nabla) \mathbf{f}_0 + \mathbf{v} \nabla \cdot \mathbf{f}_0 = 0. \quad (1.148)$$

But $\nabla \cdot \mathbf{v} = 0$ by mass conservation, and because $\mathbf{f}_0$ is constant both $\nabla \cdot \mathbf{f}_0$ and $(\mathbf{v} \cdot \nabla) \mathbf{f}_0$ vanish. Equation (1.148) thus reduces to

$$(\mathbf{f}_0 \cdot \nabla) \mathbf{v} = 0, \quad (1.149)$$

which, since $\mathbf{f}_0 = f_0 \mathbf{k}$, implies $f_0 \partial \mathbf{v} / \partial z = 0$, and in particular we have

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial v}{\partial z} = 0, \quad \frac{\partial w}{\partial z} = 0. \quad (1.150)$$

A different presentation of this argument proceeds as follows. If the flow is exactly in geostrophic and hydrostatic balance then

$$\mathbf{v} = \frac{1}{f_0} \frac{\partial \phi}{\partial x}, \quad \mathbf{u} = -\frac{1}{f_0} \frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial z} = -g. \quad (1.151a,b,c)$$

Differentiating (1.151a,b) with respect to z , and using (1.151c) yields

$$\frac{\partial v}{\partial z} = \frac{-1}{f_0} \frac{\partial g}{\partial x} = 0, \quad \frac{\partial u}{\partial z} = \frac{1}{f_0} \frac{\partial g}{\partial y} = 0. \quad (1.152)$$

Noting that the geostrophic velocities are horizontally non-divergent ($\nabla_z \cdot \mathbf{u} = 0$), and using mass continuity then gives $\partial w / \partial z = 0$, as before.

If there is a solid horizontal boundary anywhere in the fluid, for example at the surface, then $w = 0$ at that surface and thus $w = 0$ everywhere. Hence the motion occurs in planes that lie perpendicular to the axis of rotation, and the flow is effectively two dimensional. This result is known as the *Taylor–Proudman effect*, namely that for constant density flow in geostrophic and hydrostatic balance the vertical derivatives of the horizontal and the vertical velocities are zero. At zero Rossby number, if the vertical velocity is zero somewhere in the flow, it is zero everywhere in that vertical column; furthermore, the horizontal flow has no vertical shear, and the fluid moves like a slab. The effects of rotation have provided a *stiffening* of the fluid in the vertical.

In neither the atmosphere nor the ocean do we observe precisely such vertically coherent flow, mainly because of the effects of stratification. However, it is typical of geophysical fluid dynamics that the assumptions underlying a derivation are not fully satisfied, yet there are manifestations of it in real flow. Thus, one might have naïvely expected, because $\partial w / \partial z = -\nabla_z \cdot \mathbf{u}$, that the scales of the various variables would be related by $W/H \sim U/L$. However, if the flow is rapidly rotating we expect that the horizontal flow will be in near geostrophic balance and therefore nearly divergence free; thus $\nabla_z \cdot \mathbf{u} \ll U/L$, and $W \ll HU/L$.

1.10.4 Thermal wind balance

Thermal wind balance arises by combining the geostrophic and hydrostatic approximations, and this is most easily done in the context of the anelastic (or Boussinesq) equations, or in pressure coordinates. For the anelastic equations, geostrophic balance may be written

$$-fv_g = -\frac{\partial \phi}{\partial x} = -\frac{1}{a \cos \vartheta} \frac{\partial \phi}{\partial \lambda}, \quad fu_g = -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}. \quad (1.153a,b)$$

Combining these relations with hydrostatic balance, $\partial \phi / \partial z = b$, gives

$$-f \frac{\partial v_g}{\partial z} = -\frac{\partial b}{\partial x} = -\frac{1}{a \cos \lambda} \frac{\partial b}{\partial \lambda}, \quad (1.154a)$$

$$f \frac{\partial u_g}{\partial z} = -\frac{\partial b}{\partial y} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta}. \quad (1.154b)$$

These equations represent *thermal wind balance*, and the vertical derivative of the geostrophic wind is the ‘thermal wind’. Eq. (1.154b) may be written in terms of the zonal angular momentum as

$$\frac{\partial m_g}{\partial z} = -\frac{a}{2\Omega \tan \vartheta} \frac{\partial b}{\partial y}, \quad (1.155)$$

where $m_g = (u_g + \Omega a \cos \vartheta)a \cos \vartheta$. Potentially more accurate than geostrophic balance is the so-called *cyclostrophic balance* (or *gradient wind balance*), which retains a centrifugal term in the

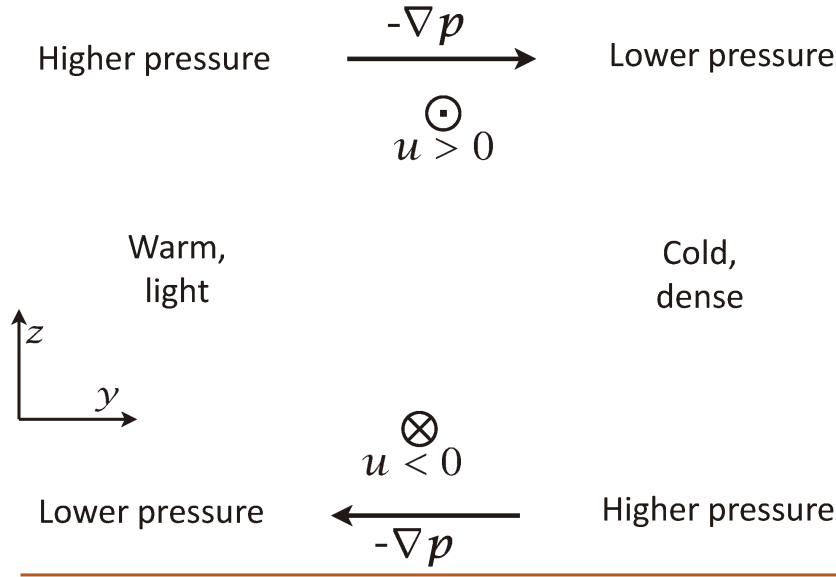


Figure 1.4 The mechanism of thermal wind. A cold fluid is denser than a warm fluid, so by hydrostasy the vertical pressure gradient is greater where the fluid is cold. Thus, the pressure gradients form as shown, where ‘higher’ and ‘lower’ mean relative to the average at that height. The horizontal pressure gradients are balanced by the Coriolis force, producing (for $f > 0$) the horizontal winds shown ($\otimes$ into the paper, and $\odot$ out of the paper). Only the wind *shear* is given by the thermal wind.

momentum equation. Thus, we omit only the material derivative in the meridional momentum equation (1.44b) and obtain

$$2u\Omega \sin \vartheta + \frac{u^2}{a} \tan \vartheta \approx -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}. \quad (1.156)$$

For large-scale flow this only differs significantly from geostrophic balance very close to the equator. Taking the vertical derivative of (1.156) and using the hydrostatic relation $\partial \phi / \partial z = b$ gives a modified thermal wind relation, and this may be put in the simple form

$$\frac{\partial m^2}{\partial z} = -\frac{a^3 \cos^3 \vartheta}{\sin \vartheta} \frac{\partial b}{\partial y}, \quad (1.157)$$

where $m = (u + \Omega a \cos \vartheta)a \cos \vartheta$ is the annular angular momentum.

If the density or buoyancy is constant then there is no shear and (1.154b) or (1.157) give the Taylor–Proudman result. But suppose that the temperature falls in the poleward direction. Then thermal wind balance implies that the (eastward) wind will increase with height — just as is observed in the atmosphere! In general, a vertical shear of the horizontal wind is associated with a horizontal temperature gradient, and this is one of the most simple and far-reaching effects in geophysical fluid dynamics. The underlying physical mechanism is illustrated in Fig. 1.4.

Geostrophic and thermal wind balance in pressure coordinates

In pressure coordinates geostrophic balance is just

$$\mathbf{f} \times \mathbf{u}_g = -\nabla_p \Phi, \quad (1.158)$$

where Φ is the geopotential and ∇_p is the gradient operator taken at constant pressure. If f is constant, it follows from (1.158) that the geostrophic wind is non-divergent on pressure surfaces. Taking the vertical derivative of (1.158) (that is, its derivative with respect to p) and using the hydrostatic equation, $\partial\Phi/\partial p = -\alpha$, gives the thermal wind equation

$$\mathbf{f} \times \frac{\partial \mathbf{u}_g}{\partial p} = \nabla_p \alpha = \frac{R}{p} \nabla_p T, \quad (1.159)$$

where the last equality follows using the ideal gas equation and because the horizontal derivative is at constant pressure. In component form this is

$$-f \frac{\partial v_g}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial x}, \quad f \frac{\partial u_g}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial y}. \quad (1.160)$$

In log-pressure coordinates, with $Z = -H \ln(p/p_R)$, thermal wind is

$$\mathbf{f} \times \frac{\partial \mathbf{u}_g}{\partial Z} = -\frac{R}{H} \nabla_Z T. \quad (1.161)$$

The physical meaning in all these cases is the same: a horizontal temperature gradient, or a temperature gradient along an isobaric surface, is accompanied by a vertical shear of the horizontal wind.

1.10.5 Vertical velocity and hydrostatic balance

Scaling for vertical velocity

If the Coriolis parameter is constant then the horizontal component of flows that are in geostrophic balance have zero horizontal divergence ($\nabla_x \cdot \mathbf{u} = 0$) and zero vertical velocity. We can therefore expect that any flow with small Rossby number will have a ‘small’ vertical velocity. Let us make this statement more precise using the rotating Boussinesq equations, (1.125) with constant Coriolis parameter. Let $\mathbf{u} = \mathbf{u}_g + \mathbf{u}_a$ where the geostrophic flow satisfies $\mathbf{f}_0 \times \mathbf{u}_g = -\nabla\phi$. The horizontal momentum equation, with corresponding scales for each term, then becomes

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + w \frac{\partial \mathbf{u}}{\partial z} + \mathbf{f}_0 \times \mathbf{u}_a = 0, \quad (1.162)$$

$$\frac{U^2}{L} \quad \frac{U^2}{L} \quad \frac{WU}{H} \quad f_0 U_a. \quad (1.163)$$

This equation suggests a scaling for the ageostrophic flow of

$$U_a = \frac{U}{f_0 L} U = Ro U. \quad (1.164)$$

That is, the ageostrophic flow is Rossby number smaller (at least) than the geostrophic flow. To obtain a scaling for the vertical velocity we look to the mass continuity equation written in the form

$$\frac{\partial w}{\partial z} = -\nabla \cdot \mathbf{u}_a, \quad (1.165)$$

since only the ageostrophic flow has a divergence. Equations (1.164) and (1.165) suggest the scaling

$$W = Ro \frac{HU}{L}. \quad (1.166)$$

That is, the vertical velocity is order Rossby number smaller than an estimate based purely on the mass continuity equation would suggest.

If the Coriolis parameter is not constant then the geostrophic flow itself is divergent and this induces a vertical velocity, as in (1.146). The scaling for vertical velocity is now

$$W = \frac{\beta}{f} HU = Ro_\beta \frac{HU}{L}. \quad (1.167)$$

where $Ro_\beta = \beta L / f$ is the *beta Rossby number*. It is less than one for all flows except those with a truly global scale.

Scaling for hydrostatic balance

Let us non-dimensionalize the rotating Boussinesq equations, (1.125), by writing

$$\begin{aligned} (x, y) &= L(\hat{x}, \hat{y}), & z &= H\hat{z}, & \mathbf{u} &= U\hat{\mathbf{u}}, & t &= T\hat{t} = \frac{L}{U}\hat{t}, & \mathbf{f} &= f_0\hat{\mathbf{f}}, \\ w &= \frac{\epsilon HU}{L}\hat{w}, & \phi &= \Phi\hat{\phi} = f_0 UL\hat{\phi}, & b &= B\hat{b} = \frac{f_0 UL}{H}\hat{b}, \end{aligned} \quad (1.168)$$

These relations are almost the same as (1.131), except for the factor of ϵ in the scaling of w . If the Coriolis parameter is constant or nearly so then, from (1.166), $\epsilon = Ro$, whereas if the Coriolis parameter varies then $\epsilon = Ro_\beta$, as in (1.166). The scaling for ϕ and b' are suggested by geostrophic and thermal wind balance with f_0 a representative value of f . Substituting these values into (1.125) we obtain the following scaled momentum equations:

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi}, \quad Ro\epsilon\alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} - \hat{b}. \quad (1.169a,b)$$

where $D/D\hat{t} = \partial/\partial\hat{t} + \hat{\mathbf{u}} \cdot \nabla_z + \epsilon\hat{w}\partial/\partial\hat{z}$. There are two notable aspects to these equations. First and most obviously, when $Ro \ll 1$, (1.169a) reduces to geostrophic balance, $\mathbf{f} \times \mathbf{u} \approx -\nabla\phi$. Second, the material derivative in (1.169b) is multiplied by three non-dimensional parameters, and we can understand the appearance of each as follows.

- (i) The aspect ratio dependence (α^2) arises in the same way as for non-rotating flows — that is, because of the presence of w and z in the vertical momentum equation as opposed to (u, v) and (x, y) in the horizontal equations.

- (ii) The Rossby number dependence (Ro) arises because in rotating flow the pressure gradient is balanced by the Coriolis force, which is Rossby number larger than the advective terms.
- (iii) The factor ϵ arises because in rotating flow w is smaller than u by ϵ times the aspect ratio. The factor may be the Rossby number itself, or the beta Rossby number.

The factor $Ro\epsilon\alpha^2$ is very small for large-scale flow; the reader is invited to calculate representative values. Evidently, a rapidly rotating fluid is more likely to be in hydrostatic balance than a non-rotating fluid, other conditions being equal. The combined effects of rotation and stratification are, not surprisingly, quite subtle and we leave that topic for chapter 3.

CHAPTER 2

SHALLOW WATER SYSTEMS

WEEKS 4 TO 6

2.1 DYNAMICS OF A SINGLE, SHALLOW LAYER

Consider fluid in a container above which is another fluid of negligible density (and therefore negligible inertia) relative to the fluid of interest, as illustrated in Fig. 2.1. Our notation is that $\mathbf{v} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$ is the three-dimensional velocity and $\mathbf{u} = u\mathbf{i} + v\mathbf{j}$ is the horizontal velocity. $h(x, y)$ is the thickness of the liquid column, H is its mean height, and η is the height of the free surface. In a flat-bottomed container $\eta = h$, whereas in general $h = \eta - \eta_b$, where η_b is the height of the floor of the container.

2.1.1 Momentum equations

The vertical momentum equation is just the hydrostatic equation,

$$\frac{\partial p}{\partial z} = -\rho_0 g, \quad (2.1)$$

and, because density is assumed constant, we may integrate this to

$$p(x, y, z, t) = -\rho_0 g z + p_o. \quad (2.2)$$

At the top of the fluid, $z = \eta$, the pressure is determined by the weight of the overlying fluid and this is assumed to be negligible. Thus, $p = 0$ at $z = \eta$, giving

$$p(x, y, z, t) = \rho_0 g (\eta(x, y, t) - z). \quad (2.3)$$

The consequence of this is that the horizontal gradient of pressure is independent of height. That is

$$\nabla_z p = \rho_0 g \nabla_z \eta, \quad (2.4)$$

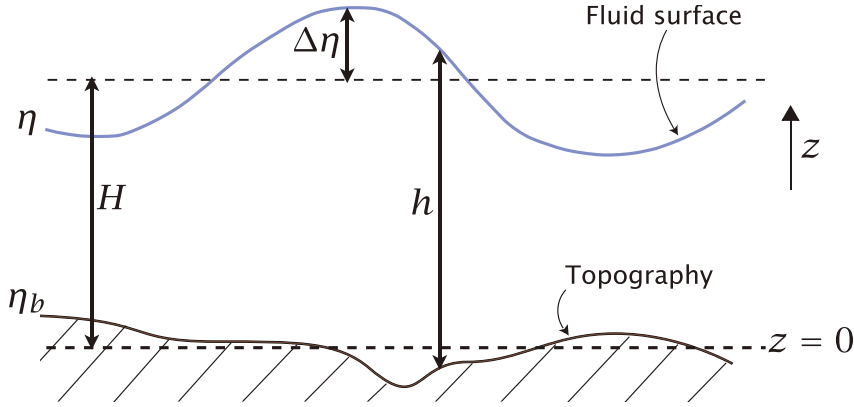


Figure 2.1 A shallow water system. h is the thickness of a water column, H its mean thickness, η the height of the free surface and η_b is the height of the lower, rigid, surface, above some arbitrary origin, typically chosen such that the average of η_b is zero. $\Delta\eta$ is the deviation free surface height, so we have $\eta = \eta_b + h = H + \Delta\eta$.

where

$$\nabla_z = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} \quad (2.5)$$

is the gradient operator at constant z . (In the rest of this chapter we will drop the subscript z unless that causes ambiguity. The three-dimensional gradient operator will be denoted by ∇_3 . We will also mostly use Cartesian coordinates, but the shallow water equations may certainly be applied over a spherical planet — indeed, ‘Laplace’s tidal equations’ are essentially the shallow water equations on a sphere.) The horizontal momentum equations therefore become

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho_0} \nabla p = -g \nabla \eta. \quad (2.6)$$

The right-hand side of this equation is independent of the vertical coordinate z . Thus, if the flow is initially independent of z , it must stay so. (This z independence is unrelated to that arising from the rapid rotation necessary for the Taylor–Proudman effect.) The velocities u and v are functions of x , y and t only, and the horizontal momentum equation is therefore

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + u \frac{\partial \mathbf{u}}{\partial x} + v \frac{\partial \mathbf{u}}{\partial y} = -g \nabla \eta. \quad (2.7)$$

That the horizontal velocity is independent of z is a consequence of the hydrostatic equation, which ensures that the horizontal pressure gradient is independent of height. (Another starting point would be to take this independence of the horizontal motion with height as the *definition* of shallow water flow. In real physical situations such independence does not hold exactly — for example, friction at the bottom may induce a vertical dependence of the flow in a boundary

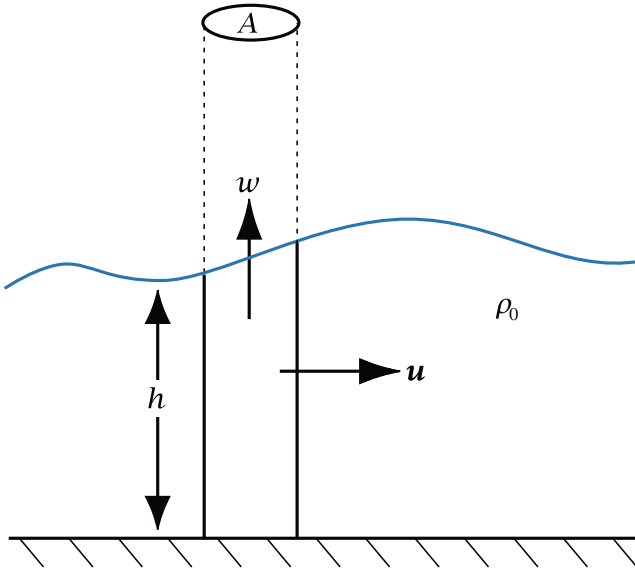


Figure 2.2 The mass budget for a column of area A in a shallow water system. The fluid leaving the column is $\oint \rho_0 h \mathbf{u} \cdot \mathbf{n} dl$ where $\mathbf{n}$ is the unit vector normal to the boundary of the fluid column. There is a non-zero vertical velocity at the top of the column if the mass convergence into the column is non-zero.

layer.) In the presence of rotation, (2.7) easily generalizes to

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (2.8)$$

where $\mathbf{f} = f\mathbf{k}$. Just as with the primitive equations, f may be constant or may vary with latitude, so that on a spherical planet $f = 2\Omega \sin \vartheta$ and on the β -plane $f = f_0 + \beta y$.

2.1.2 Mass continuity equation

The mass contained in a fluid column of height h and cross-sectional area A is given by $\int_A \rho_0 h dA$ (see Fig. 2.2). If there is a net flux of fluid across the column boundary (by advection) then this must be balanced by a net increase in the mass in A , and therefore a net increase in the height of the water column. The mass convergence into the column is given by

$$F_m = \text{mass flux in} = - \int_S \rho_0 \mathbf{u} \cdot d\mathbf{S}, \quad (2.9)$$

where S is the area of the vertical boundary of the column. The surface area of the column is composed of elements of area $h\mathbf{n} \delta l$, where δl is a line element circumscribing the column and $\mathbf{n}$ is a unit vector perpendicular to the boundary, pointing outwards. Thus (2.9) becomes

$$F_m = - \oint \rho_0 h \mathbf{u} \cdot \mathbf{n} dl. \quad (2.10)$$

Using the divergence theorem in two dimensions, (2.10) simplifies to

$$F_m = - \int_A \nabla \cdot (\rho_0 h \mathbf{u}) dA, \quad (2.11)$$

where the integral is over the cross-sectional area of the fluid column (looking down from above). This is balanced by the local increase in height of the water column, given by

$$F_m = \frac{d}{dt} \int \rho_0 dV = \frac{d}{dt} \int_A \rho_0 h dA = \int_A \rho_0 \frac{\partial h}{\partial t} dA. \quad (2.12)$$

Because ρ_0 is constant, the balance between (2.11) and (2.12) leads to

$$\int_A \left[\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) \right] dA = 0, \quad (2.13)$$

and because the area is arbitrary the integrand itself must vanish, whence,

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = 0, \quad (2.14)$$

or equivalently

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0. \quad (2.15)$$

This derivation holds whether or not the lower surface is flat. If it is, then $h = \eta$, and if not $h = \eta - \eta_b$.

From the 3D mass conservation equation

Since the fluid is incompressible, the three-dimensional mass continuity equation is just $\nabla \cdot \mathbf{v} = 0$. Writing this out in component form

$$\frac{\partial w}{\partial z} = - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = - \nabla \cdot \mathbf{u}. \quad (2.16)$$

Integrating this from the bottom of the fluid to the top, and using the boundary conditions of w (express w in terms of h) gives (2.15). Details left to the reader.

2.2 REDUCED GRAVITY EQUATIONS

Consider now a single shallow moving layer of fluid on top of a deep, quiescent fluid layer (Fig. 2.3), and beneath a fluid of negligible inertia. This configuration is often used as a model of the upper ocean: the upper layer represents flow in perhaps the upper few hundred metres of the ocean, the lower layer being the near-stagnant abyss. If we turn the model upside-down we have a perhaps slightly less realistic model of the atmosphere: the lower layer represents motion in the troposphere above which lies an inactive stratosphere. The equations of motion are virtually the same in both cases.

The Shallow Water Equations

For a single-layer fluid, and including the Coriolis term, the inviscid shallow water equations are

$$\text{momentum:} \quad \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta. \quad (\text{SW.1})$$

$$\text{mass continuity:} \quad \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0 \quad \text{or} \quad \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \quad (\text{SW.2})$$

where $\mathbf{u}$ is the horizontal velocity, h is the total fluid thickness, η is the height of the upper free surface and η_b is the height of the lower surface (the bottom topography). Thus, $h(x, y, t) = \eta(x, y, t) - \eta_b(x, y)$. The material derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y}, \quad (\text{SW.3})$$

with the rightmost expression holding in Cartesian coordinates.

2.2.1 Pressure gradient in the active layer

We will derive the equations for the oceanic case (active layer on top) in two cases, which differ slightly in the assumption made about the upper surface.

I Free upper surface

The pressure in the upper layer is given by integrating the hydrostatic equation down from the upper surface. Thus, at a height z in the upper layer

$$p_1(z) = g\rho_1(\eta_0 - z), \quad (2.17)$$

where η_0 is the height of the upper surface. Hence, everywhere in the upper layer,

$$\frac{1}{\rho_1} \nabla p_1 = g\nabla\eta_0, \quad (2.18)$$

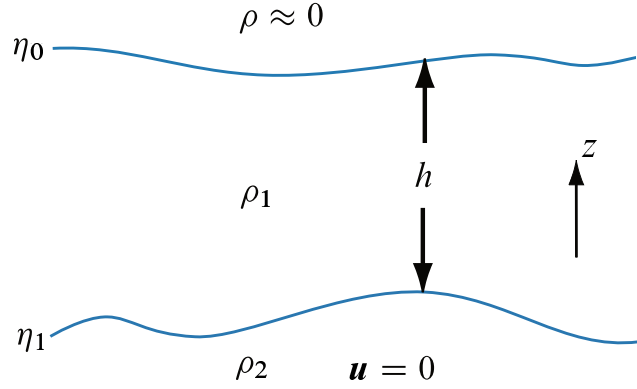
and the momentum equation is

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta_0. \quad (2.19)$$

In the lower layer the pressure is also given by the weight of the fluid above it. Thus, at some level z in the lower layer,

$$p_2(z) = \rho_1 g(\eta_0 - \eta_1) + \rho_2 g(\eta_1 - z). \quad (2.20)$$

Figure 2.3 The reduced gravity shallow water system. An active layer lies over a deep, more dense, quiescent layer. In a common variation the upper surface is held flat by a rigid lid, and $\eta_0 = 0$.



But if this layer is motionless the horizontal pressure gradient in it is zero and therefore

$$\rho_1 g \eta_0 = -\rho_1 g' \eta_1 + \text{constant}, \quad (2.21)$$

where $g' = g(\rho_2 - \rho_1)/\rho_1$ is the *reduced gravity*. The momentum equation becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = g' \nabla \eta_1. \quad (2.22)$$

The equations are completed by the usual mass conservation equation,

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0, \quad (2.23)$$

where $h = \eta_0 - \eta_1$. Because $g \gg g'$, (2.21) shows that surface displacements are *much smaller* than the displacements at the interior interface. We see this in the real ocean where the mean interior isopycnal displacements may be several tens of metres but variations in the mean height of ocean surface are of the order of centimetres.

II The rigid lid approximation

The smallness of the upper surface displacement suggests that we will make little error if we impose a *rigid lid* at the top of the fluid. Displacements are no longer allowed, but the lid will in general impart a pressure force to the fluid. Suppose that this is $P(x, y, t)$, then the horizontal pressure gradient in the upper layer is simply

$$\nabla p_1 = \nabla P. \quad (2.24)$$

The pressure in the lower layer is again given by hydrostasy, and is

$$p_2 = -\rho_1 g \eta_1 + \rho_2 g(\eta_1 - z) + P = \rho_1 g h - \rho_2 g(h + z) + P, \quad (2.25)$$

so that

$$\nabla p_2 = -g(\rho_2 - \rho_1) \nabla h + \nabla P. \quad (2.26)$$

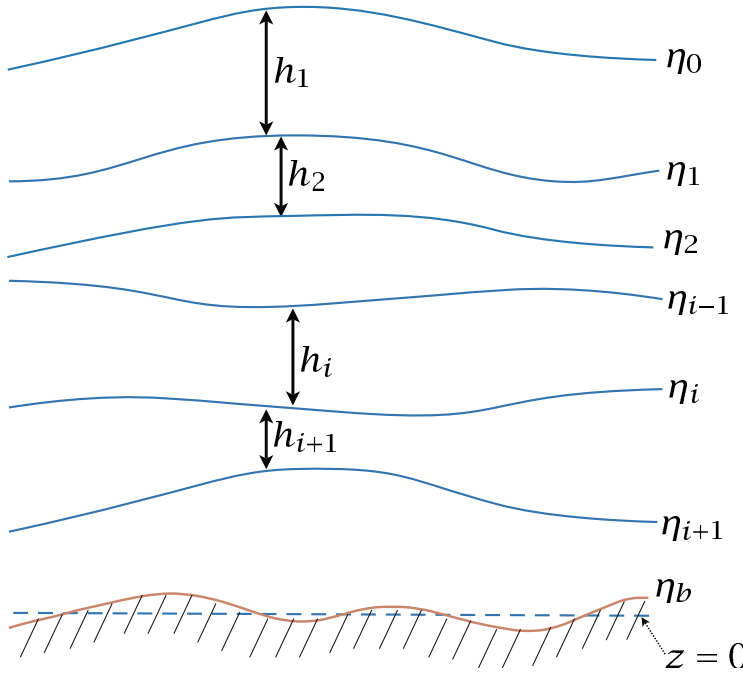


Figure 2.4 The multi-layer shallow water system. The layers are numbered from the top down. The coordinates of the interfaces are denoted by η , and the layer thicknesses by h , so that $h_i = \eta_i - \eta_{i-1}$.

Then if $\nabla p_2 = 0$ we have

$$g(\rho_2 - \rho_1)\nabla h = \nabla P, \quad (2.27)$$

and the momentum equation for the upper layer is just

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g'_1 \nabla h. \quad (2.28)$$

where $g' = g(\rho_2 - \rho_1)/\rho_1$. These equations differ from the usual shallow water equations only in the use of a reduced gravity g' in place of g itself. It is the density *difference* between the two layers that is important. Similarly, if we take a shallow water system, with the moving layer on the bottom, and we suppose that overlying it is a stationary fluid of finite density, then we would easily find that the fluid equations for the moving layer are the same as if the fluid on top had zero inertia, except that g would be replaced by an appropriate reduced gravity (problem 2.??).

2.3 ♦ MULTI-LAYER SHALLOW WATER EQUATIONS

We now consider the dynamics of multiple layers of fluid stacked on top of each other. This is a crude representation of continuous stratification, but it turns out to be a powerful model of many geophysically interesting phenomena as well as being physically realizable in the laboratory. The pressure is continuous across the interface, but the density jumps discontinuously and this allows the horizontal velocity to have a corresponding discontinuity. The set up is illustrated in Fig. 2.4.

In each layer pressure is given by the hydrostatic approximation, and so anywhere in the interior we can find the pressure by integrating down from the top. Thus, at a height z in the first layer we have

$$p_1 = \rho_1 g(\eta_0 - z), \quad (2.29)$$

and in the second layer,

$$p_2 = \rho_1 g(\eta_0 - \eta_1) + \rho_2 g(\eta_1 - z) = \rho_1 g\eta_0 + \rho_1 g'_1 \eta_1 - \rho_2 g z, \quad (2.30)$$

where $g'_1 = g(\rho_2 - \rho_1)/\rho_1$, and so on. The term involving z is irrelevant for the dynamics, because only the horizontal derivative enters the equation of motion. Omitting this term, for the n th layer the dynamical pressure is given by the sum from the top down:

$$p_n = \rho_1 \sum_{i=0}^{n-1} g'_i \eta_i, \quad (2.31)$$

where $g'_i = g(\rho_{i+1} - \rho_i)/\rho_1$ (but $g_0 = g$). The interface displacements may be expressed in terms of the layer thicknesses by summing from the bottom up:

$$\eta_n = \eta_b + \sum_{i=n+1}^{i=N} h_i. \quad (2.32)$$

The momentum equation for each layer may then be written, in general,

$$\frac{D\mathbf{u}_n}{Dt} + \mathbf{f} \times \mathbf{u}_n = -\frac{1}{\rho_n} \nabla p_n, \quad (2.33)$$

where the pressure is given by (2.31) and in terms of the layer depths using (2.32). If we make the Boussinesq approximation then ρ_n on the right-hand side of (2.33) is replaced by ρ_1 .

The mass conservation equation for each layer has the same form as the single-layer case, and is

$$\frac{Dh_n}{Dt} + h_n \nabla \cdot \mathbf{u}_n = 0. \quad (2.34)$$

The two-layer case

The two-layer model (Fig. 2.5) is the simplest model to capture the effects of stratification. Evaluating the pressures using (2.31) and (2.32) we find:

$$p_1 = \rho_1 g\eta_0 = \rho_1 g(h_1 + h_2 + \eta_b) \quad (2.35a)$$

$$p_2 = \rho_1 [g\eta_0 + g'_1 \eta_1] = \rho_1 [g(h_1 + h_2 + \eta_b) + g'_1(h_2 + \eta_b)]. \quad (2.35b)$$

The momentum equations for the two layers are then

$$\frac{D\mathbf{u}_1}{Dt} + \mathbf{f} \times \mathbf{u}_1 = -g\nabla\eta_0 = -g\nabla(h_1 + h_2 + \eta_b). \quad (2.36a)$$

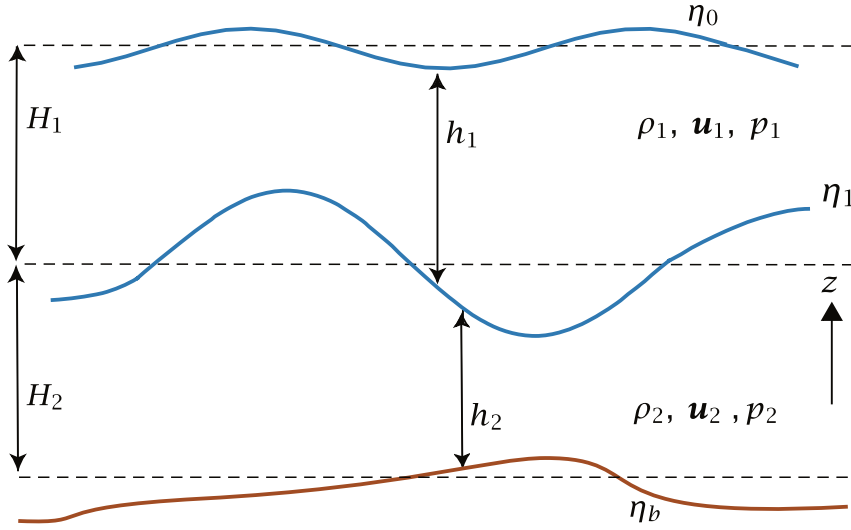


Figure 2.5 The two-layer shallow water system. A fluid of density ρ_1 lies over a denser fluid of density ρ_2 . In the reduced gravity case the lower layer may be arbitrarily thick and is assumed stationary and so has no horizontal pressure gradient. In the ‘rigid-lid’ approximation the top surface displacement is neglected, but there is then a non-zero pressure gradient induced by the lid.

and in the bottom layer

$$\begin{aligned} \frac{D\mathbf{u}_2}{Dt} + \mathbf{f} \times \mathbf{u}_2 &= -\frac{\rho_1}{\rho_2} (g\nabla\eta_0 + g'_1\nabla\eta_1) \\ &= -\frac{\rho_1}{\rho_2} [g\nabla(\eta_b + h_1 + h_2) + g'_1\nabla(h_2 + \eta_b)]. \end{aligned} \quad (2.36b)$$

In the Boussinesq approximation ρ_1/ρ_2 is replaced by unity.

The mass conservation equation for each layer, i , of a multi-layer model is

$$\frac{\partial h_i}{\partial t} + \nabla \cdot (\mathbf{u}_i h_i) = 0. \quad (2.37)$$

2.4 GEOSTROPHIC AND THERMAL-WIND BALANCE

Geostrophic balance occurs in the shallow water equations, just as in the continuously stratified equations, when the Rossby number U/fL is small and the Coriolis term dominates the advective terms in the momentum equation. In the single-layer shallow water equations the geostrophic flow is:

$$\mathbf{f} \times \mathbf{u}_g = -g\nabla\eta. \quad (2.38)$$

Thus, the geostrophic velocity is proportional to the slope of the surface, as sketched in Fig. 2.6. (For the rest of this section, we will drop the subscript g , and take all velocities to be geostrophic.)

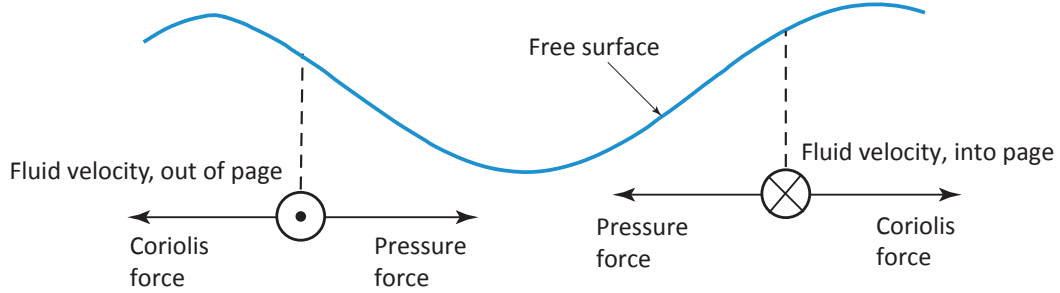


Figure 2.6 Geostrophic flow in a shallow water system, with a positive value of the Coriolis parameter f , as in the Northern Hemisphere. The pressure force is directed down the gradient of the height field, and this can be balanced by the Coriolis force if the fluid velocity is at right angles to it. If f were negative, the geostrophic flow would be reversed.

In both the single-layer and multi-layer cases, the slope of an interfacial surface is directly related to the difference in pressure gradient on either side and so, by geostrophic balance, to the shear of the flow. This is the shallow water analogue of the thermal wind relation. To obtain an expression for this, consider the interface, η , between two layers labelled 1 and 2. The pressure in two layers is given by the hydrostatic relation and so,

$$p_1 = A(x, y) - \rho_1 g z \quad (\text{at some } z \text{ in layer 1}) \quad (2.39a)$$

$$\begin{aligned} p_2 &= A(x, y) - \rho_1 g \eta + \rho_2 g (\eta - z) \\ &= A(x, y) + \rho_1 g'_1 \eta - \rho_2 g z \quad (\text{at some } z \text{ in layer 2}) \end{aligned} \quad (2.39b)$$

where $A(x, y)$ is a function of integration. Thus we find

$$\frac{1}{\rho_1} \nabla(p_1 - p_2) = -g'_1 \nabla \eta. \quad (2.40)$$

If the flow is geostrophically balanced and Boussinesq then, in each layer, the velocity obeys

$$f \mathbf{u}_i = \frac{1}{\rho_i} \mathbf{k} \times \nabla p_i. \quad (2.41)$$

Using (2.40) then gives

$$f(\mathbf{u}_1 - \mathbf{u}_2) = -\mathbf{k} \times g'_1 \nabla \eta, \quad (2.42)$$

or in general

$$f(\mathbf{u}_n - \mathbf{u}_{n+1}) = -\mathbf{k} \times g'_n \nabla \eta. \quad (2.43)$$

This is the thermal wind equation for the shallow water system. It applies at any interface, and it implies *the shear is proportional to the interface slope*, a result known as the ‘Margules relation’ (Fig. 2.7).

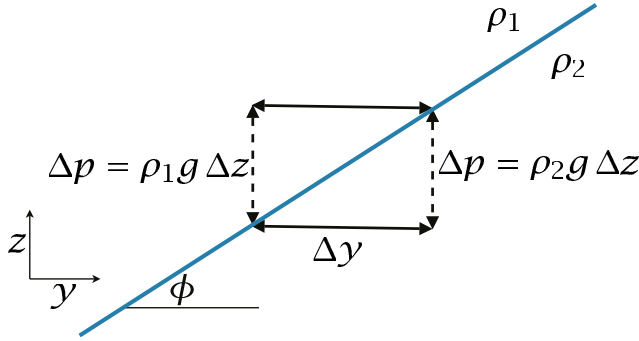


Figure 2.7 Margules' relation: using hydrostasy, the difference in the horizontal pressure gradient between the upper and the lower layer is given by $-g' \rho_1 s$, where $s = \tan \phi = \Delta z / \Delta y$ is the interface slope and $g' = g(\rho_2 - \rho_1) / \rho_1$. Geostrophic balance then gives $f(u_1 - u_2) = g's$, which is a special case of (2.43).

Suppose that we represent the atmosphere by two layers of fluid; a meridionally decreasing temperature may then be represented by an interface that slopes upwards toward the pole. Then, in either hemisphere, we have

$$u_1 - u_2 = \frac{g'_1}{f} \frac{\partial \eta}{\partial y} > 0, \quad (2.44)$$

and the temperature gradient is associated with a positive shear (see problem 2.??).

2.5 CONSERVATION PROPERTIES OF SHALLOW WATER SYSTEMS

There are two common types of conservation property in fluids: (i) material invariants and (ii) integral invariants. Material invariance occurs when a property (ϕ say) is conserved on each fluid element, and so obeys the equation $D\phi/Dt = 0$. An integral invariant is one that is conserved following an integration over some, usually closed, volume; energy is an example.

2.5.1 Potential vorticity: a material invariant

The vorticity of a fluid, denoted $\boldsymbol{\omega}$, is defined to be the curl of the velocity field. Let us also define the shallow water vorticity, $\boldsymbol{\omega}^*$, as the curl of the horizontal velocity. We therefore have:

$$\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}, \quad \boldsymbol{\omega}^* \equiv \nabla \times \mathbf{u}. \quad (2.45)$$

Because $\partial u / \partial z = \partial v / \partial z = 0$, only the vertical component of $\boldsymbol{\omega}^*$ is non-zero and

$$\boldsymbol{\omega}^* = \mathbf{k} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \equiv \mathbf{k} \zeta. \quad (2.46)$$

Considering first the non-rotating case, we use the vector identity

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \frac{1}{2} \nabla (\mathbf{u} \cdot \mathbf{u}) - \mathbf{u} \times (\nabla \times \mathbf{u}), \quad (2.47)$$

to write the momentum equation, (2.8) with $f = 0$, as

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega}^* \times \mathbf{u} = -\nabla \left(g\eta + \frac{1}{2} \mathbf{u}^2 \right). \quad (2.48)$$

To obtain an evolution equation for the vorticity we take the curl of (2.48), and make use of the vector identity

$$\begin{aligned}\nabla \times (\boldsymbol{\omega}^* \times \mathbf{u}) &= (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}^* - (\boldsymbol{\omega}^* \cdot \nabla) \mathbf{u} + \boldsymbol{\omega}^* \nabla \cdot \mathbf{u} - \mathbf{u} \nabla \cdot \boldsymbol{\omega}^* \\ &= (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}^* + \boldsymbol{\omega}^* \nabla \cdot \mathbf{u},\end{aligned}\quad (2.49)$$

using the fact that $\nabla \cdot \boldsymbol{\omega}^*$ is the divergence of a curl and therefore zero, and $(\boldsymbol{\omega}^* \cdot \nabla) \mathbf{u} = 0$ because $\boldsymbol{\omega}^*$ is perpendicular to the surface in which $\mathbf{u}$ varies. Taking the curl of (2.48) gives

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla) \zeta = -\zeta \nabla \cdot \mathbf{u}, \quad (2.50)$$

where $\zeta = \mathbf{k} \cdot \boldsymbol{\omega}^*$. Now, the mass conservation equation may be written as

$$-\zeta \nabla \cdot \mathbf{u} = \frac{\zeta}{h} \frac{Dh}{Dt}, \quad (2.51)$$

and using this (2.50) becomes

$$\frac{D\zeta}{Dt} = \frac{\zeta}{h} \frac{Dh}{Dt}, \quad (2.52)$$

which simplifies to

$$\frac{D}{Dt} \left(\frac{\zeta}{h} \right) = 0. \quad (2.53)$$

The important quantity ζ/h , often denoted by Q , is known as the *potential vorticity*, and (2.53) is the potential vorticity equation. We re-derive this conservation law in a more general way in section ??

Because Q is conserved on parcels, then so is any function of Q ; that is, $F(Q)$ is a material invariant, where F is any function. To see this algebraically, multiply (2.53) by $F'(Q)$, the derivative of F with respect to Q , giving

$$F'(Q) \frac{DQ}{Dt} = \frac{D}{Dt} F(Q) = 0. \quad (2.54)$$

Since F is arbitrary there are an infinite number of material invariants corresponding to different choices of F .

Effects of rotation

In a rotating frame of reference, the shallow water momentum equation is

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (2.55)$$

where (as before) $\mathbf{f} = f\mathbf{k}$. This may be written in vector invariant form as

$$\frac{\partial \mathbf{u}}{\partial t} + (\boldsymbol{\omega}^* + \mathbf{f}) \times \mathbf{u} = -\nabla \left(g\eta + \frac{1}{2} \mathbf{u}^2 \right), \quad (2.56)$$

and taking the curl of this gives the vorticity equation

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)(\zeta + f) = -(f + \zeta)\nabla \cdot \mathbf{u}. \quad (2.57)$$

This is the same as the shallow water vorticity equation in a non-rotating frame, save that ζ is replaced by $\zeta + f$, the reason for this being that f is the vorticity that the fluid has by virtue of the background rotation. Thus, (2.57) is simply the equation of motion for the total or absolute vorticity, $\boldsymbol{\omega}_a = \boldsymbol{\omega}^* + \mathbf{f} = (\zeta + f)\mathbf{k}$.

The potential vorticity equation in the rotating case follows, much as in the non-rotating case, by combining (2.57) with the mass conservation equation, giving

$$\frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = 0. \quad (2.58)$$

That is, $Q \equiv (\zeta + f)/h$, the potential vorticity in a rotating shallow system, is a material invariant.

Vorticity and circulation

Although vorticity itself is not a material invariant, its integral over a horizontal material area is invariant. To demonstrate this in the non-rotating case, consider the integral

$$C = \int_A \zeta \, dA = \int_A Qh \, dA, \quad (2.59)$$

over a surface A , the cross-sectional area of a column of height h (as in Fig. 2.2). Taking the material derivative of this gives

$$\frac{DC}{Dt} = \int_A \frac{DQ}{Dt} h \, dA + \int_A Q \frac{D}{Dt} (h \, dA). \quad (2.60)$$

The first term is zero, by (2.52); the second term is just the derivative of the volume of a column of fluid and it too is zero, by mass conservation. Thus,

$$\frac{DC}{Dt} = \frac{D}{Dt} \int_A \zeta \, dA = 0. \quad (2.61)$$

Thus, the integral of the vorticity over a some cross-sectional area of the fluid is unchanging, although both the vorticity and area of the fluid may individually change. Using Stokes' theorem, it may be written as

$$\frac{DC}{Dt} = \frac{D}{Dt} \oint \mathbf{u} \cdot d\mathbf{l}, \quad (2.62)$$

where the line integral is around the boundary of A . This is an example of Kelvin's circulation theorem, which we shall meet again in a more general form in chapter ??, where we also consider the rotating case.

A slight generalization of (2.61) is possible. Consider the integral $I = \int F(Q)h \, dA$ where again F is any differentiable function of its argument. It is clear that

$$\frac{D}{Dt} \int_A F(Q)h \, dA = 0. \quad (2.63)$$

If the area of integration in (2.48) or (2.63) is the whole domain (enclosed by frictionless walls, for example) then it is clear that the integral of $hF(Q)$ is a constant, including as a special case the integral of ζ .

2.5.2 Energy conservation: an integral invariant

Since we have made various simplifications in deriving the shallow water system, it is not self-evident that energy should be conserved, or indeed what form the energy takes. The kinetic energy density (KE), that is the kinetic energy per unit area, is $\rho_0 h \mathbf{u}^2 / 2$. The potential energy density of the fluid is

$$PE = \int_0^h \rho_0 g z \, dz = \frac{1}{2} \rho_0 g h^2. \quad (2.64)$$

The factor ρ_0 appears in both kinetic and potential energies and, because it is a constant, we will omit it. For algebraic simplicity we also assume the bottom is flat, at $z = 0$.

Using the mass conservation equation (2.15) we obtain an equation for the evolution of potential energy density, namely

$$\frac{D}{Dt} \frac{gh^2}{2} + gh^2 \nabla \cdot \mathbf{u} = 0 \quad (2.65a)$$

or

$$\frac{\partial}{\partial t} \frac{gh^2}{2} + \nabla \cdot \left(\mathbf{u} \frac{gh^2}{2} \right) + \frac{gh^2}{2} \nabla \cdot \mathbf{u} = 0. \quad (2.65b)$$

From the momentum and mass continuity equations we obtain an equation for the evolution of kinetic energy density, namely

$$\frac{D}{Dt} \frac{h\mathbf{u}^2}{2} + \frac{\mathbf{u}^2 h}{2} \nabla \cdot \mathbf{u} = -g\mathbf{u} \cdot \nabla \frac{h^2}{2} \quad (2.66a)$$

or

$$\frac{\partial}{\partial t} \frac{h\mathbf{u}^2}{2} + \nabla \cdot \left(\mathbf{u} \frac{h\mathbf{u}^2}{2} \right) + g\mathbf{u} \cdot \nabla \frac{h^2}{2} = 0. \quad (2.66b)$$

Adding (2.65b) and (2.66b) we obtain

$$\frac{\partial}{\partial t} \frac{1}{2} (h\mathbf{u}^2 + gh^2) + \nabla \cdot \left[\frac{1}{2} \mathbf{u} (gh^2 + h\mathbf{u}^2 + gh^2) \right] = 0, \quad (2.67)$$

or

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{F} = 0, \quad (2.68)$$

where $E = KE + PE = (h\mathbf{u}^2 + gh^2)/2$ is the density of the total energy and $\mathbf{F} = \mathbf{u}(h\mathbf{u}^2/2 + gh^2)$ is the energy flux. If the fluid is confined to a domain bounded by rigid walls, on which the normal component of velocity vanishes, then on integrating (2.67) over that area and using Gauss's theorem, the total energy is seen to be conserved; that is

$$\frac{d\hat{E}}{dt} = \frac{1}{2} \frac{d}{dt} \int_A (h\mathbf{u}^2 + gh^2) dA = 0. \quad (2.69)$$

Such an energy principle also holds in the case with bottom topography. Note that, as we found in the case for a compressible fluid in chapter ??, the energy flux in (2.68) is not just the energy density multiplied by the velocity; it contains an additional term $guh^2/2$, and this represents the energy transfer occurring when the fluid does work against the pressure force (see problem 2.??).

2.6 SHALLOW WATER WAVES

Let us now look at the gravity waves that occur in shallow water. To isolate the essence of the phenomena, we will consider waves in a single fluid layer, with a flat bottom and a free upper surface, in which gravity provides the sole restoring force.

2.6.1 Non-rotating shallow water waves

Given a flat bottom the fluid thickness is equal to the free surface displacement (Fig. 2.1), and taking the basic state of the fluid to be at rest we let

$$h(x, y, t) = H + h'(x, y, t) = H + \eta'(x, y, t), \quad (2.70a)$$

$$\mathbf{u}(x, y, t) = \mathbf{u}'(x, y, t). \quad (2.70b)$$

The mass conservation equation, (2.15), then becomes

$$\frac{\partial \eta'}{\partial t} + (H + \eta') \nabla \cdot \mathbf{u}' + \mathbf{u}' \cdot \nabla \eta' = 0, \quad (2.71)$$

and neglecting squares of small quantities this yields the linear equation

$$\frac{\partial \eta'}{\partial t} + H \nabla \cdot \mathbf{u}' = 0. \quad (2.72)$$

Similarly, linearizing the momentum equation, (2.8) with $\mathbf{f} = 0$, yields

$$\frac{\partial \mathbf{u}'}{\partial t} = -g \nabla \eta'. \quad (2.73)$$

Eliminating velocity by differentiating (2.72) with respect to time and taking the divergence of (2.73) leads to

$$\frac{\partial^2 \eta'}{\partial t^2} - gH \nabla^2 \eta' = 0, \quad (2.74)$$

which may be recognized as a wave equation. We can find the dispersion relationship for this by substituting the trial solution

$$\eta' = \text{Re } \tilde{\eta} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (2.75)$$

where $\tilde{\eta}$ is a complex constant, $\mathbf{k} = i\mathbf{k} + j\mathbf{l}$ is the horizontal wavenumber and Re indicates that the real part of the solution should be taken. If, for simplicity, we restrict attention to the one-dimensional problem, with no variation in the y -direction, then substituting into (2.74) leads to the dispersion relationship

$$\omega = \pm ck, \quad (2.76)$$

where $c = \sqrt{gH}$; that is, the wave speed is proportional to the square root of the mean fluid depth and is independent of the wavenumber — the waves are dispersionless. The general solution is a superposition of all such waves, with the amplitudes of each wave (or Fourier component) being determined by the Fourier decomposition of the initial conditions.

Because the waves are dispersionless, the general solution can be written as

$$\eta'(x, t) = \frac{1}{2} [F(x - ct) + F(x + ct)], \quad (2.77)$$

where $F(x)$ is the height field at $t = 0$. From this, it is easy to see that the shape of an initial disturbance is preserved as it propagates both to the right and to the left at speed c , (see also problem 2.??).

2.6.2 Rotating shallow water (Poincaré) waves

We now consider the effects of rotation on shallow water waves. Linearizing the rotating, flat-bottomed f -plane shallow water equations [i.e., (SW.1) and (SW.2) on page 39] about a state of rest we obtain

$$\frac{\partial u'}{\partial t} - f_0 v' = -g \frac{\partial \eta'}{\partial x}, \quad \frac{\partial v'}{\partial t} + f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad \frac{\partial \eta'}{\partial t} + H \left(\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) = 0. \quad (2.78a,b,c)$$

To obtain a dispersion relationship we let

$$(u, v, \eta) = (\tilde{u}, \tilde{v}, \tilde{\eta}) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (2.79)$$

and substitute into (2.78), giving

$$\begin{pmatrix} -i\omega & -f_0 & i g k \\ f_0 & -i\omega & i g l \\ i H k & i H l & -i\omega \end{pmatrix} \begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{\eta} \end{pmatrix} = 0. \quad (2.80)$$

This homogeneous equation has non-trivial solutions only if the determinant of the matrix vanishes. This condition gives

$$\omega(\omega^2 - f_0^2 - c^2 K^2) = 0. \quad (2.81)$$

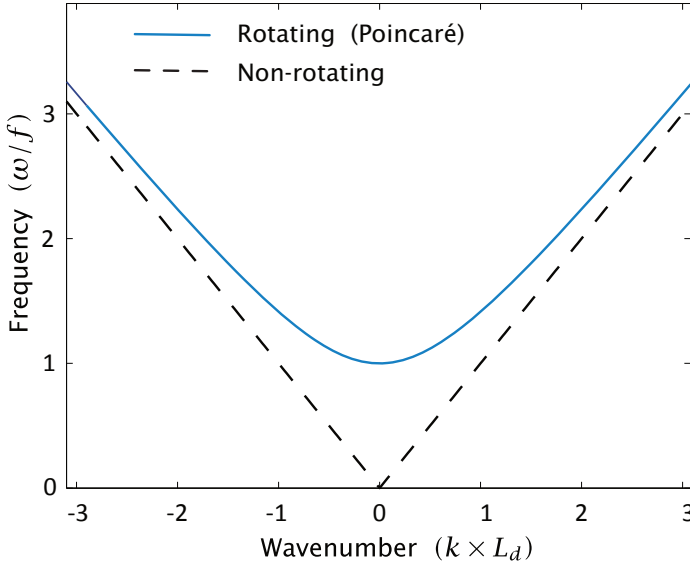


Figure 2.8 Dispersion relation for Poincaré waves (solid) and non-rotating shallow water waves (dashed). Frequency is scaled by the Coriolis frequency f , and wavenumber by the inverse deformation radius $\sqrt{gH}/f$. For small wavenumbers the frequency is approximately f ; for high wavenumbers it asymptotes to that of non-rotating waves.

where $K^2 = k^2 + l^2$ and $c^2 = gH$. There are two classes of solution to (2.81). The first is simply $\omega = 0$, i.e., time-independent flow corresponding to geostrophic balance in (2.78). Because geostrophic balance gives a divergence-free velocity field for a constant Coriolis parameter the equations are satisfied by a time-independent solution. The second set of solutions gives the dispersion relation

$$\omega^2 = f_0^2 + c^2(k^2 + l^2), \quad (2.82)$$

or

$$\omega^2 = f_0^2 + gH(k^2 + l^2). \quad (2.83)$$

The corresponding waves are known as *Poincaré waves*,² and the dispersion relationship is illustrated in Fig. 2.8. Note that the frequency is always greater than the Coriolis frequency f_0 . There are two interesting limits.

(i) *The short wave limit.* If

$$K^2 \gg \frac{f_0^2}{gH}, \quad (2.84)$$

where $K^2 = k^2 + l^2$, then the dispersion relationship reduces to that of the non-rotating case (2.76). This condition is equivalent to requiring that the wavelength be much shorter than the *deformation radius*, $L_d \equiv \sqrt{gH}/f$. Specifically, if $l = 0$ and $\lambda = 2\pi/k$ is the wavelength, the condition is

$$\lambda^2 \ll L_d^2 (2\pi)^2. \quad (2.85)$$

The numerical factor of $(2\pi)^2$ is more than an order of magnitude, so care must be taken when deciding if the condition is satisfied in particular cases. Furthermore, the wavelength

must still be longer than the depth of the fluid, otherwise the shallow water condition is not met.

(ii) *The long wave limit.* If

$$K^2 \ll \frac{f_0^2}{gH}, \quad (2.86)$$

that is if the wavelength is much longer than the deformation radius L_d , then the dispersion relationship is

$$\omega = f_0. \quad (2.87)$$

These are known as *inertial oscillations*. The equations of motion giving rise to them are

$$\frac{\partial u'}{\partial t} - f_0 v' = 0, \quad \frac{\partial v'}{\partial t} + f_0 u' = 0, \quad (2.88)$$

which are equivalent to material equations for free particles in a rotating frame, unconstrained by pressure forces, namely

$$\frac{d^2 x}{dt^2} - f_0 v = 0, \quad \frac{d^2 y}{dt^2} + f_0 u = 0. \quad (2.89)$$

2.6.3 Kelvin waves

The Kelvin wave is a particular type of gravity wave that exists in the presence of both rotation and a lateral boundary. Suppose there is a solid boundary at $y = 0$; clearly harmonic solutions in the y -direction are not allowable, as these would not satisfy the condition of no normal flow at the boundary. Do any wave-like solutions exist? The affirmative answer to this question was provided by Kelvin and the associated waves are now eponymously known as *Kelvin waves*.³ We begin with the linearized shallow water equations, namely

$$\frac{\partial u'}{\partial t} - f_0 v' = -g \frac{\partial \eta'}{\partial x}, \quad \frac{\partial v'}{\partial t} + f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad \frac{\partial \eta'}{\partial t} + H \left(\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) = 0. \quad (2.90a,b,c)$$

The fact that $v' = 0$ at $y = 0$ suggests that we look for a solution with $v' = 0$ everywhere, whence these equations become

$$\frac{\partial u'}{\partial t} = -g \frac{\partial \eta'}{\partial x}, \quad f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad \frac{\partial \eta'}{\partial t} + H \frac{\partial u'}{\partial x} = 0. \quad (2.91a,b,c)$$

Equations (2.91a) and (2.91c) lead to the standard wave equation

$$\frac{\partial^2 u'}{\partial t^2} = c^2 \frac{\partial^2 u'}{\partial x^2}, \quad (2.92)$$

where $c = \sqrt{gH}$, the usual wave speed of shallow water waves. The solution of (2.92) is

$$u' = F_1(x + ct, y) + F_2(x - ct, y), \quad (2.93)$$

with corresponding surface displacement

$$\eta' = \sqrt{H/g} [-F_1(x + ct, y) + F_2(x - ct, y)]. \quad (2.94)$$

The solution represents the superposition of two waves, one (F_1) travelling in the negative x -direction, and the other in the positive x -direction. To obtain the y dependence of these functions we use (2.91b) which gives

$$\frac{\partial F_1}{\partial y} = \frac{f_0}{\sqrt{gH}} F_1, \quad \frac{\partial F_2}{\partial y} = -\frac{f_0}{\sqrt{gH}} F_2, \quad (2.95)$$

with solutions

$$F_1 = F(x + ct)e^{y/L_d} \quad F_2 = G(x - ct)e^{-y/L_d}, \quad (2.96)$$

where $L_d = \sqrt{gH}/f_0$ is the radius of deformation. If we consider flow in the half-plane in which $y > 0$, then for positive f_0 the solution F_1 grows exponentially away from the wall, and so fails to satisfy the condition of boundedness at infinity. It thus must be eliminated, leaving the general solution

$$\begin{aligned} u' &= e^{-y/L_d} G(x - ct), & v' &= 0, \\ \eta' &= \sqrt{H/g} e^{-y/L_d} G(x - ct). \end{aligned} \quad (2.97a,b,c)$$

These are Kelvin waves, and they decay exponentially away from the boundary. In general, for f_0 positive the boundary is to the right of an observer moving with the wave. Given a constant Coriolis parameter, we could equally well have obtained a solution on a meridional wall, in which case we would find that the wave again moves such that the wall is to the right of the wave direction. (This is obvious once it is realized that f -plane dynamics are isotropic in x and y .) Thus, in the Northern Hemisphere the wave moves anticlockwise round a basin, and conversely in the Southern Hemisphere, and in both hemispheres the direction is cyclonic.

2.7 GEOSTROPHIC ADJUSTMENT

Large-scale, extratropical circulation of the atmosphere is in near-geostrophic balance. Why is this? Why should the Rossby number be small? It turns out there is in fact a powerful and ubiquitous process whereby a fluid in an initially unbalanced state naturally evolves toward a state of geostrophic balance, namely *geostrophic adjustment*. This process occurs quite generally in rotating fluids, whether stratified or not. We consider the free evolution of a single shallow layer of fluid whose initial state is manifestly unbalanced, and we will suppose that surface

displacements are small so that the evolution of the system is described by the linearized shallow equations of motion. These are

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad \frac{\partial \eta}{\partial t} + H\nabla \cdot \mathbf{u} = 0, \quad (2.98a,b)$$

where η is the free surface displacement and H is the mean fluid depth, and we omit the primes on the linearized variables.

2.7.1 Non-rotating flow

We consider first the non-rotating problem set, with little loss of generality, in one dimension. We suppose that initially the fluid is at rest but with a simple discontinuity in the height field so that

$$\eta(x, t = 0) = \begin{cases} +\eta_0 & x < 0 \\ -\eta_0 & x > 0 \end{cases} \quad (2.99)$$

and $u(x, t = 0) = 0$ everywhere. We can realize these initial conditions physically by separating two fluid masses of different depths by a thin dividing wall, and then quickly removing the wall. What is the subsequent evolution of the fluid? The general solution to the linear problem is given by (2.77) where the functional form is determined by the initial conditions so that here

$$F(x) = \eta(x, t = 0) = -\eta_0 \operatorname{sgn}(x). \quad (2.100)$$

Equation (2.77) states that this initial pattern is propagated to the right and to the left. That is, two discontinuities in fluid height move to the right and left at a speed $c = \sqrt{gH}$. Specifically, the solution is

$$\eta(x, t) = -\frac{1}{2}\eta_0[\operatorname{sgn}(x + ct) + \operatorname{sgn}(x - ct)]. \quad (2.101)$$

The initial conditions may be much more complex than a simple front, but, because the waves are dispersionless, the solution is still simply a sum of the translation of those initial conditions to the right and to the left at speed c . The velocity field in this class of problem is obtained from

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x}, \quad (2.102)$$

which gives, using (2.77),

$$u = -\frac{g}{2c}[F(x + ct) - F(x - ct)]. \quad (2.103)$$

Consider the case with initial conditions given by (2.99). At a given location, away from the initial disturbance, the fluid remains at rest and undisturbed until the front arrives. After the front has passed, the fluid surface is again undisturbed and the velocity is uniform and non zero. Specifically:

$$\eta = \begin{cases} -\eta_0 \operatorname{sgn}(x) \\ 0 \end{cases} \quad u = \begin{cases} 0 & |x| > ct \\ (\eta_0 g/c) & |x| < ct. \end{cases} \quad (2.104)$$

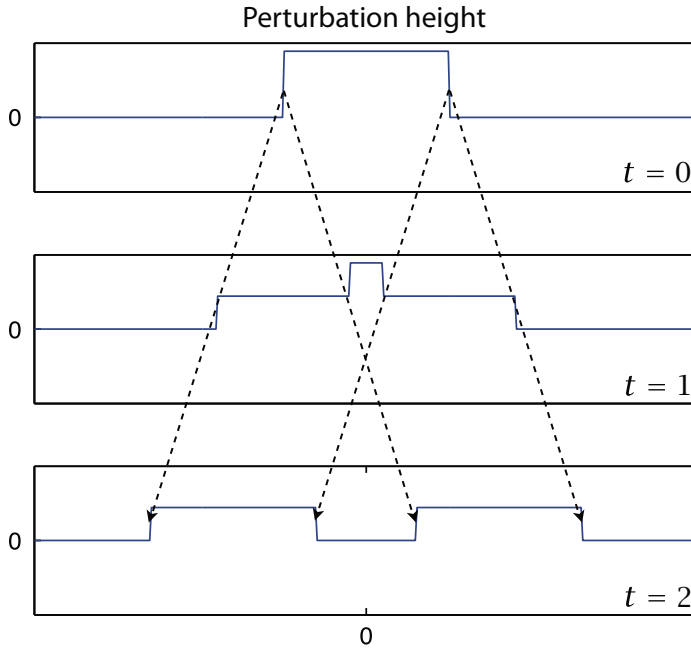


Figure 2.9 The time development of an initial ‘top hat’ height disturbance, with zero initial velocity, in non-rotating flow. Fronts propagate in both directions, and the velocity is non-zero between fronts, but ultimately the disturbance are radiated away to infinity, and the fluid is left at rest with zero perturbation height.

The solution with ‘top-hat’ initial conditions in the height field, and zero initial velocity, is a superposition two discontinuities similar to (2.104) and is illustrated in Fig. 2.9. Two fronts propagate in either direction from each discontinuity and, in this case, the final velocity, as well as the fluid displacement, is zero after all the fronts have passed. That is, the disturbance is radiated completely away.

2.7.2 Rotating flow

Rotation makes a profound difference to the adjustment problem of the shallow water system, because a steady, adjusted, solution can exist with non-zero gradients in the height field — the associated pressure gradients being balanced by the Coriolis force — and potential vorticity conservation provides a powerful constraint on the fluid evolution.⁴ In a rotating shallow fluid that conservation is represented by

$$\frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q = 0, \quad (2.105)$$

where $Q = (\zeta + f)/h$. In the linear case with constant Coriolis parameter, (2.105) becomes

$$\frac{\partial q}{\partial t} = 0, \quad q = \left(\zeta - f_0 \frac{\eta}{H} \right). \quad (2.106)$$

This equation may be obtained either from the linearized velocity and mass conservation equations, (2.98), or from (2.105) directly. In the latter case, we write

$$Q = \frac{\zeta + f_0}{H + \eta} \approx \frac{1}{H}(\zeta + f_0) \left(1 - \frac{\eta}{H}\right) \approx \frac{1}{H} \left(f_0 + \zeta - f_0 \frac{\eta}{H}\right) = \frac{f_0}{H} + \frac{q}{H} \quad (2.107)$$

having used $f_0 \gg |\zeta|$ and $H \gg |\eta|$. The term f_0/H is a constant and so dynamically unimportant, as is the H^{-1} factor multiplying q . Further, the advective term $\mathbf{u} \cdot \nabla Q$ becomes $\mathbf{u} \cdot \nabla q$, and this is second order in perturbed quantities and so is neglected. Thus, making these approximations, (2.105) reduces to (2.106). The potential vorticity field is therefore fixed in space! Of course, this was also true in the non-rotating case where the fluid is initially at rest. Then $q = \zeta = 0$ and the fluid remains irrotational throughout the subsequent evolution of the flow. However, this is rather a weak constraint on the subsequent evolution of the fluid; it does nothing, for example, to prevent the conversion of all the potential energy to kinetic energy. In the rotating case the potential vorticity is non-zero, and potential vorticity conservation and geostrophic balance are all we need to infer the final steady state, assuming it exists, without solving for the details of the flow evolution, as we now see.

With an initial condition for the height field given by (2.99), the initial potential vorticity is given by

$$q(x, y) = \begin{cases} -f_0\eta_0/H & x < 0 \\ f_0\eta_0/H & x > 0, \end{cases} \quad (2.108)$$

and this remains unchanged throughout the adjustment process. The final steady state is then the solution of the equations

$$\zeta - f_0 \frac{\eta}{H} = q(x, y), \quad f_0 u = -g \frac{\partial \eta}{\partial y}, \quad f_0 v = g \frac{\partial \eta}{\partial x}, \quad (2.109a,b,c)$$

where $\zeta = \partial v / \partial x - \partial u / \partial y$. Because the Coriolis parameter is constant, the velocity field is horizontally non-divergent and we may define a streamfunction $\psi = g\eta / f_0$. Equations (2.109) then reduce to

$$\left(\nabla^2 - \frac{1}{L_d^2} \right) \psi = q(x, y), \quad (2.110)$$

where $L_d = \sqrt{gH} / f_0$ is known as the *Rossby radius of deformation* or often just the ‘deformation radius’ or the ‘Rossby radius’. It is a naturally occurring length scale in problems involving both rotation and gravity, and arises in a slightly different form in stratified fluids.

The initial conditions (2.108) admit of a nice analytic solution, for the flow will remain uniform in y , and (2.110) reduces to

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{1}{L_d^2} \psi = \frac{f_0 \eta_0}{H} \text{sgn}(x). \quad (2.111)$$

We solve this separately for $x > 0$ and $x < 0$ and then match the solutions and their first derivatives at $x = 0$, also imposing the condition that the velocity decays to zero as $x \rightarrow \pm\infty$.

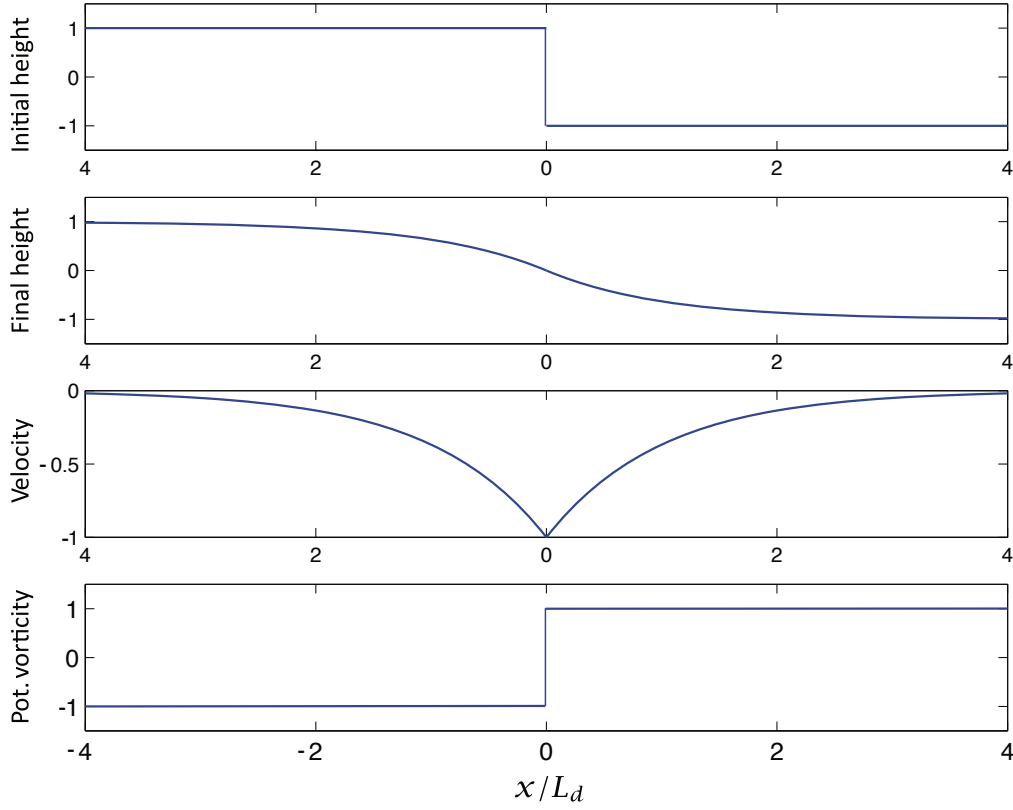


Figure 2.10 Solutions of a linear geostrophic adjustment problem. Top panel: the initial height field, given by (2.99) with $\eta_0 = 1$. Second panel: equilibrium (final) height field, η given by (2.112) and $\eta = f_0 \psi / g$. Third panel: equilibrium geostrophic velocity (normal to the gradient of height field), given by (2.113). Bottom panel: potential vorticity, given by (2.108), and this does not evolve. The distance, x is non-dimensionalized by the deformation radius $L_d = \sqrt{gH}/f_0$, and the velocity by $\eta_0(g/f_0 L_d)$. Changes to the initial state occur only within $\mathcal{O}(L_d)$ of the initial discontinuity; and as $x \rightarrow \pm\infty$ the initial state is unaltered.

The solution is

$$\psi = \begin{cases} -(g\eta_0/f_0)(1 - e^{-x/L_d}) & x > 0 \\ +(g\eta_0/f_0)(1 - e^{x/L_d}) & x < 0. \end{cases} \quad (2.112)$$

The velocity field associated with this is obtained from (2.109b,c), and is

$$u = 0, \quad v = -\frac{g\eta_0}{f_0 L_d} e^{-|x|/L_d}. \quad (2.113)$$

The velocity is perpendicular to the slope of the free surface, and a jet forms along the initial discontinuity, as illustrated in Fig. 2.10.

The important point of this problem is that the variations in the height and field are not radiated away to infinity, as in the non-rotating problem. Rather, potential vorticity conservation

constrains the influence of the adjustment to within a deformation radius (we see now why this name is appropriate) of the initial disturbance. This property is a general one in geostrophic adjustment — it also arises if the initial condition consists of a velocity jump, as considered in problem 2.??.

2.7.3 ♦ Energetics of adjustment

How much of the initial potential energy of the flow is lost to infinity by gravity wave radiation, and how much is converted to kinetic energy? The linear equations (2.98) lead to

$$\frac{1}{2} \frac{\partial}{\partial t} (H\mathbf{u}^2 + g\eta^2) + gH\nabla \cdot (\mathbf{u}\eta) = 0, \quad (2.114)$$

so that energy conservation holds in the form

$$E = \frac{1}{2} \int (H\mathbf{u}^2 + g\eta^2) d\mathbf{x}, \quad \frac{dE}{dt} = 0, \quad (2.115)$$

provided the integral of the divergence term vanishes, as it normally will in a closed domain. The fluid has a non-zero potential energy, $(1/2) \int_{-\infty}^{\infty} g\eta^2 dx$, if there are variations in fluid height, and with the initial conditions (2.99) the initial potential energy is

$$PE_I = \int_0^{\infty} g\eta_0^2 dx. \quad (2.116)$$

This is nominally infinite if the fluid has no boundaries, and the initial potential energy density is $g\eta_0^2/2$ everywhere.

In the non-rotating case, and with initial conditions (2.99), after the front has passed, the potential energy density is zero and the kinetic energy density is $H\mathbf{u}^2/2 = g\eta_0^2/2$, using (2.104) and $c^2 = gH$. Thus, all the potential energy is locally converted to kinetic energy as the front passes, and eventually the kinetic energy is distributed uniformly along the line. In the case illustrated in Fig. 2.9, the potential energy and kinetic energy are both radiated away from the initial disturbance. (Note that although we can superpose the solutions from different initial conditions, we cannot superpose their potential and kinetic energies.) The general point is that the evolution of the disturbance is not confined to its initial location.

In contrast, in the rotating case the conversion from potential to kinetic energy is *largely confined to within a deformation radius of the initial disturbance*, and at locations far from the initial disturbance the initial state is essentially unaltered. The conservation of potential vorticity has prevented the complete conversion of potential energy to kinetic energy, a result that is not sensitive to the precise form of the initial conditions (see also problem 2.??).

In fact, in the rotating case, some of the initial potential energy is converted to kinetic energy, some remains as potential energy and some is lost to infinity; let us calculate these amounts. The final potential energy, after adjustment, is, using (2.112),

$$PE_F = \frac{1}{2} g\eta_0^2 \left[\int_0^{\infty} (1 - e^{-x/L_d})^2 dx + \int_{-\infty}^0 (1 - e^{x/L_d})^2 dx \right]. \quad (2.117)$$

This is nominally infinite, but the change in potential energy is finite and is given by

$$PE_I - PE_F = g\eta_0^2 \int_0^\infty (2e^{-x/L_d} - e^{-2x/L_d}) dx = \frac{3}{2}g\eta_0^2 L_d. \quad (2.118)$$

The initial kinetic energy is zero, because the fluid is at rest, and its final value is, using (2.113),

$$KE_F = \frac{1}{2}H \int \mathbf{u}^2 dx = H \left(\frac{g\eta_0}{fL_d} \right)^2 \int_0^\infty e^{-2x/L_d} dx = \frac{g\eta_0^2 L_d}{2}. \quad (2.119)$$

Thus one-third of the difference between the initial and final potential energies is converted to kinetic energy, and this is trapped within a distance of the order of a deformation radius of the disturbance; the remainder, an amount $gL_d\eta_0^2$ is radiated away and lost to infinity. In any finite region surrounding the initial discontinuity the final energy is less than the initial energy.

2.7.4 A variational perspective

In the non-rotating problem, all of the initial potential energy is eventually radiated away to infinity. In the rotating problem, the final state contains both potential and kinetic energy. Why is the energy not all radiated away to infinity? It is because potential vorticity conservation on parcels prevents all of the energy being dispersed. This suggests that it may be informative to think of the geostrophic adjustment problem as a *variational problem*: we seek to minimize the energy consistent with the conservation of potential vorticity. We stay in the linear approximation in which, because the advection of potential vorticity is neglected, potential vorticity remains constant at each point.

The energy of the flow is given by the sum of potential and kinetic energies, namely

$$\text{energy} = \int (H\mathbf{u}^2 + g\eta^2) dA, \quad (2.120)$$

(where $dA \equiv dx dy$) and the potential vorticity field is

$$q = \zeta - f_0 \frac{\eta}{H} = (v_x - u_y) - f_0 \frac{\eta}{H}, \quad (2.121)$$

where the subscripts x, y denote derivatives. The problem is then to extremize the energy subject to potential vorticity conservation. This is a constrained problem in the calculus of variations, sometimes called an *isoperimetric* problem because of its origins in maximizing the area of a surface for a given perimeter.⁵ The mathematical problem is to extremize the integral

$$I = \int \{H(u^2 + v^2) + g\eta^2 + \lambda(x, y)[(v_x - u_y) - f_0\eta/H]\} dA, \quad (2.122)$$

where $\lambda(x, y)$ is a Lagrange multiplier, undetermined at this stage. It is a function of space: if it were a constant, the integral would merely extremize energy subject to a given integral of

potential vorticity, and rearrangements of potential vorticity (which here we wish to disallow) would leave the integral unaltered.

As there are three independent variables there are three Euler–Lagrange equations that must be solved in order to minimize I . These are

$$\begin{aligned} \frac{\partial L}{\partial \eta} - \frac{\partial}{\partial x} \frac{\partial L}{\partial \eta_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial \eta_y} &= 0, \\ \frac{\partial L}{\partial u} - \frac{\partial}{\partial x} \frac{\partial L}{\partial u_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial u_y} &= 0, \quad \frac{\partial L}{\partial v} - \frac{\partial}{\partial x} \frac{\partial L}{\partial v_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial v_y} = 0, \end{aligned} \quad (2.123)$$

where L is the integrand on the right-hand side of (2.122). Substituting the expression for L into (2.123) gives, after a little algebra,

$$2g\eta - \frac{\lambda f_0}{H} = 0, \quad 2Hu + \frac{\partial \lambda}{\partial y} = 0, \quad 2Hv - \frac{\partial \lambda}{\partial x} = 0, \quad (2.124)$$

and then eliminating λ gives the simple relationships

$$u = -\frac{g}{f_0} \frac{\partial \eta}{\partial y}, \quad v = \frac{g}{f_0} \frac{\partial \eta}{\partial x}, \quad (2.125)$$

which are the equations of geostrophic balance. Thus, in the linear approximation, *geostrophic balance is the minimum energy state for a given field of potential vorticity*.

CHAPTER 3

GEOSTROPHIC THEORY

WEEKS 7 TO 8

This chapter is concerned with flows that are close to geostrophic balance, with the specific goal of deriving equation sets that exploit this closeness and that are simpler than the original, ‘primitive’ equations. We will in particular derive the quasi-geostrophic and planetary-geostrophic sets of equations. We do this first for shallow water and then for the stratified three-dimensional equations.

3.1 GEOSTROPHIC SCALING

3.1.1 Scaling in the shallow water equations

With the odd exception, we will denote the scales of variables by capital letters; thus, if L is a typical length scale of the motion we wish to describe, and U is a typical velocity scale, and assuming the scales are horizontally isotropic, we write

$$\begin{aligned} (x, y) &\sim L & \text{or} & & (x, y) &= \mathcal{O}(L) \\ (u, v) &\sim U & \text{or} & & (u, v) &= \mathcal{O}(U). \end{aligned} \quad (3.1)$$

and similarly for other variables. We may then non-dimensionalize the variables by writing

$$(x, y) = L(\hat{x}, \hat{y}), \quad (u, v) = U(\hat{u}, \hat{v}), \quad (3.2)$$

where the hatted variables are non-dimensional and, by supposition, are $\mathcal{O}(1)$. The various terms in the momentum equation then scale as:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta, \quad (3.3a)$$

$$\frac{U}{T} \quad \frac{U^2}{L} \quad fU \quad g \frac{\mathcal{H}}{L}, \quad (3.3b)$$

where the ∇ operator acts in the x, y plane and $\mathcal{H}$ is the amplitude of the variations in the surface displacement. (We use η to denote the height of the free surface above some arbitrary reference level, as in Fig. 2.1. Thus, $\eta = H + \Delta\eta$, where $\Delta\eta$ denotes the variation of η about its mean position.)

The ratio of the advective term to the rotational term in the momentum equation (3.3) is $(U^2/L)/(fU) = U/fL$; this is the Rossby number, first encountered in chapter ??.⁶ Using values typical of the large-scale circulation (e.g., from Table 1.1) we find that $Ro \approx 0.1$ for the atmosphere and $Ro \approx 0.01$ for the ocean: small in both cases. If we are interested in motion that has the advective time scale $T = L/U$ then we scale time by L/U so that

$$t = \frac{L}{U} \hat{t}, \quad (3.4)$$

and the local time derivative and the advective term then both scale as U^2/L , and both are smaller than the rotation term by a factor of the order of the Rossby number. Then, either the Coriolis term is the dominant term in the equation, in which case we have a state of no motion with $-fv = 0$, or else the Coriolis force is balanced by the pressure force, and the dominant balance is

$$-fv = -g \frac{\partial \eta}{\partial x}, \quad (3.5)$$

namely *geostrophic balance*. If we make this non-trivial choice, then variations in η (i.e., $\Delta\eta$) scale according to

$$\Delta\eta \sim \mathcal{H} = \frac{fUL}{g}. \quad (3.6)$$

We can also write $\mathcal{H}$ as

$$\mathcal{H} = Ro \frac{f^2 L^2}{g} = Ro H \frac{L^2}{L_d^2}, \quad (3.7)$$

where $L_d = \sqrt{gH}/f$ is the deformation radius and H is the mean depth of the fluid. The variations in fluid height thus scale as

$$\frac{\Delta\eta}{H} \sim Ro \frac{L^2}{L_d^2}, \quad (3.8)$$

and the height of the fluid may be written as

$$\eta = H \left(1 + Ro \frac{L^2}{L_d^2} \hat{\eta} \right) \quad \text{and} \quad \Delta\eta = Ro \frac{L^2}{L_d^2} H \hat{\eta}, \quad (3.9)$$

where $\hat{\eta}$ is the $\mathcal{O}(1)$ non-dimensional value of the surface height deviation.

Non-dimensional momentum equation

If we use (3.9) to scale height variations, (3.2) to scale lengths and velocities, and (3.4) to scale time, then the momentum equation (3.3) becomes

$$Ro \left[\frac{\partial \hat{\mathbf{u}}}{\partial \hat{t}} + (\hat{\mathbf{u}} \cdot \nabla) \hat{\mathbf{u}} \right] + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\eta}, \quad (3.10)$$

where $\hat{\mathbf{f}} = \mathbf{k}\hat{f} = \mathbf{k}f/f_0$, where f_0 is a representative value of the Coriolis parameter. (If f is a constant, then $\hat{f} = 1$, but it is informative to explicitly write $\hat{f}$ in the equations. Also, where the operator ∇ operates on a non-dimensional variable then the differentials are taken with respect to the non-dimensional variables $\hat{x}, \hat{y}$.) All the variables in (3.10) will be assumed to be of order unity, and the Rossby number multiplying the local time derivative and the advective terms indicates the smallness of those terms. By construction, the dominant balance in this equation is the geostrophic balance between the last two terms.

Non-dimensional mass continuity (height) equation

The (dimensional) mass continuity equation can be written as

$$\frac{1}{H} \frac{D\eta}{Dt} + \left(1 + \frac{\Delta\eta}{H}\right) \nabla \cdot \mathbf{u} = 0, \quad (3.11)$$

Using (3.2), (3.4) and (3.9) this equation may be written

$$Ro \left(\frac{L}{L_d}\right)^2 \frac{D\hat{\eta}}{D\hat{t}} + \left[1 + Ro \left(\frac{L}{L_d}\right)^2 \hat{\eta}\right] \nabla \cdot \hat{\mathbf{u}} = 0. \quad (3.12)$$

Equations (3.10) and (3.12) are the non-dimensional versions of the full shallow water equations of motion. Evidently, some terms in the equations of motion are small and may be eliminated with little loss of accuracy, and the way this is done will depend on the size of the second non-dimensional parameter, $(L/L_d)^2$. We explore this in sections 3.2 and 3.3.

Froude and Burger numbers

The Froude number may be generally defined as the ratio of a fluid particle speed to a wave speed. In a shallow water system this gives

$$Fr \equiv \frac{U}{\sqrt{gH}} = \frac{U}{f_0 L_d} = Ro \frac{L}{L_d}. \quad (3.13)$$

The Burger number is a useful measure of the scale of motion of the fluid, relative to the deformation radius, and may be defined by

$$Bu \equiv \left(\frac{L_d}{L}\right)^2 = \frac{gH}{f_0^2 L^2} = \left(\frac{Ro}{Fr}\right)^2. \quad (3.14)$$

It is also useful to define the parameter $F \equiv Bu^{-1}$, which is like the square of a Froude number but uses the rotational speed fL instead of U in the numerator.

3.2 THE PLANETARY-GEOSTROPHIC EQUATIONS

3.2.1 Informal derivation

The advection and time derivative terms in the momentum equation (3.10) are order Rossby number smaller than the Coriolis and pressure terms (the term in square brackets is multiplied by Ro), and therefore let us neglect them. The momentum equation straightforwardly becomes

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\eta}. \quad (3.15)$$

The mass conservation equation (3.12), contains two non-dimensional parameters, $Ro = U/(f_0 L)$ (the Rossby number), and $F = (L/L_d)^2$ (the ratio of the length scale of the motion to the deformation scale; $F = Bu^{-1}$) and we must make a choice as to the relationship between these two numbers. We will choose

$$FRo = \mathcal{O}(1), \quad (3.16)$$

which implies

$$L^2 \gg L_d^2 \quad \text{or equivalently} \quad F \gg 1, \quad Bu \ll 1. \quad (3.17)$$

That is to say, we suppose that the scales of motion are much larger than the deformation scale. Given this choice, all the terms in the mass conservation equation, (3.12), are of roughly the same size, and we retain them all. Thus, the shallow water planetary geostrophic equations are the full mass continuity equation along with geostrophic balance and a geometric relationship between the height field and the fluid thickness, and in dimensional form these are:

$$\begin{aligned} \frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} &= 0 \\ \mathbf{f} \times \mathbf{u} &= -g \nabla \eta, \quad \eta = h + \eta_b. \end{aligned} \quad (3.18a,b)$$

We emphasize that *the planetary-geostrophic equations are only valid for scales of motion much larger than the deformation radius*. The height variations are then as large as the mean height field itself; that is, using (3.8), $\Delta \eta / H = \mathcal{O}(1)$.

Formal derivation

We make the following assumptions.

- (i) The Rossby number is small. $Ro = U/f_0 L \ll 1$.
- (ii) The scale of the motion is significantly larger than the deformation scale. That is, (3.16) holds or

$$F = Bu^{-1} = \left(\frac{L}{L_d} \right)^2 \gg 1 \quad (3.19)$$

and in particular

$$FRo = \mathcal{O}(1). \quad (3.20)$$

- (iii) Time scales advectively, so that $T = L/U$.

We expand the non-dimensional variables velocity and height fields in an asymptotic series with the Rossby number as the small parameter, substitute into the equations of motion and derive a simpler set of equations. It is a nearly trivial exercise in this instance, and so it illustrates the methodology well. The expansions are

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_0 + Ro \hat{\mathbf{u}}_1 + Ro^2 \hat{\mathbf{u}}_2 + \dots \quad (3.21a)$$

and

$$\hat{\eta} = \hat{\eta}_0 + Ro \hat{\eta}_1 + Ro^2 \hat{\eta}_2 + \dots \quad (3.21b)$$

Then substituting (3.21a) and (3.21b) into the momentum equation gives

$$Ro \left[\frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{t}} + \hat{\mathbf{u}}_0 \cdot \nabla \hat{\mathbf{u}}_0 + \hat{\mathbf{f}} \times \hat{\mathbf{u}}_1 \right] + \hat{\mathbf{f}} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\eta}_0 - Ro [\nabla \hat{\eta}_1] + \mathcal{O}(Ro^2) \quad (3.22)$$

The Rossby number is an asymptotic ordering parameter; thus, the sum of all the terms at any particular order in Rossby number must vanish. At lowest order we obtain the simple expression

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\eta}_0. \quad (3.23)$$

Note that although f_0 is a representative value of f , we have made no assumptions about the constancy of f . In particular, f is allowed to vary by an order one amount, provided that it does not become so small that the Rossby number $U/(f_0 L)$ is not small.

The appropriate height (mass conservation) equation is similarly obtained by substituting (3.21a) and (3.21b) into the shallow water mass conservation equation. Because $F Ro = \mathcal{O}(1)$ at lowest order we simply retain all the terms in the equation to give

$$F Ro \left[\frac{\partial \hat{\eta}_0}{\partial \hat{t}} + \hat{\mathbf{u}}_0 \cdot \nabla \hat{\eta}_0 \right] + [1 + F Ro \hat{\eta}] \nabla \cdot \hat{\mathbf{u}}_0 = 0. \quad (3.24)$$

Equations (3.23) and (3.24) are a closed set, and constitute the non-dimensional planetary-geostrophic equations. The dimensional forms of these equations are just (3.18).

Variation of the Coriolis parameter

Suppose then that f is a constant (f_0). Then, from the curl of (3.23), $\nabla \cdot \mathbf{u}_0 = 0$. This means that we can define a streamfunction for the flow and, from geostrophic balance, the height field is just that streamfunction. That is, in dimensional form,

$$\psi = \frac{g}{f_0} \eta, \quad \mathbf{u} = \mathbf{k} \times \nabla \psi, \quad (3.25a,b)$$

and (3.24) becomes, in dimensional form,

$$\frac{\partial \eta}{\partial t} + \mathbf{u} \cdot \nabla \eta = 0 \quad \text{or} \quad \frac{\partial \eta}{\partial t} + J(\psi, \eta) = 0, \quad (3.26)$$

where $J(a, b) \equiv a_x b_y - a_y b_x$. But since $\eta \propto \psi$ the advective term is proportional to $J(\psi, \psi)$, which is zero. Thus, the flow does not evolve at this order. The planetary-geostrophic equations are *uninteresting* if the scale of the motion is such that the Coriolis parameter is not variable. On Earth, the scale of motion on which this parameter regime exists is rather limited, since the planetary-geostrophic equations require that the scale of motion also be larger than the deformation radius. In the Earth's atmosphere, any scale that is larger than the deformation radius will be such that the Coriolis parameter varies significantly over it, and we do not encounter this parameter regime. On the other hand, in the Earth's ocean the deformation radius is relatively small and there exists a small parameter regime that has scales larger than the deformation radius but smaller than that on which the Coriolis parameter varies.⁷

Potential vorticity

The shallow water PG equations may be written as an evolution equation for an approximated potential vorticity. A little manipulation reveals that (3.18) are equivalent to:

$$\begin{aligned} \frac{DQ}{Dt} &= 0 \\ Q &= \frac{f}{h}, \quad \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad \eta = h + \eta_b. \end{aligned} \tag{3.27}$$

Thus, potential vorticity is a material invariant in the approximate equation set, just as it is in the full equations. The other variables — the free surface height and the velocity — are diagnosed from it, a process known as *potential vorticity inversion*. In the planetary geostrophic approximation, the inversion proceeds using the approximate form f/h rather than the full potential vorticity, $(f + \zeta)/h$. (Strictly speaking, we do not approximate potential vorticity, because this is the evolving variable. Rather, we approximate the inversion relations from which we derive the height and velocity fields.) The simplest way of all to derive the shallow water PG equations is to *begin* with the conservation of potential vorticity, and to note that at small Rossby number the expression $(\zeta + f)/h$ may be approximated by f/h . Then, noting in addition that the flow is geostrophic, (3.27) immediately emerges. *Every* approximate set of equations that we derive in this chapter may be expressed as the evolution of potential vorticity, with the other fields being obtained diagnostically from it.

3.3 THE SHALLOW WATER QUASI-GEOSTROPHIC EQUATIONS

We now derive a set of geostrophic equations that is valid (unlike the PG equations) when the horizontal scale of motion is similar to that of the deformation radius. These equations are called the *quasi-geostrophic* equations, and are perhaps the most widely used set of equations for theoretical studies of the atmosphere and ocean. The specific assumptions we make are as follows.

- (i) The Rossby number is small, so that the flow is in near-geostrophic balance.

- (ii) The scale of the motion is not significantly larger than the deformation scale. Specifically, we shall require that

$$Ro \left(\frac{L}{L_d} \right)^2 = \mathcal{O}(Ro). \quad (3.28)$$

For the shallow water equations, this assumption implies, using (3.9), that the variations in fluid depth are small compared to its total depth. For the continuously stratified system it implies, using (3.70), that the variations in stratification are small compared to the background stratification.

- (iii) Variations in the Coriolis parameter are small; that is, $|\beta L| \ll |f_0|$ where L is the length scale of the motion.

- (iv) Time scales advectively; that is, the scaling for time is given by $T = L/U$.

The second and third of these differ from the planetary-geostrophic counterparts: we make the second assumption because we wish to explore a different parameter regime, and we then find that the third assumption is necessary to avoid a rather trivial state [i.e., a leading order balance of $\beta v = 0$, see the discussion surrounding (3.52)]. All of the assumptions are the same whether we consider the shallow water equations or a continuously stratified flow, and in this section we consider the former.

3.3.1 Single-layer shallow water quasi-geostrophic equations

The algorithm is, again, to expand the variables $\hat{u}, \hat{v}, \hat{\eta}$ in an asymptotic series with the Rossby number as the small parameter, substitute into the equations of motion, and derive a simpler set of equations. Thus we let

$$\hat{u} = \hat{u}_0 + Ro \hat{u}_1 + Ro^2 \hat{u}_2 + \dots, \quad \hat{v} = \hat{v}_0 + Ro \hat{v}_1 + Ro^2 \hat{v}_2 + \dots \quad (3.29a)$$

$$\hat{\eta} = \hat{\eta}_0 + Ro \hat{\eta}_1 + Ro^2 \hat{\eta}_2 + \dots. \quad (3.29b)$$

We will recognize the smallness of β compared to f_0/L by letting $\beta = \hat{\beta}U/L^2$, where $\hat{\beta}$ is assumed to be a parameter of order unity. Then the expression $f = f_0 + \beta y$ becomes

$$\hat{f} = f/f_0 = \hat{f}_0 + Ro \hat{\beta} \hat{y}. \quad (3.30)$$

where $\hat{f}_0$ is the non-dimensional value of f_0 ; its value is unity, but it is helpful to denote it explicitly. Substitute (3.29) into the non-dimensional momentum equation (3.10), and equate powers of Ro . At lowest order we obtain

$$\hat{f}_0 \hat{u}_0 = -\frac{\partial \hat{\eta}_0}{\partial \hat{y}}, \quad \hat{f}_0 \hat{v}_0 = \frac{\partial \hat{\eta}_0}{\partial \hat{x}}. \quad (3.31)$$

Cross-differentiating gives

$$\nabla \cdot \hat{\mathbf{u}}_0 = 0, \quad (3.32)$$

where, when ∇ operates on a non-dimensional variable, the derivatives are taken with respect to the non-dimensional variables $\hat{x}$ and $\hat{y}$. From (3.32) we see that the velocity field is divergence-free, and that this arises from the momentum equation rather than the mass conservation equation.

The mass conservation equation is also, at lowest order, $\nabla \cdot \hat{\mathbf{u}}_0 = 0$, and at next order we have

$$F \frac{\partial \hat{\eta}_0}{\partial \hat{t}} + F \hat{\mathbf{u}}_0 \cdot \nabla \hat{\eta}_0 + \nabla \cdot \hat{\mathbf{u}}_1 = 0 \quad (3.33)$$

This equation is not closed, because the evolution of the zeroth-order term involves evaluation of a first-order quantity. For closure, we go to the next order in the momentum equation,

$$\frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla) \hat{\mathbf{u}}_0 + \hat{\beta} \hat{y} \mathbf{k} \times \hat{\mathbf{u}}_0 + \hat{f}_0 \mathbf{k} \times \hat{\mathbf{u}}_1 = -\nabla \hat{\eta}_1, \quad (3.34)$$

and take its curl to give the vorticity equation:

$$\frac{\partial \hat{\zeta}_0}{\partial \hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \hat{\beta} \hat{y}) = -\hat{f}_0 \nabla \cdot \hat{\mathbf{u}}_1. \quad (3.35)$$

The term on the right-hand side is the *vortex stretching* term. Only vortex stretching by the background or planetary vorticity is present, because the vortex stretching by the relative vorticity is smaller by a factor of the Rossby number. Equation (3.35) is also not closed; however, we may use (3.33) to eliminate the divergence term to give

$$\frac{\partial \hat{\zeta}_0}{\partial \hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \hat{\beta} \hat{y}) = \hat{f}_0 \left(F \frac{\partial \hat{\eta}_0}{\partial \hat{t}} + F \hat{\mathbf{u}}_0 \cdot \nabla \hat{\eta}_0 \right), \quad (3.36)$$

or

$$\frac{\partial}{\partial \hat{t}}(\hat{\zeta}_0 - \hat{f}_0 F \hat{\eta}_0) + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \hat{\beta} \hat{y} - F \hat{f}_0 \hat{\eta}_0) = 0. \quad (3.37)$$

The final step is to note that the lowest-order vorticity and height fields are related through geostrophic balance, so that using (3.31) we can write

$$\hat{u}_0 = -\frac{\partial \hat{\psi}_0}{\partial \hat{y}}, \quad \hat{v}_0 = \frac{\partial \hat{\psi}_0}{\partial \hat{x}}, \quad \hat{\zeta}_0 = \nabla^2 \hat{\psi}_0, \quad (3.38)$$

where $\hat{\psi}_0 = \hat{\eta}_0 / \hat{f}_0$ is the streamfunction. Equation (3.37) can thus be written as

$$\frac{\partial}{\partial \hat{t}}(\nabla^2 \hat{\psi}_0 - \hat{f}_0^2 F \hat{\psi}_0) + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \hat{\beta} \hat{y} - \hat{f}_0^2 F \hat{\psi}_0) = 0 \quad (3.39)$$

or

$$\frac{D_0}{Dt}(\nabla^2 \hat{\psi}_0 + \hat{\beta} \hat{y} - \hat{f}_0^2 F \hat{\psi}_0) = 0, \quad (3.40)$$

where the subscript ‘0’ on the material derivative indicates that the lowest order velocity, the geostrophic velocity, is the advecting velocity. Restoring the dimensions, (3.40) becomes

$$\frac{D}{Dt} \left(\nabla^2 \psi + \beta y - \frac{1}{L_d^2} \psi \right) = 0, \quad (3.41)$$

where $\psi = (g/f_0)\eta$, $L_d^2 = gH/f_0^2$, and the advective derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u_g \frac{\partial}{\partial x} + v_g \frac{\partial}{\partial y} = \frac{\partial}{\partial t} - \frac{\partial \psi}{\partial y} \frac{\partial}{\partial x} + \frac{\partial \psi}{\partial x} \frac{\partial}{\partial y} = \frac{\partial}{\partial t} + J(\psi, \cdot). \quad (3.42)$$

Another form of (3.41) is

$$\frac{D}{Dt} \left(\zeta + \beta y - \frac{f_0}{H} \eta \right) = 0, \quad (3.43)$$

with $\zeta = (g/f_0)\nabla^2 \eta$. Equations (3.41) and (3.43) are forms of the shallow water quasi-geostrophic potential vorticity equation. The quantity

$$q \equiv \zeta + \beta y - \frac{f_0}{H} \eta = \nabla^2 \psi + \beta y - \frac{1}{L_d^2} \psi \quad (3.44)$$

is the *shallow water quasi-geostrophic potential vorticity*.

Connection to shallow water potential vorticity

The quantity q given by (3.44) is an approximation (except for dynamically unimportant constant additive and multiplicative factors) to the shallow water potential vorticity. To see the truth of this statement, begin with the expression for the shallow water potential vorticity,

$$Q = \frac{f + \zeta}{h}. \quad (3.45)$$

Now let $h = H(1 + \eta'/H)$, where η' is the perturbation of the free-surface height, and assume that η'/H is small to obtain

$$Q = \frac{f + \zeta}{H(1 + \eta'/H)} \approx \frac{1}{H} (f + \zeta) \left(1 - \frac{\eta'}{H} \right) \approx \frac{1}{H} \left(f_0 + \beta y + \zeta - f_0 \frac{\eta'}{H} \right). \quad (3.46)$$

Because f_0/H is a constant it has no effect in the evolution equation, and the quantity given by

$$q = \beta y + \zeta - f_0 \frac{\eta'}{H} \quad (3.47)$$

is materially conserved. Using geostrophic balance we have $\zeta = \nabla^2 \psi$ and $\eta' = f_0 \psi / g$ so that (3.47) is identical to (3.44). [Note that only the variation in η is important in (3.43) or (3.44).]

The approximations needed to go from (3.45) to (3.47) are the same as those used in our earlier, more long-winded, derivation of the quasi-geostrophic equations. That is, we assumed that f itself is nearly constant, and that f_0 is much larger than ζ , equivalent to a low Rossby number assumption. It was also necessary to assume that $H \gg \eta'$ to enable the expansion of the height field which, using assumption ((ii)) on page 67, is equivalent to requiring that the scale of motion not be significantly larger than the deformation scale. The derivation is completed by noting that the advection of the potential vorticity should be by the geostrophic velocity alone, and we recover (3.41) or (3.43).

Two interesting limits

There are two interesting limits to the quasi-geostrophic potential vorticity equation which, taking $\beta = 0$ for simplicity, are as follows.

- (i) *Motion on scales much smaller than the deformation radius.* That is, $L \ll L_d$ and thus $Bu \gg 1$ or $F \ll 1$. Then (3.41) becomes

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) = 0, \quad (3.48)$$

where $\zeta = \nabla^2 \psi$ and $J(\psi, \zeta) = \psi_x \zeta_y - \psi_y \zeta_x$. Thus, the motion obeys the two-dimensional vorticity equation. Physically, on small length scales the deviations in the height field are very small and may be neglected.

- (ii) *Motion on scales much larger than the deformation radius.* Although scales are not allowed to become so large that $Ro(L/L_d)^2$ is of order unity, we may, a posteriori, still have $L \gg L_d$, whence the potential vorticity equation, (3.41), becomes

$$\frac{\partial \psi}{\partial t} + J(\psi, \psi) = 0 \quad \text{or} \quad \frac{\partial \eta}{\partial t} + J(\psi, \eta) = 0, \quad (3.49)$$

because $\psi = g\eta/f_0$. The Jacobian term evidently vanishes. Thus, one is left with a trivial equation that implies there is no advective evolution of the height field. There is nothing wrong with our reasoning; the mathematics has indeed pointed out a limit interesting in its uninterestingness. From a physical point of view, however, such a lack of motion is likely to be rare, because on such large scales the Coriolis parameter varies considerably, and we are led to the planetary-geostrophic equations.

In practice, often the most severe restriction of quasi-geostrophy is that variations in layer thickness are small: what does this have to do with geostrophy? If we scale η assuming geostrophic balance then $\eta \sim fUL/g$ and $\eta/H \sim Ro(L/L_d)^2$. Thus, if Ro is to remain small, η/H can only be of order one if $(L/L_d)^2 \gg 1$. That is, the height variations must occur on a large scale, or we are led to a scaling inconsistency. Put another way, *if there are order-one height variations over a length scale of less than or of the order of the deformation scale, the Rossby number will not be small.* Large height variations are allowed if the scale of motion is large, but this contingency is described by the planetary-geostrophic equations.

Another flow regime

Although perhaps of little terrestrial interest, we can imagine a regime in which the Coriolis parameter varies fully, but the scale of motion remains no larger than the deformation radius. This parameter regime is not quasi-geostrophic, but it gives an interesting result. Because $\eta'/H \sim Ro(L/L_d)^2$ deviations of the height field are at least of order Rossby number smaller than the reference height and $|\eta'| \ll H$. The dominant balance in the height equation is then

$$H\nabla \cdot \mathbf{u} = 0, \quad (3.50)$$

presuming that time still scales advectively. This zero horizontal divergence must remain consistent with geostrophic balance

$$\mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (3.51)$$

where now f is a fully variable Coriolis parameter. Taking the curl of (that is, cross-differentiating) (3.51) gives

$$\beta v + f\nabla \cdot \mathbf{u} = 0, \quad (3.52)$$

whence, using (3.50), $v = 0$, and the flow is purely zonal. Although not at all useful as an evolution equation, this illustrates the constraining effect that differential rotation has on meridional velocity. This effect may be the cause of the banded, highly zonal flow on some of the giant planets, and we will revisit this issue in our discussion of geostrophic turbulence.

3.3.2 ♦ Two-layer and multi-layer quasi-geostrophic systems

Just as for the one-layer case, the multi-layer shallow water equations simplify to a corresponding quasi-geostrophic system in appropriate circumstances. The assumptions are virtually the same as before, although we assume that the variation in the thickness of *each* layer is small compared to its mean thickness. The basic fluid system for a two-layer case is sketched in Fig. 3.1 (and see also Fig. 2.5).

Let us proceed directly from the potential vorticity equation for each layer. We will also stay in dimensional variables, foregoing a strict asymptotic approach for the sake of informality and insight, and use the Boussinesq approximation. For each layer the potential vorticity equation is just

$$\frac{DQ_i}{Dt} = 0, \quad Q_i = \frac{\zeta_i + f}{h_i}. \quad (3.53)$$

Let $h_i = H_i + h'_i$ where $|h'_i| \ll H_i$. The potential vorticity then becomes

$$Q_i \approx \frac{1}{H_i}(\zeta_i + f) \left(1 - \frac{h'_i}{H_i} \right) \quad \text{— variations in layer thickness are small} \quad (3.54a)$$

$$\approx \frac{1}{H_i} \left(f + \zeta_i - f \frac{h'_i}{H_i} \right) \quad \text{— the Rossby number is small} \quad (3.54b)$$

$$\approx \frac{1}{H_i} \left(f + \zeta_i - f_0 \frac{h'_i}{H_i} \right) \quad \text{— variations in Coriolis parameter are small} \quad (3.54c)$$

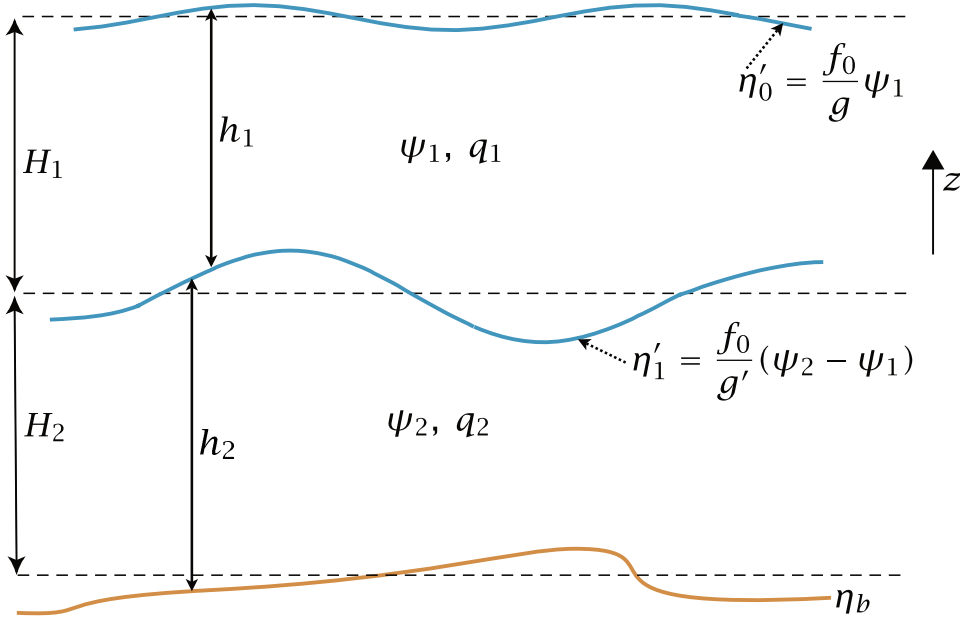


Figure 3.1 A quasi-geostrophic fluid system consisting of two immiscible fluids of different density. The quantities η' are the interface displacements from the resting basic state, denoted with dashed lines, with η_b being the bottom topography.

Now, because Q appears in the equations only as an advected quantity, it is only the *variations* in the Coriolis parameter that are important in the first term on the right-hand side of (3.54c), and given this all three terms are of the same approximate magnitude. Then, because mean layer thicknesses are constant, we can define the quasi-geostrophic potential vorticity in each layer by

$$q_i = \left(\beta y + \zeta_i - f_0 \frac{h'_i}{H_i} \right), \quad (3.55)$$

and this will evolve according to $Dq_i/Dt = 0$, where the advective derivative is by the geostrophic wind. As in the one-layer case, the quasi-geostrophic potential vorticity has different dimensions from the full shallow water potential vorticity.

Two-layer model

To obtain a closed set of equations we must obtain an advecting field from the potential vorticity. We use geostrophic balance to do this, and neglecting the advective derivative in (2.36) gives

$$f_0 \times \mathbf{u}_1 = -g \nabla \eta_0 = -g \nabla (h'_1 + h'_2 + \eta_b), \quad (3.56a)$$

$$f_0 \times \mathbf{u}_2 = -g \nabla \eta_0 - g' \nabla \eta_1 = -g \nabla (h'_1 + h'_2 + \eta_b) - g' \nabla (h'_2 + \eta_b), \quad (3.56b)$$

where $g' = g(\rho_2 - \rho_1)/\rho_1$ and η_b is the height of any bottom topography, and, because variations in the Coriolis parameter are presumptively small, we use a constant value of f (i.e., f_0) on the

left-hand side. For each layer there is therefore a streamfunction, given by

$$\psi_1 = \frac{g}{f_0}(h'_1 + h'_2 + \eta_b), \quad \psi_2 = \frac{g}{f_0}(h'_1 + h'_2 + \eta_b) + \frac{g'}{f_0}(h'_2 + \eta_b), \quad (3.57a,b)$$

and these two equations may be manipulated to give

$$h'_1 = \frac{f_0}{g'}(\psi_1 - \psi_2) + \frac{f_0}{g}\psi_1, \quad h'_2 = \frac{f_0}{g'}(\psi_2 - \psi_1) - \eta_b. \quad (3.58a,b)$$

We note as an aside that the interface displacements are given by

$$\eta'_0 = \frac{f_0}{g}\psi_1, \quad \eta'_1 = \frac{f_0}{g'}(\psi_2 - \psi_1). \quad (3.59a,b)$$

Using (3.55) and (3.58) the quasi-geostrophic potential vorticity for each layer becomes

$$\begin{aligned} q_1 &= \beta y + \nabla^2 \psi_1 + \frac{f_0^2}{g'H_1}(\psi_2 - \psi_1) - \frac{f_0^2}{gH_1}\psi_1, \\ q_2 &= \beta y + \nabla^2 \psi_2 + \frac{f_0^2}{g'H_2}(\psi_1 - \psi_2) + f_0 \frac{\eta_b}{H_2}. \end{aligned} \quad (3.60a,b)$$

In the rigid-lid approximation the last term in (3.60a) is neglected. The potential vorticity in each layer is advected by the geostrophic velocity, so that the evolution equation for each layer is just

$$\frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0, \quad i = 1, 2. \quad (3.61)$$

3.4 GEOSTROPHIC SCALING IN THE STRATIFIED EQUATIONS

We use the hydrostatic anelastic equations, which we write as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad (3.62a)$$

$$\frac{\partial \phi}{\partial z} = b, \quad (3.62b)$$

$$\frac{Db}{Dt} = 0, \quad (3.62c)$$

$$\nabla \cdot (\tilde{\rho} \mathbf{v}) = 0. \quad (3.62d)$$

where b is the buoyancy and $\tilde{\rho}$ is a reference density profile. Anticipating that the average stratification may not scale in the same way as the deviation from it, let us separate out the contribution of the advection of a reference stratification in (3.62c) by writing

$$b = \tilde{b}(z) + b'(x, y, z, t). \quad (3.63)$$

Then the thermodynamic equation becomes

$$\frac{Db'}{Dt} + N^2 w = 0, \quad (3.64)$$

where $N^2 \equiv \partial \tilde{b} / \partial z$ (and the advective derivative is still three-dimensional). We then let $\phi = \tilde{\phi}(z) + \phi'$, where $\tilde{\phi}$ is hydrostatically balanced by $\tilde{b}$, and the hydrostatic equation becomes

$$\frac{\partial \phi'}{\partial z} = b'. \quad (3.65)$$

Equations (3.64) and (3.65) replace (3.62c) and (3.62b), and ϕ' is used in (3.62a).

3.4.1 Non-dimensional equations

We scale the basic variables by supposing that

$$(x, y) \sim L, \quad (u, v) \sim U, \quad t \sim \frac{L}{U}, \quad z \sim H, \quad f \sim f_0, \quad (3.66)$$

where the scaling variables (capitalized, except for f_0) are chosen to be such that the non-dimensional variables have magnitudes of the order of unity. We presume that the scales chosen are such that the Rossby number is small; that is $Ro = U/(f_0 L) \ll 1$. In the momentum equation the pressure term then balances the Coriolis force,

$$|\mathbf{f} \times \mathbf{u}| \sim |\nabla \phi'| \quad (3.67)$$

and so the pressure scales as

$$\phi' \sim \Phi = f_0 UL. \quad (3.68)$$

Using the hydrostatic relation, (3.68) implies that the buoyancy scales as

$$b' \sim B = \frac{f_0 UL}{H}, \quad (3.69)$$

and from this we obtain

$$\frac{(\partial b' / \partial z)}{N^2} \sim Ro \frac{L^2}{L_d^2}, \quad (3.70)$$

where $L_d = NH/f_0$ is the deformation radius in the continuously stratified fluid, analogous to the quantity $\sqrt{gH}/f_0$ in the shallow water system, and we use the same symbol, L_d , for both. In the continuously stratified system, *if the scale of motion is the same as or smaller than the deformation radius, and the Rossby number is small, then the variations in stratification are small.* The choice of scale is the key difference between the planetary-geostrophic and quasi-geostrophic equations.

Finally, we will non-dimensionalize the vertical velocity by using the mass conservation equation,

$$\frac{1}{\bar{\rho}} \frac{\partial \bar{\rho} w}{\partial z} = - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right), \quad (3.71)$$

and we suppose that this implies

$$w \sim W = \frac{UH}{L}. \quad (3.72)$$

This is a naïve scaling for rotating flow: if the Coriolis parameter is nearly constant the geostrophic velocity is nearly horizontally non-divergent and the right-hand side of (3.71) is small, and $W \ll UH/L$. We might then estimate w by cross-differentiating geostrophic balance (with $\bar{\rho}$ constant for simplicity) to obtain the linear geostrophic vorticity equation and corresponding scaling:

$$\beta v \approx f \frac{\partial w}{\partial z}, \quad w \sim W = \frac{\beta UH}{f_0}. \quad (3.73a,b)$$

However, rather than using (3.73b) from the outset, we will use (3.72) and let the asymptotics guide us to a proper scaling in the fullness of time. Note that if variations in the Coriolis parameter are large and $\beta \sim f_0/L$, then (3.73b) is the same as (3.72).

Given the scalings above [using (3.72) for w] we non-dimensionalize by setting

$$\begin{aligned} (\hat{x}, \hat{y}) &= L^{-1}(x, y), & \hat{z} &= H^{-1}z, & (\hat{u}, \hat{v}) &= U^{-1}(u, v), & \hat{t} &= \frac{U}{L}t, \\ \hat{w} &= \frac{L}{UH}w, & \hat{f} &= f_0^{-1}f, & \hat{\phi} &= \frac{\phi'}{f_0UL}, & \hat{b} &= \frac{H}{f_0UL}b', \end{aligned} \quad (3.74)$$

where the hatted variables are non-dimensional. The horizontal momentum and hydrostatic equations then become

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi}, \quad (3.75)$$

and

$$\frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}. \quad (3.76)$$

The non-dimensional mass conservation equation is simply

$$\frac{1}{\bar{\rho}} \nabla \cdot (\bar{\rho} \hat{\mathbf{v}}) = \left(\frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{1}{\bar{\rho}} \frac{\partial \bar{\rho} \hat{w}}{\partial \hat{z}} \right) = 0, \quad (3.77)$$

and the non-dimensional thermodynamic equation is

$$\frac{f_0UL}{H} \frac{U}{L} \frac{D\hat{b}}{D\hat{t}} + N^2 \frac{HU}{L} \hat{w} = 0, \quad (3.78)$$

or

$$Ro \frac{D\hat{b}}{D\hat{t}} + \left(\frac{L_d}{L} \right)^2 \hat{w} = 0. \quad (3.79)$$

The non-dimensional primitive equations are summarized in the box on the following page.

Non-dimensional Primitive Equations

$$\text{Horizontal momentum:} \quad Ro \frac{D\hat{\mathbf{u}}}{Dt} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi} \quad (\text{PE.1})$$

$$\text{Hydrostatic:} \quad \frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b} \quad (\text{PE.2})$$

$$\text{Mass continuity:} \quad \left(\frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{1}{\hat{\rho}} \frac{\partial \hat{\rho} \hat{w}}{\partial \hat{z}} \right) = 0 \quad (\text{PE.3})$$

$$\text{Thermodynamic:} \quad Ro \frac{D\hat{b}}{Dt} + \left(\frac{L_d}{L} \right)^2 \hat{w} = 0 \quad (\text{PE.4})$$

These equations are written for the anelastic equations in a rotating frame of reference. The Boussinesq equations result if we take $\hat{\rho} = 1$. The equations in pressure coordinates also have a similar form — see section 1.8.2.

3.5 PLANETARY-GEOSTROPHIC EQUATIONS FOR STRATIFIED FLOW

To explore the stratified system we will use the inviscid and adiabatic Boussinesq equations of motion with the hydrostatic approximation. The derivation carries through easily enough using the anelastic or pressure-coordinate equations, but as the PG equations have more oceanographic than atmospheric importance, using the incompressible equations is quite appropriate.

3.5.1 Simplifying the equations

The non-dimensional equations we begin with are (3.75)–(3.79). As in the shallow water case we expand these in a series in the Rossby number, so that:

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_0 + Ro \hat{\mathbf{u}}_1 + Ro^2 \hat{\mathbf{u}}_2 + \dots, \quad \hat{b} = \hat{b}_0 + Ro \hat{b}_1 + Ro^2 \hat{b}_2 + \dots, \quad (3.80)$$

and similarly for $\hat{v}$, $\hat{w}$ and $\hat{\phi}$. Substituting into the non-dimensional equations of motion (above) and equating powers of Ro gives the lowest-order momentum, hydrostatic, and mass conservation equations:

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\phi}_0, \quad \frac{\partial \hat{\phi}_0}{\partial \hat{z}} = \hat{b}_0, \quad \nabla \cdot \hat{\mathbf{u}}_0 = 0. \quad (3.81a,b,c)$$

If we also assume that $L_d/L = \mathcal{O}(1)$, then the thermodynamic equation (3.79) becomes

$$\left(\frac{L_d}{L} \right)^2 \hat{w}_0 = 0. \quad (3.82)$$

Of course we have neglected any diabatic terms in this equation, which would in general provide a non-zero right-hand side. Nevertheless, this is not a useful equation, because the set of the

equations we have derived, (3.81) and (3.82), can no longer evolve: all the time derivatives have been scaled away! Thus, although instructive, these equations are not very useful. If instead we assume that the scale of motion is much larger than the deformation scale then the other terms in the thermodynamic equation will become equally important. Thus, we suppose that $L_d^2 \ll L^2$ or, more formally, that $L^2 = \mathcal{O}(Ro^{-1})L_d^2$, and then all the terms in the thermodynamic equation are retained. A closed set of equations is then given by (3.81) and the thermodynamic equation (3.79).

3.5.2 Dimensional equations

Restoring the dimensions, dropping the asymptotic subscripts, and allowing for the possibility of a source term, denoted by $S[b']$, in the thermodynamic equation, the *planetary-geostrophic* equations of motion are:

$$\begin{aligned} \frac{Db'}{Dt} + wN^2 &= S[b'] \\ \mathbf{f} \times \mathbf{u} &= -\nabla\phi', \quad \frac{\partial\phi'}{\partial z} = b', \quad \nabla \cdot \mathbf{v} = 0 \end{aligned} \quad (3.83)$$

The thermodynamic equation may also be written simply as

$$\frac{Db}{Dt} = \dot{b}, \quad (3.84)$$

where b now represents the total stratification. The relevant pressure, ϕ , is then the pressure that is in hydrostatic balance with b , so that geostrophic and hydrostatic balance are most usefully written as

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi, \quad \frac{\partial\phi}{\partial z} = b. \quad (3.85a,b)$$

3.5.3 Potential vorticity

Manipulation of (3.83) reveals that we can equivalently write the equations as an evolution equation for potential vorticity. Thus, the evolution equations may be written as

$$\begin{aligned} \frac{DQ}{Dt} &= \dot{Q} \\ Q &= f \frac{\partial b}{\partial z} \end{aligned} \quad (3.86)$$

where $\dot{Q} = f \partial \dot{b} / \partial z$, and the inversion — i.e., the diagnosis of velocity, pressure and buoyancy — is carried out using the hydrostatic, geostrophic and mass conservation equations.

3.5.4 Applicability to the ocean and atmosphere

In the atmosphere a typical deformation radius NH/f is about 1000 km. The constraint that the scale of motion be much larger than the deformation radius is thus quite hard to satisfy, since one quickly runs out of room on a planet whose equator-to-pole distance is 10 000 km. Thus, only the largest planetary waves can satisfy the planetary-geostrophic scaling in the atmosphere and we should then also write the equations in spherical coordinates. In the ocean the deformation radius is about 100 km, so there is lots of room for the planetary-geostrophic equations to hold, and indeed much of the theory of the large-scale structure of the ocean involves the planetary-geostrophic equations.

3.6 THE CONTINUOUSLY STRATIFIED QUASI-GEOSTROPHIC SYSTEM

We now consider the quasi-geostrophic equations for the continuously stratified hydrostatic system. The primitive equations of motion are given by (3.62), and we extract the mean stratification so that the thermodynamic equation is given by (3.64). We also stay on the β -plane for simplicity. Readers who wish for a briefer, more informal derivation may peruse the box on page 85; however, it is important to realize that there is a systematic asymptotic derivation of the quasi-geostrophic equations, for it is this that ensures that the resulting equations have good conservation properties, as explained in section ??.

3.6.1 Scaling and assumptions

The scaling assumptions we make are just those we made for the shallow water system on page 66, with a deformation radius now given by $L_d = NH/f_0$. The non-dimensionalization and scaling are initially precisely that of section 3.4, and we obtain the following non-dimensional equations:

$$\text{horizontal momentum:} \quad Ro \frac{D\hat{\mathbf{u}}}{Dt} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla_z \hat{\phi}, \quad (3.87)$$

$$\text{hydrostatic:} \quad \frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}, \quad (3.88)$$

$$\text{mass continuity:} \quad \frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{1}{\bar{\rho}} \frac{\partial \bar{\rho} \hat{w}}{\partial \hat{z}} = 0, \quad (3.89)$$

$$\text{thermodynamic:} \quad Ro \frac{D\hat{b}}{Dt} + \left(\frac{L_d}{L} \right)^2 \hat{w} = 0. \quad (3.90)$$

In Cartesian coordinates we may express the Coriolis parameter as

$$\mathbf{f} = f_0 + \beta y \mathbf{k}, \quad (3.91)$$

where $f_0 = f_0 \mathbf{k}$. The variation of the Coriolis parameter is assumed to be small (this is a key difference between the quasi-geostrophic system and the planetary-geostrophic system), and in

particular we shall assume that βy is approximately the size of the relative vorticity, and so is much smaller than f_0 itself.⁸ Thus,

$$\beta y \sim \frac{U}{L}, \quad \beta \sim \frac{U}{L^2}, \quad (3.92)$$

and so we define an $\mathcal{O}(1)$ non-dimensional beta parameter by

$$\hat{\beta} = \frac{\beta L^2}{U} = \frac{\beta L}{Ro f_0}. \quad (3.93)$$

From this it follows that if $f = f_0 + \beta y$, the corresponding non-dimensional version is

$$\hat{f} = \hat{f}_0 + Ro \hat{\beta} \hat{y}. \quad (3.94)$$

where $\hat{f} = f/f_0$ and $\hat{f}_0 = f_0/f_0 = 1$.

3.6.2 Asymptotics

We now expand the non-dimensional dependent variables in an asymptotic series in Rossby number, and write

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_0 + Ro \hat{\mathbf{u}}_1 + \dots, \quad \hat{\phi} = \hat{\phi}_0 + Ro \hat{\phi}_1 + \dots, \quad \hat{b} = \hat{b}_0 + Ro \hat{b}_1 + \dots. \quad (3.95)$$

Substituting these into the equations of motion, the lowest-order momentum equation is simply geostrophic balance,

$$\hat{\mathbf{f}}_0 \times \hat{\mathbf{u}}_0 = -\nabla \hat{\phi}_0 \quad (3.96)$$

with a *constant* value of the Coriolis parameter. (For the rest of this chapter we drop the subscript z from the ∇ operator.) From (3.96) it is evident that

$$\nabla \cdot \hat{\mathbf{u}}_0 = 0. \quad (3.97)$$

Thus, the horizontal flow is, to leading order, non-divergent; this is a consequence of geostrophic balance, and is *not* a mass conservation equation. Using (3.97) in the mass conservation equation, (3.89), gives

$$\frac{\partial}{\partial \hat{z}}(\hat{\rho} \hat{w}_0) = 0, \quad (3.98)$$

which implies that if w_0 is zero somewhere (e.g., at a solid surface) then w_0 is zero everywhere (essentially the Taylor–Proudman effect). A physical way of saying this is that the scaling estimate $W = UH/L$ is an overestimate of the size of the vertical velocity, because even though $\partial w / \partial z \approx -\nabla \cdot \mathbf{u}$, the horizontal divergence of the geostrophic flow is small if f is nearly constant and $|\nabla \cdot \mathbf{u}| \ll U/L$. We might have anticipated this from the outset, and scaled w differently, perhaps using the geostrophic vorticity balance estimate, $w \sim \beta UH/f_0 = Ro UH/L$ as the scaling factor for w , but there is no a priori guarantee that this would be correct.

At next order the momentum equation is

$$\frac{D_0 \hat{\mathbf{u}}_0}{D\hat{t}} + \hat{\beta} \hat{\mathbf{y}} \mathbf{k} \times \hat{\mathbf{u}}_0 + \hat{\mathbf{f}} \times \hat{\mathbf{u}}_1 = -\nabla \hat{\phi}_1, \quad (3.99)$$

where $D_0/Dt = \partial/\partial\hat{t} + (\hat{\mathbf{u}}_0 \cdot \nabla)$, and the next order mass conservation equation is

$$\nabla_z \cdot (\tilde{\rho} \hat{\mathbf{u}}_1) + \frac{\partial}{\partial z} (\tilde{\rho} \hat{w}_1) = 0. \quad (3.100)$$

From (3.90), the lowest-order thermodynamic equation is just

$$\left(\frac{L_d}{L} \right)^2 \hat{w}_0 = 0 \quad (3.101)$$

provided that, as we have assumed, the scales of motion are not sufficiently large that $Ro(L/L_d)^2 = \mathcal{O}(1)$. (This is a key difference between quasi-geostrophy and planetary geostrophy.) At next order we obtain an evolution equation for the buoyancy, and this is

$$\frac{D_0 \hat{b}_0}{D\hat{t}} + \hat{w}_1 \left(\frac{L_d}{L} \right)^2 = 0. \quad (3.102)$$

3.6.3 The potential vorticity equation

To obtain a single evolution equation for lowest-order quantities we eliminate w_1 between the thermodynamic and momentum equations. Cross-differentiating the first-order momentum equation (3.99) gives the vorticity equation,

$$\frac{\partial \hat{\zeta}_0}{\partial \hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla) \hat{\zeta}_0 + \hat{v}_0 \hat{\beta} = -\hat{f}_0 \nabla_z \cdot \hat{\mathbf{u}}_1. \quad (3.103)$$

(In dimensional terms, the divergence on the right-hand side is small, but is multiplied by the large term f_0 , and their product is of the same order as the terms on the left-hand side.) Using the mass conservation equation (3.100), (3.103) becomes

$$\frac{D_0}{D\hat{t}} (\zeta_0 + \hat{f}) = \frac{\hat{f}_0}{\tilde{\rho}} \frac{\partial}{\partial z} (w_1 \tilde{\rho}). \quad (3.104)$$

Combining (3.104) and (3.102) gives

$$\frac{D_0}{D\hat{t}} (\zeta_0 + \hat{f}) = -\frac{\hat{f}_0}{\tilde{\rho}} \frac{\partial}{\partial \hat{z}} \left[\frac{D_0}{D\hat{t}} (F \tilde{\rho} \hat{b}_0) \right], \quad (3.105)$$

where $F \equiv (L/L_d)^2$. The right-hand side of this equation is

$$\frac{\partial}{\partial \hat{z}} \left(\frac{D_0 \hat{b}_0}{D\hat{t}} \right) = \frac{D_0}{D\hat{t}} \left(\frac{\partial \hat{b}_0}{\partial \hat{z}} \right) + \frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{z}} \cdot \nabla \hat{b}_0. \quad (3.106)$$

The second term on the right-hand side vanishes identically using the thermal wind equation

$$\mathbf{k} \times \frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{z}} = -\frac{1}{\hat{f}_0} \nabla \hat{b}_0, \quad (3.107)$$

and so (3.105) becomes

$$\frac{D_0}{D\hat{t}} \left[\hat{\zeta}_0 + \hat{f} + \frac{\hat{f}_0}{\hat{\rho}} \frac{\partial}{\partial \hat{z}} (\hat{\rho} F \hat{b}_0) \right] = 0, \quad (3.108)$$

or, after using the hydrostatic equation,

$$\frac{D_0}{D\hat{t}} \left[\hat{\zeta}_0 + \hat{f} + \frac{\hat{f}_0}{\hat{\rho}} \frac{\partial}{\partial \hat{z}} \left(\hat{\rho} F \frac{\partial \hat{\phi}_0}{\partial \hat{z}} \right) \right] = 0. \quad (3.109)$$

Since the lowest-order horizontal velocity is divergence-free, we can define a streamfunction $\hat{\psi}$ such that

$$\hat{u}_0 = -\frac{\partial \hat{\psi}}{\partial \hat{y}}, \quad \hat{v}_0 = \frac{\partial \hat{\psi}}{\partial \hat{x}} \quad (3.110)$$

where also, using (3.96), $\phi_0 = \hat{f}_0 \hat{\psi}$. The vorticity is then given by $\hat{\zeta}_0 = \nabla^2 \hat{\psi}$ and (3.109) becomes a single equation in a single unknown, to wit

$$\frac{D_0}{D\hat{t}} \left[\nabla^2 \hat{\psi} + \hat{\beta} \hat{y} + \frac{\hat{f}_0^2}{\hat{\rho}} \frac{\partial}{\partial \hat{z}} \left(\hat{\rho} F \frac{\partial \hat{\psi}}{\partial \hat{z}} \right) \right] = 0, \quad (3.111)$$

where the material derivative is evaluated using $\hat{\mathbf{u}}_0 = \mathbf{k} \times \nabla \hat{\psi}$. This is the non-dimensional form of the quasi-geostrophic potential vorticity equation, one of the most important equations in dynamical meteorology and oceanography. In deriving it we have reduced the Navier–Stokes equations, which are six coupled nonlinear partial differential equations in six unknowns (u, v, w, T, p, ρ) to a single (albeit nonlinear) first-order partial differential equation in a single unknown.⁹

3.6.4 Dimensional equations

The dimensional version of the quasi-geostrophic potential vorticity equation may be written as

$$\begin{aligned} \frac{Dq}{Dt} &= 0, \\ q &= \nabla^2 \psi + f + \frac{f_0^2}{\bar{\rho}} \frac{\partial}{\partial z} \left(\frac{\bar{\rho}}{N^2} \frac{\partial \psi}{\partial z} \right), \end{aligned} \quad (3.112a,b)$$

where only the variable part of f (e.g., βy) is relevant in the second term on the right-hand side of the expression for q . The quantity q is known as the *quasi-geostrophic potential vorticity*. It is analogous to the exact (Ertel) potential vorticity (see section ?? for more about this), and it is conserved when advected by the *horizontal* geostrophic flow. All the other dynamical variables may be obtained from potential vorticity as follows.

- (i) Streamfunction, using (3.112b).
- (ii) Velocity: $\mathbf{u} = \mathbf{k} \times \nabla \psi [\equiv \nabla^\perp \psi = -\nabla \times (\mathbf{k} \psi)]$.
- (iii) Relative vorticity: $\zeta = \nabla^2 \psi$.
- (iv) Perturbation pressure: $\phi = f_0 \psi$.
- (v) Perturbation buoyancy: $b' = f_0 \partial \psi / \partial z$.

The length scale $L_d = NH/f_0$, emerges naturally from the QG dynamics. It is the scale at which buoyancy and relative vorticity effects contribute equally to the potential vorticity, and is called the *deformation radius*; it is analogous to the quantity $\sqrt{gH}/f_0$ arising in shallow water theory. In the upper ocean, with $N \approx 10^{-2} \text{ s}^{-1}$, $H \approx 10^3 \text{ m}$ and $f_0 \approx 10^{-4} \text{ s}^{-1}$, then $L_d \approx 100 \text{ km}$. At high latitudes the ocean is much less stratified and f is somewhat larger, and the deformation radius may be as little as 30 km (see Fig. ?? on page ??, where the deformation radius is defined slightly differently). In the atmosphere, with $N \approx 10^{-2} \text{ s}^{-1}$, $H \approx 10^4 \text{ m}$, then $L_d \approx 1000 \text{ km}$. It is this order of magnitude difference in the deformation scales that accounts for a great deal of the quantitative difference in the dynamics of the ocean and the atmosphere. If we take the limit $L_d \rightarrow \infty$ then the stratified quasi-geostrophic equations reduce to

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + f. \quad (3.113)$$

This is the two-dimensional vorticity equation, identical to (??). The high stratification of this limit has suppressed all vertical motion, and variations in the flow become confined to the horizontal plane. Finally, we note that it is typical in quasi-geostrophic applications to omit the prime on the buoyancy perturbations, and write $b = f_0 \partial \psi / \partial z$; however, we will keep the prime in this chapter.

3.6.5 Buoyancy advection at the surface

The solution of the elliptic equation in (3.112) requires vertical boundary conditions on ψ at the ground and at the top of the atmosphere, and these are given by use of the thermodynamic equation. For a flat, slippery, rigid surface the vertical velocity is zero so that the thermodynamic equation may be written as

$$\frac{Db'}{Dt} = 0, \quad b' = f_0 \frac{\partial \psi}{\partial z}. \quad (3.114)$$

We apply this at the ground and at the tropopause, treating the latter as a lid on the lower atmosphere. In the presence of friction and topography the vertical velocity is not zero, but is given by

$$w = r \nabla^2 \psi + \mathbf{u} \cdot \nabla \eta_b \quad (3.115)$$

where the first term represents Ekman friction (with the constant r proportional to the thickness of the Ekman layer) and the second term represents topographic forcing. The boundary condition becomes

$$\frac{\partial}{\partial t} \left(f_0 \frac{\partial \psi}{\partial z} \right) + \mathbf{u} \cdot \nabla \left(f_0 \frac{\partial \psi}{\partial z} + N^2 \eta_b \right) + N^2 r \nabla^2 \psi = 0, \quad (3.116)$$

where all the fields are evaluated at $z = 0$ or $z = H$, the height of the lid. Thus, the quasi-geostrophic system is characterized by the horizontal advection of potential vorticity in the interior and the advection of buoyancy at the boundary. Instead of a lid at the top, then in a compressible fluid such as the atmosphere we may suppose that all disturbances tend to zero as $z \rightarrow \infty$.

3.6.6 Quasi-geostrophy in pressure coordinates

The derivation of the quasi-geostrophic system in pressure coordinates is very similar to that in height coordinates, with the main difference coming at the boundaries, and we give only the results. The starting point is the primitive equations in pressure coordinates, (P.1). In pressure coordinates, the quasi-geostrophic potential vorticity is found to be

$$q = f + \nabla^2 \psi + \frac{\partial}{\partial p} \left(\frac{f_0^2}{S^2} \frac{\partial \psi}{\partial p} \right), \quad (3.117)$$

where $\psi = \Phi/f_0$ is the streamfunction and Φ the geopotential, and

$$S^2 \equiv -\frac{R}{p} \left(\frac{p}{p_R} \right)^\kappa \frac{d\tilde{\theta}}{dp} = -\frac{1}{\rho\theta} \frac{d\tilde{\theta}}{dp}, \quad (3.118)$$

where $\tilde{\theta}$ is a reference profile and a function of pressure only. In log-pressure coordinates, with $Z = -H \ln p$, the potential vorticity may be written as

$$q = f + \nabla^2 \psi + \frac{1}{\rho_*} \frac{\partial}{\partial Z} \left(\frac{\rho_* f_0^2}{N_Z^2} \frac{\partial \psi}{\partial Z} \right), \quad (3.119)$$

where

$$N_Z^2 = S^2 \left(\frac{p}{H} \right)^2 = -\left(\frac{R}{H} \right) \left(\frac{p}{p_R} \right)^\kappa \frac{d\tilde{\theta}}{dZ} \quad (3.120)$$

is the buoyancy frequency and $\rho_* = \exp(-z/H)$. Temperature and potential temperature are related to the streamfunction by

$$T = -\frac{f_0 p}{R} \frac{\partial \psi}{\partial p} = \frac{H f_0}{R} \frac{\partial \psi}{\partial Z}, \quad (3.121a)$$

$$\theta = -\left(\frac{p_R}{p} \right)^\kappa \left(\frac{f_0 p}{R} \right) \frac{\partial \psi}{\partial p} = \left(\frac{p_R}{p} \right)^\kappa \left(\frac{H f_0}{R} \right) \frac{\partial \psi}{\partial Z}. \quad (3.121b)$$

In pressure or log-pressure coordinates, potential vorticity is advected along isobaric surfaces, analogous to the horizontal advection in height coordinates.

The surface boundary condition again is derived from the thermodynamic equation. In log-pressure coordinates this is

$$\frac{D}{Dt} \left(\frac{\partial \psi}{\partial Z} \right) + \frac{N_Z^2}{f_0} W = 0, \quad (3.122)$$

where $W = DZ/Dt$. This is not the real vertical velocity, w , but it is related to it by

$$w = \frac{f_0}{g} \frac{\partial \psi}{\partial t} + \frac{RT}{gH} W. \quad (3.123)$$

Thus, choosing $H = RT(0)/g$, we have, at $Z = 0$,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi}{\partial Z} - \frac{N^2}{g} \psi \right) + \mathbf{u} \cdot \nabla \frac{\partial \psi}{\partial Z} = -\frac{N^2}{f_0} w, \quad (3.124)$$

where

$$w = \mathbf{u} \cdot \nabla \eta_b + r \nabla^2 \psi. \quad (3.125)$$

This differs from the expression in height coordinates only by the second term in the local time derivative. In applications where accuracy is not the main issue the simpler boundary condition $D(\partial_Z \psi)/Dt = 0$ is sometimes used.

3.6.7 ♦ The two-level quasi-geostrophic system

The quasi-geostrophic system has, in general, continuous variation in the vertical direction (and horizontal, of course). By finite-differencing the continuous equations we can obtain a *multi-level* model, and a crude but important special case of this is the *two-level* model, also known as the Phillips model.¹⁰ To obtain the equations of motion one way to proceed is to take a crude finite difference of the continuous relation between potential vorticity and streamfunction given in (3.112b). In the Boussinesq case (or in pressure coordinates, with a slight reinterpretation of the meaning of the symbols) the continuous expression for potential vorticity is

$$q = \zeta + f + \frac{\partial}{\partial z} \left(\frac{f_0 b'}{N^2} \right), \quad (3.126)$$

where $b' = f_0 \partial \psi / \partial z$. In the case with a flat bottom and rigid lid at the top (and incorporating topography is an easy extension) the boundary condition of $w = 0$ is satisfied by $D\partial_z \psi / Dt = 0$ at the top and bottom. An obvious finite-differencing of (3.126) in the vertical direction (see Fig. 3.2) then gives

$$q_1 = \zeta_1 + f + \frac{2f_0^2}{N^2 H_1 H} (\psi_2 - \psi_1), \quad q_2 = \zeta_2 + f + \frac{2f_0^2}{N^2 H_2 H} (\psi_1 - \psi_2). \quad (3.127)$$

In atmospheric problems it is common to choose $H_1 = H_2$, whereas in oceanic problems we might choose to have a thinner upper layer, representing the flow above the main thermocline. Note that the boundary conditions of $w = 0$ at the top and bottom are already taken care of in (3.127): *they are incorporated into the definition of the potential vorticity* — a finite-difference analogue of the delta-function construction of section 3.6.5. At each level the potential vorticity is advected by the streamfunction so that the evolution equation for each level is:

$$\frac{Dq_i}{Dt} = \frac{\partial q_i}{\partial t} + \mathbf{u}_i \cdot \nabla q_i = \frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0, \quad i = 1, 2. \quad (3.128)$$

Informal Derivation of Stratified QG Equations

We will use the Boussinesq equations, but similar derivations could be given using the anelastic equations or pressure coordinates. The first ingredient is the vertical component of the vorticity equation, (??); in the Boussinesq version (or the pressure coordinate or anelastic versions) there is no baroclinic term and we have

$$\frac{D_3}{Dt}(\zeta + f) = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right). \quad (\text{QG.1})$$

We now apply the assumptions on page 66. The advection and the vorticity on the left-hand side are geostrophic, but we keep the horizontal divergence (which is small) on the right-hand side where it is multiplied by the big term f . Furthermore, because f is nearly constant we replace it with f_0 except where it is differentiated. The second term (tilting) on the right-hand side is smaller than the advection terms on the left-hand side by the ratio $[UW/(HL)]/[U^2/L^2] = [W/H]/[U/L] \ll 1$, because w is small ($\partial w/\partial z$ equals the divergence of the ageostrophic velocity). We therefore neglect it, and given all this (QG.1) becomes

$$\frac{D_g}{Dt}(\zeta_g + f) = -f_0 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = f_0 \frac{\partial w}{\partial z}, \quad (\text{QG.2})$$

where the second equality uses mass continuity and $D_g/Dt = \partial/\partial t + \mathbf{u}_g \cdot \nabla$.

The second ingredient is the three-dimensional thermodynamic equation,

$$\frac{D_3 b}{Dt} = 0. \quad (\text{QG.3})$$

The stratification is assumed to be nearly constant, so we write $b = \tilde{b}(z) + b'(x, y, z, t)$, where $\tilde{b}$ is the basic state buoyancy. Furthermore, because w is small it only advects the basic state, and with $N^2 = \partial \tilde{b}/\partial z$ (QG.3) becomes

$$\frac{D_g b'}{Dt} + w N^2 = 0. \quad (\text{QG.4})$$

Hydrostatic and geostrophic wind balance enable us to write the geostrophic velocity, vorticity, and buoyancy in terms of streamfunction ψ [$= p/(f_0 \rho_0)$]:

$$\mathbf{u}_g = \mathbf{k} \times \nabla \psi, \quad \zeta_g = \nabla^2 \psi, \quad b' = f_0 \partial \psi / \partial z. \quad (\text{QG.5})$$

The quasi-geostrophic potential vorticity equation is obtained by eliminating w between (QG.2) and (QG.4), and this gives

$$\frac{D_g q}{Dt} = 0, \quad q = \zeta_g + f + \frac{\partial}{\partial z} \left(\frac{f_0 b'}{N^2} \right). \quad (\text{QG.6})$$

This equation is the Boussinesq version of (3.112), and using (QG.5) it may be expressed entirely in terms of the streamfunction, with $D_g \cdot /Dt = \partial/\partial t + J(\psi, \cdot)$. The vertical boundary conditions, at $z = 0$ and $z = H$ say, are given by (QG.4) with $w = 0$, with straightforward generalizations if topography or friction are present.

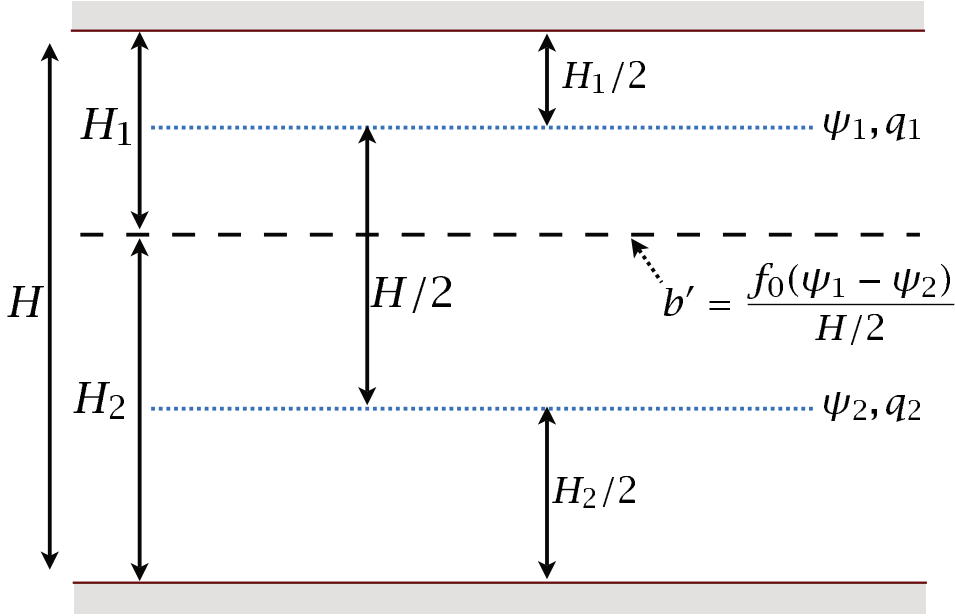


Figure 3.2 The two-level quasi-geostrophic system with a flat bottom and rigid lid at which $w = 0$.

Models with more than two levels can be constructed by extending the finite-differencing procedure in a natural way.

Connection to the layered system

The two-level expressions, (3.127), have an obvious similarity to the *two-layer* expressions, (3.60). Noting that $N^2 = \partial \hat{b} / \partial z$ and that $b = -g\delta\rho/\rho_0$ it is natural to let

$$N^2 = -\frac{g}{\rho_0} \frac{\rho_1 - \rho_2}{H/2} = \frac{g'}{H/2}. \quad (3.129)$$

With this identification we find that (3.127) becomes

$$q_1 = \zeta_1 + f + \frac{f_0^2}{g'H_1}(\psi_2 - \psi_1), \quad q_2 = \zeta_2 + f + \frac{f_0^2}{g'H_2}(\psi_1 - \psi_2). \quad (3.130)$$

These expressions are identical to (3.60) in the flat-bottomed, rigid lid case. Similarly, a multi-layered system with n layers is equivalent to a finite-difference representation with n levels. It should be said, though, that in the pantheon of quasi-geostrophic models the two-level and two-layer models hold distinguished places.

3.7 ENERGETICS OF QUASI-GEOSTROPHY

If the quasi-geostrophic set of equations is to represent a real fluid system in a physically meaningful way, then it should have a consistent set of energetics. In particular, the total energy should be conserved, and there should be analogs of kinetic and potential energy and conversion between the two. We now show that such energetic properties do hold, using the Boussinesq set as an example.

Let us write the governing equations as a potential vorticity equation in the interior,

$$\frac{D}{Dt} \left[\nabla^2 \psi + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) \right] + \beta \frac{\partial \psi}{\partial x} = 0, \quad 0 < z < 1, \quad (3.131)$$

and buoyancy advection at the boundary,

$$\frac{D}{Dt} \left(\frac{\partial \psi}{\partial z} \right) = 0, \quad z = 0, 1. \quad (3.132)$$

For lateral boundary conditions we may assume that $\psi = \text{constant}$, or impose periodic conditions. If we multiply (3.131) by $-\psi$ and integrate over the domain, using the boundary conditions, we easily find

$$\frac{d\hat{E}}{dt} = 0, \quad \hat{E} = \frac{1}{2} \int_V \left[(\nabla \psi)^2 + \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 \right] dV. \quad (3.133a,b)$$

The term involving β makes no direct contribution to the energy budget. Equation (3.133) is the fundamental energy equation for quasi-geostrophic motion, and it states that in the absence of viscous or diabatic terms the total energy is conserved. The two terms in (3.133b) can be identified as the kinetic energy (KE) and available potential energy (APE) of the flow, where

$$KE = \frac{1}{2} \int_V (\nabla \psi)^2 dV, \quad APE = \frac{1}{2} \int_V \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV. \quad (3.134a,b)$$

The available potential energy may also be written as

$$APE = \frac{1}{2} \int_V \frac{H^2}{L_d^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV, \quad (3.135)$$

where L_d is the deformation radius NH/f_0 and we may choose H such that $z \sim H$. At some scale L the ratio of the kinetic energy to the potential energy is thus, roughly,

$$\frac{KE}{APE} \sim \frac{L_d^2}{L^2}. \quad (3.136)$$

For scales much larger than L_d the potential energy dominates the kinetic energy, and contrariwise.

3.7.1 Conversion between APE and KE

Let us return to the vorticity and thermodynamic equations,

$$\frac{D\zeta}{Dt} = f \frac{\partial w}{\partial z} \quad (3.137)$$

where $\zeta = \nabla^2 \psi$, and

$$\frac{Db'}{Dt} + N^2 w = 0 \quad (3.138)$$

where $b' = f_0 \partial \psi / \partial z$. From (3.137) we form a kinetic energy equation namely

$$\frac{1}{2} \frac{d}{dt} \int_V (\nabla \psi)^2 dV = - \int_V f_0 \frac{\partial w}{\partial z} \psi dV = \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (3.139)$$

From (3.138) we form a potential energy equation, namely

$$\frac{d}{dt} \frac{1}{2} \int_V \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV = - \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (3.140)$$

Thus, the *conversion* from APE to KE is represented by

$$\frac{d}{dt} KE = - \frac{d}{dt} APE = \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (3.141)$$

Because the buoyancy is proportional to $\partial \psi / \partial z$, when warm fluid rises there is a correlation between w and $\partial \psi / \partial z$ and APE is converted to KE. Whether such a phenomenon occurs depends of course on the dynamics of the flow; however, such a conversion *is*, in fact, a common feature of geophysical flows, and in particular of baroclinic instability, as we shall see in chapter ??.

3.7.2 Energetics of two-layer flows

Two-layer or two-level flows are an important special case. For layers of equal thickness let us write the evolution equations as

$$\frac{D}{Dt} \left[\nabla^2 \psi_1 - \frac{1}{2} k_d^2 (\psi_1 - \psi_2) \right] + \beta \frac{\partial \psi_1}{\partial x} = 0 \quad (3.142a)$$

$$\frac{D}{Dt} \left[\nabla^2 \psi_2 + \frac{1}{2} k_d^2 (\psi_1 - \psi_2) \right] + \beta \frac{\partial \psi_2}{\partial x} = 0 \quad (3.142b)$$

where $k_d^2/2 = (2f_0/NH)^2$. On multiplying these two equations by $-\psi_1$ and $-\psi_2$, respectively, and integrating over the horizontal domain, the advective term in the material derivatives and the beta term all vanish, and we obtain

$$\int_A \left[\frac{d}{dt} \frac{1}{2} (\nabla \psi_1)^2 + \frac{1}{2} k_d^2 \psi_1 \frac{d}{dt} (\psi_1 - \psi_2) \right] dA = 0, \quad (3.143a)$$

$$\int_A \left[\frac{d}{dt} \frac{1}{2} (\nabla \psi_2)^2 - \frac{1}{2} k_d^2 \psi_2 \frac{d}{dt} (\psi_1 - \psi_2) \right] dA = 0. \quad (3.143b)$$

Adding these gives

$$\frac{d}{dt} \int_A \left[\frac{1}{2} (\nabla \psi_1)^2 + \frac{1}{2} (\nabla \psi_2)^2 + \frac{k_d^2}{4} (\psi_1 - \psi_2)^2 \right] dA = 0. \quad (3.144)$$

This is the energy conservation statement for the two layer model. The first two terms represent the kinetic energy and the last term represents the available potential energy.

Energy in the baroclinic and barotropic modes

A useful partitioning of the energy is between the energy in the barotropic and baroclinic modes. The barotropic streamfunction, $\bar{\psi}$, is the vertically averaged streamfunction and the baroclinic mode is the difference between the streamfunctions in the two layers. That is, for equal layer thicknesses,

$$\bar{\psi} \equiv \frac{1}{2} (\psi_1 + \psi_2), \quad \tau \equiv \frac{1}{2} (\psi_1 - \psi_2). \quad (3.145)$$

Substituting (3.145) into (3.144) reveals that

$$\frac{d}{dt} \int_A \left[(\nabla \bar{\psi})^2 + (\nabla \tau)^2 + k_d^2 \tau^2 \right] dA = 0. \quad (3.146)$$

The energy density in the barotropic mode is thus just $(\nabla \bar{\psi})^2$, and that in the baroclinic mode is $(\nabla \tau)^2 + k_d^2 \tau^2$. This partitioning will prove particularly useful when we consider baroclinic turbulence in chapter ??.

3.7.3 Enstrophy conservation

Potential vorticity is advected only by the horizontal flow, and thus it is materially conserved on the horizontal surface at every height and

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0. \quad (3.147)$$

Furthermore, the advecting flow is divergence-free so that $\mathbf{u} \cdot \nabla q = \nabla \cdot (\mathbf{u}q)$. Thus, on multiplying (3.147) by q and integrating over a horizontal domain, A , using either no-normal flow or periodic boundary conditions, we straightforwardly obtain

$$\frac{d\widehat{Z}}{dt} = 0, \quad \widehat{Z} = \frac{1}{2} \int_A q^2 dA. \quad (3.148)$$

The quantity $\widehat{Z}$ is known as the enstrophy, and this is conserved at each height as well as, naturally, over the entire volume.

The enstrophy is just one of an infinity of invariants in quasi-geostrophic flow. Because the potential vorticity of a fluid element is conserved, *any* function of the potential vorticity must be a material invariant and we can immediately write

$$\frac{D}{Dt}F(q) = 0. \quad (3.149)$$

To verify that this is true, simply note that (3.149) implies that $(dF/dq)Dq/Dt = 0$, which is true by virtue of (3.147). [However, by virtue of the material advection, the function $F(q)$ need not be differentiable in order for (3.149) to hold.] Each of the material invariants corresponding to different choices of $F(q)$ has a corresponding integral invariant; that is,

$$\frac{d}{dt} \int_A F(q) dA = 0. \quad (3.150)$$

The enstrophy invariant corresponds to choosing $F(q) = q^2$; it plays a particularly important role because, like energy, it is a quadratic invariant, and its presence profoundly alters the behaviour of two-dimensional and quasi-geostrophic flow compared to three-dimensional flow.

3.8 EKMAN LAYERS

We now consider the effects of friction on geostrophic flow, and in particular the generation of boundary layers known as *Ekman layers*.

The fluid fields in the interior of a domain are often set by different physical processes than those occurring at a boundary, and consequently often change rapidly in a thin *boundary layer*, as in Fig. 3.3. Such boundary layers nearly always involve one or both of viscosity and diffusion, because these appear in the terms of highest differential order in the equations of motion, and so are responsible for the number and type of boundary conditions that the equations must satisfy — for example, the presence of molecular viscosity leads to the condition that the tangential flow (as well as the normal flow) must vanish at a rigid surface. In many boundary layers in non-rotating flow the dominant balance in the momentum equation is between the advective and viscous terms. In some contrast, in large-scale atmospheric and oceanic flow the effects of rotation are large, and this results in a boundary layer, known as the *Ekman layer*, in which the dominant balance is between Coriolis and frictional or stress terms.

The atmospheric Ekman layer occurs near the ground, and the stress at the ground itself is due to the surface wind (and its vertical variation). In the ocean the main Ekman layer is near the surface, and the stress at ocean surface is largely due to the presence of the overlying wind. There is also a weak Ekman layer at the bottom of the ocean, analogous to the atmospheric Ekman layer. To analyze all these layers, let us assume the following.

- The Ekman layer is Boussinesq. This is a very good assumption for the ocean, and a reasonable one for the atmosphere if the boundary layer is not too deep.
- The Ekman layer has a finite depth that is less than the total depth of the fluid, this depth being given by the level at which the frictional stresses essentially vanish. Within the Ekman layer, frictional terms are important, whereas geostrophic balance holds beyond it.

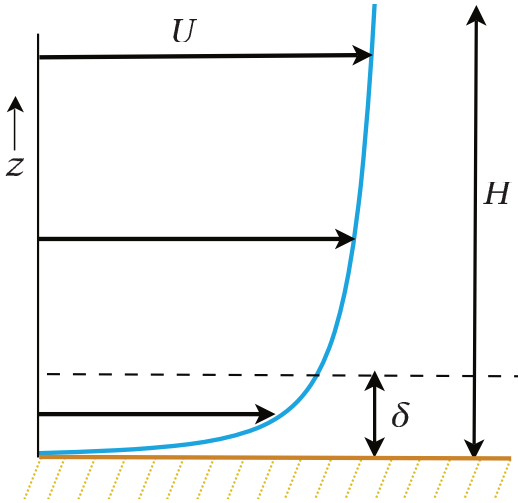


Figure 3.3 An idealized boundary layer. The values of a field, such as velocity, U , may vary rapidly in a boundary in order to satisfy the boundary conditions at a rigid surface. The parameter δ is a measure of the boundary layer thickness, H is a typical scale of variation away from the boundary, and typically a boundary layer has $\delta \ll H$.

- The nonlinear and time-dependent terms in the equations of motion are negligible, hydrostatic balance holds in the vertical, and buoyancy is constant, not varying in the horizontal.
- As needed, we shall assume that friction can be parameterized by a viscous term of the form $\rho_0^{-1} \partial \boldsymbol{\tau} / \partial z = A \partial^2 \mathbf{u} / \partial z^2$, where A is constant and $\boldsymbol{\tau}$ is the stress. [In general, stress is a tensor, τ_{ij} , with an associated force given by $F_i = \partial \tau_{ij} / \partial x_j$, summing over the repeated index. It is common in geophysical fluid dynamics that the vertical derivative dominates, and in this case the force is $\mathbf{F} = \partial \boldsymbol{\tau} / \partial z$. We still use the word stress for $\boldsymbol{\tau}$, but it now refers to a vector whose derivative in a particular direction (z in this case) is the force on a fluid.] In laboratory settings A may be the molecular viscosity, whereas in the atmosphere and ocean it is a so-called *eddy viscosity*.

3.8.1 Equations of motion and scaling

Frictional–geostrophic balance in the horizontal momentum equation is:

$$\mathbf{f} \times \mathbf{u} = -\nabla_z \phi + \frac{\partial \tilde{\boldsymbol{\tau}}}{\partial z}, \quad (3.151)$$

where $\tilde{\boldsymbol{\tau}} \equiv \boldsymbol{\tau} / \rho_0$ is the kinematic stress and $\mathbf{f} = f \mathbf{k}$, where the Coriolis parameter f is allowed to vary with latitude. If we model the stress with an eddy viscosity, (3.151) becomes

$$\mathbf{f} \times \mathbf{u} = -\nabla_z \phi + A \frac{\partial^2 \mathbf{u}}{\partial z^2}. \quad (3.152)$$

The vertical momentum equation is $\partial \phi / \partial z = b$, i.e., hydrostatic balance, and, because buoyancy is constant, we may without loss of generality write this as

$$\frac{\partial \phi}{\partial z} = 0. \quad (3.153)$$

The equation set is completed by the mass continuity equation, $\nabla \cdot \mathbf{v} = 0$.

The Ekman number

We non-dimensionalize the equations by setting

$$(u, v) = U(\hat{u}, \hat{v}), \quad (x, y) = L(\hat{x}, \hat{y}), \quad f = f_0 \hat{f}, \quad z = H\hat{z}, \quad \phi = \Phi \hat{\phi}, \quad (3.154)$$

where hatted variables are non-dimensional. H is a scaling for the height, and at this stage we will suppose it to be some height scale in the free atmosphere or ocean, not the height of the Ekman layer itself. Geostrophic balance suggests that $\Phi = f_0 UL$. Substituting (3.154) into (3.152) we obtain

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\hat{\nabla} \hat{\phi} + Ek \frac{\partial^2 \hat{\mathbf{u}}}{\partial \hat{z}^2}, \quad (3.155)$$

where the parameter

$$Ek \equiv \left(\frac{A}{f_0 H^2} \right), \quad (3.156)$$

is the *Ekman number*, and it determines the importance of frictional terms in the horizontal momentum equation. If $Ek \ll 1$ then the friction is small in the flow interior where $\hat{z} = \mathcal{O}(1)$. However, the friction term cannot necessarily be neglected in the boundary layer because it is of the highest differential order in the equation, and so determines the boundary conditions; if Ek is small the vertical scales become small and the second term on the right-hand side of (3.155) remains finite. The case when this term is simply omitted from the equation is therefore a *singular limit*, meaning that it differs from the case with $Ek \rightarrow 0$. If $Ek \geq 1$ friction is important everywhere, but it is usually the case that Ek is small for atmospheric and oceanic large-scale flow, and the interior flow is very nearly geostrophic. (In part this is because A itself is only large near a rigid surface where the presence of a shear creates turbulence and a significant eddy viscosity.)

Momentum balance in the Ekman layer

For definiteness, suppose the fluid lies above a rigid surface at $z = 0$. Sufficiently far away from the boundary the velocity field is known, and we suppose this flow to be in geostrophic balance. We then write the velocity field and the pressure field as the sum of the interior geostrophic part, plus a boundary layer correction:

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_g + \hat{\mathbf{u}}_E, \quad \hat{\phi} = \hat{\phi}_g + \hat{\phi}_E, \quad (3.157)$$

where the Ekman layer corrections, denoted with a subscript E , are negligible away from the boundary layer. Now, in the fluid interior we have, by hydrostatic balance, $\partial \hat{\phi}_g / \partial \hat{z} = 0$. In the boundary layer we still have $\partial \hat{\phi}_g / \partial \hat{z} = 0$ so that, to satisfy hydrostasy, $\partial \hat{\phi}_E / \partial \hat{z} = 0$. But because $\hat{\phi}_E$ vanishes away from the boundary we have $\hat{\phi}_E = 0$ everywhere. Thus, *there is no boundary layer in the pressure field*. Note that this is a much stronger result than saying that pressure is continuous, which is nearly always true in fluids; rather, it is a special result for Ekman layers.

Using (3.157) with $\widehat{\phi}_E = 0$, the dimensional horizontal momentum equation (3.151) becomes, in the Ekman layer,

$$\mathbf{f} \times \mathbf{u}_E = \frac{\partial \boldsymbol{\tau}}{\partial z}. \quad (3.158)$$

The dominant force balance in the Ekman layer is thus between the Coriolis force and the friction. We can determine the thickness of the Ekman layer if we model the stress with an eddy viscosity so that

$$\mathbf{f} \times \mathbf{u}_E = A \frac{\partial^2 \mathbf{u}_E}{\partial z^2}, \quad (3.159)$$

or, non-dimensionally,

$$\widehat{\mathbf{f}} \times \widehat{\mathbf{u}}_E = Ek \frac{\partial^2 \widehat{\mathbf{u}}_E}{\partial \widehat{z}^2}. \quad (3.160)$$

It is evident this equation can only be satisfied if $\widehat{z} \neq \mathcal{O}(1)$, implying that H is not a proper scaling for z in the boundary layer. Rather, if the vertical scale in the Ekman layer is $\widehat{\delta}$ (meaning $\widehat{z} \sim \widehat{\delta}$) we must have $\widehat{\delta} \sim Ek^{1/2}$. In dimensional terms this means the thickness of the Ekman layer is

$$\delta = H \widehat{\delta} = HEk^{1/2} \quad (3.161)$$

or

$$\delta = \left(\frac{A}{f_0} \right)^{1/2}. \quad (3.162)$$

[This estimate also emerges directly from (3.159).] Note that (3.161) can be written as

$$Ek = \left(\frac{\delta}{H} \right)^2. \quad (3.163)$$

That is, the Ekman number is equal to the square of the ratio of the depth of the Ekman layer to an interior depth scale of the fluid motion. In laboratory flows where A is the molecular viscosity we can thus estimate the Ekman layer thickness, and if we know the eddy viscosity of the ocean or atmosphere we can estimate their respective Ekman layer thicknesses. We can invert this argument and obtain an estimate of A if we know the Ekman layer depth. In the atmosphere, deviations from geostrophic balance are very small in the atmosphere above 1 km, and using this gives $A \approx 10^2 \text{ m}^2 \text{ s}^{-1}$. In the ocean Ekman depths are often 50 m or less, and eddy viscosities are about $0.1 \text{ m}^2 \text{ s}^{-1}$.

3.8.2 Integral properties of the Ekman layer

What can we deduce about the Ekman layer without specifying the detailed form of the frictional term? Using dimensional notation we recall frictional–geostrophic balance,

$$\mathbf{f} \times \mathbf{u} = -\nabla \phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad (3.164)$$

where $\boldsymbol{\tau}$ is zero at the edge of the Ekman layer. In the Ekman layer itself we have

$$\mathbf{f} \times \mathbf{u}_E = \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z}. \quad (3.165)$$

Consider either a top or bottom Ekman layer, and integrate over its thickness. From (3.165) we obtain

$$\mathbf{f} \times \mathbf{M}_E = \boldsymbol{\tau}_T - \boldsymbol{\tau}_B, \quad (3.166)$$

where

$$\mathbf{M}_E = \int_{Ek} \rho_0 \mathbf{u}_E \, dz \quad (3.167)$$

is the ageostrophic mass transport in the Ekman layer, and $\boldsymbol{\tau}_T$ and $\boldsymbol{\tau}_B$ are the respective stresses at the top and the bottom of the Ekman layer at hand. The stress at the top (bottom) will be zero in a bottom (top) Ekman layer and therefore, from (3.166),

top Ekman layer:	$\mathbf{M}_E = -\frac{1}{f} \mathbf{k} \times \boldsymbol{\tau}_T$	(3.168a,b)
bottom Ekman layer:	$\mathbf{M}_E = \frac{1}{f} \mathbf{k} \times \boldsymbol{\tau}_B$	

The transport is thus at right angles to the stress at the surface, and proportional to the magnitude of the stress. These properties have a simple physical explanation: integrated over the depth of the Ekman layer the surface stress must be balanced by the Coriolis force, which in turn acts at right angles to the mass transport. A consequence of (3.168) is that the mass transports in adjacent oceanic and atmospheric Ekman layers are equal and opposite, because the stress is continuous across the ocean–atmosphere interface. Equation (3.168a) is particularly useful in the ocean, where the stress at the surface is primarily due to the wind, and is largely independent of the interior oceanic flow. In the atmosphere, the surface stress mainly arises as a result of the interior atmospheric flow, and to calculate it we need to parameterize the stress in terms of the flow.

Finally, we obtain an expression for the vertical velocity induced by an Ekman layer. The mass conservation equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \quad (3.169)$$

Integrating this over an Ekman layer gives

$$\frac{1}{\rho_0} \nabla \cdot \mathbf{M}_T = -(w_T - w_B), \quad (3.170)$$

where $\mathbf{M}_T$ is the total (Ekman plus geostrophic) mass transport in the Ekman layer,

$$\mathbf{M}_T = \int_{Ek} \rho_0 \mathbf{u} \, dz = \int_{Ek} \rho_0 (\mathbf{u}_g + \mathbf{u}_E) \, dz \equiv \mathbf{M}_g + \mathbf{M}_E, \quad (3.171)$$

and w_T and w_B are the vertical velocities at the top and bottom of the Ekman layer; the former (latter) is zero in a top (bottom) Ekman layer. Equations (3.171) and (3.166) give

$$\mathbf{k} \times (\mathbf{M}_T - \mathbf{M}_g) = \frac{1}{f}(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B). \quad (3.172)$$

Taking the curl of this (i.e., cross-differentiating) gives

$$\nabla \cdot (\mathbf{M}_T - \mathbf{M}_g) = \text{curl}_z[(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B)/f], \quad (3.173)$$

where the curl_z operator on a vector $\mathbf{A}$ is defined by $\text{curl}_z \mathbf{A} \equiv \partial_x A_y - \partial_y A_x$. Using (3.170) we obtain, for top and bottom Ekman layers respectively,

$$w_B = \frac{1}{\rho_0} \left(\text{curl}_z \frac{\boldsymbol{\tau}_T}{f} + \nabla \cdot \mathbf{M}_g \right), \quad w_T = \frac{1}{\rho_0} \left(\text{curl}_z \frac{\boldsymbol{\tau}_B}{f} - \nabla \cdot \mathbf{M}_g \right), \quad (3.174a,b)$$

where $\nabla \cdot \mathbf{M}_g = -(\beta/f)\mathbf{M}_g \cdot \mathbf{j}$ is the divergence of the geostrophic transport in the Ekman layer, and this is often small compared to the other terms in these equations. Thus, friction induces a vertical velocity at the edge of the Ekman layer, proportional to the curl of the stress at the surface, and this is perhaps the most used result in Ekman layer theory. Numerical models sometimes do not have the vertical resolution to explicitly resolve an Ekman layer, and (3.174) provides a means of *parameterizing* the Ekman layer in terms of resolved or known fields. It is particularly useful for the top Ekman layer in the ocean, where the stress can be regarded as a given function of the overlying wind.

3.8.3 Sverdrup Balance

In this section we rederive the above results in a slightly more direct way, and also obtain a result for the total transport induced by a windstress. To this end, consider an ocean forced by a windstress at the top that satisfies the Ekman-layer equations

$$-fv = -\frac{\partial \phi}{\partial x} + \frac{\partial \tilde{\tau}_x}{\partial z}, \quad fu = -\frac{\partial \phi}{\partial y} + \frac{\partial \tilde{\tau}_y}{\partial z}. \quad (3.175)$$

where $\tilde{\tau} = \tau/\rho_0$. Equivalently we have

$$f(v_g - v) = \frac{\partial \tilde{\tau}_x}{\partial z}, \quad f(u - u_g) = \frac{\partial \tilde{\tau}_y}{\partial z}. \quad (3.176)$$

We note that the geostrophic velocity field satisfies,

$$f \left(\frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} \right) = -\beta v_g. \quad (3.177)$$

If we integrate the mass continuity equation over the depth of the Ekman layer, the vertical velocity at its base is given by

$$w_E = \int_{-H_E}^0 \left(\frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y} + \frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} \right). \quad (3.178)$$

The divergence of the geostrophic velocity is given by (3.177), and that of the ageostrophic velocity is obtained from (3.176). We thus obtain

$$w_E = \left[\frac{\partial}{\partial x} \left(\frac{\tilde{\tau}_{y0}}{f} \right) - \frac{\partial}{\partial y} \left(\frac{\tilde{\tau}_{x0}}{f} \right) \right] - \int_{-H_E}^0 \frac{\beta}{f} v_g dz, \quad (3.179)$$

where $\tilde{\tau}_{x0}, \tilde{\tau}_{y0}$ are the components of the stress at the surface. This equation is essentially the same as (3.174a).

If we go back to (3.175), cross differentiate and integrate from the top down we obtain an expression for the vertical velocity at the base of the Ekman layer in terms of the stress and the *total* velocity,

$$w_E = \frac{1}{f} \left[\frac{\partial \tilde{\tau}_{y0}}{\partial x} - \frac{\partial \tilde{\tau}_{x0}}{\partial y} \right] - \int_{-H_E}^0 \frac{\beta}{f} v dz. \quad (3.180)$$

If we let the integral go over the entire depth of the ocean, and assume that the vertical velocity is zero at the bottom, we obtain

$$\int \beta v dz = \frac{\partial \tilde{\tau}_{y0}}{\partial x} - \frac{\partial \tilde{\tau}_{x0}}{\partial y}. \quad (3.181)$$

This is known as the Sverdrup relation, and is a relation between the stress at the surface and the total meridional transport in the ocean.

3.8.4 ♦ Frictional parameterization of the Ekman layer

Suppose that the free atmosphere is described by the quasi-geostrophic vorticity equation,

$$\frac{D\zeta_g}{Dt} = f_0 \frac{\partial w}{\partial z}, \quad (3.182)$$

where ζ_g is the geostrophic relative vorticity. Let us further model the atmosphere as a single homogeneous layer of thickness H lying above an Ekman layer of thickness $d \ll H$. If the vertical velocity is negligible at the top of the layer (at $z = H + d$) the equation of motion becomes

$$\frac{D\zeta_g}{Dt} = \frac{f_0[w(H+d) - w(d)]}{H} = -\frac{f_0 d}{2H} \zeta_g \quad (3.183)$$

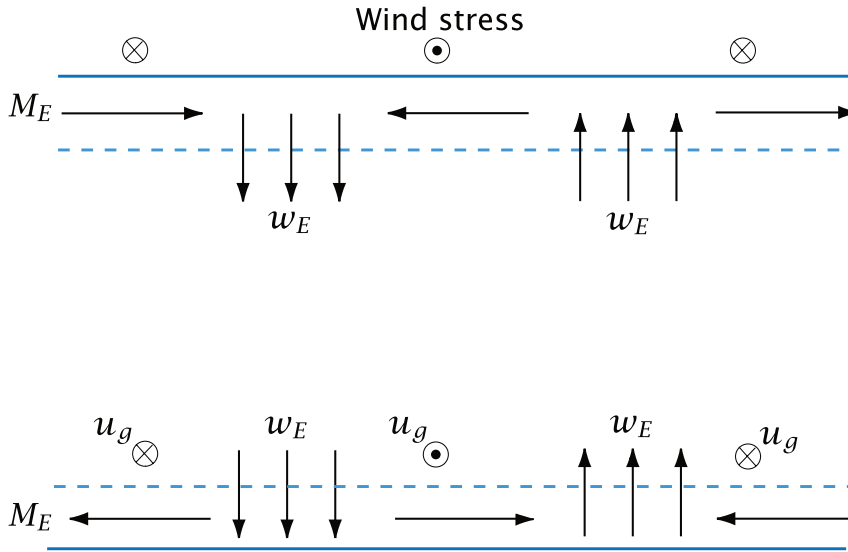


Figure 3.4 Upper and lower Ekman layers. The upper Ekman layer in the ocean is primarily driven by an imposed wind stress, whereas the lower Ekman layer in the atmosphere or ocean largely results from the interaction of interior geostrophic velocity and a rigid lower surface. The upper part of figure shows the vertical Ekman ‘pumping’ velocities that result from the given wind stress, and the lower part of the figure shows the Ekman pumping velocities given the interior geostrophic flow.

using (??). This equation shows that the Ekman layer acts as a *linear drag* on the interior flow, with a drag coefficient r equal to $f_0 d / 2H$ and with associated time scale T_{Ek} given by

$$T_{Ek} = \frac{2H}{f_0 d} = \frac{2H}{\sqrt{2} f_0 A}. \quad (3.184)$$

In the oceanic case the corresponding vorticity equation for the interior flow is

$$\frac{D\zeta_g}{Dt} = \frac{1}{H} \text{curl}_z \tau_s, \quad (3.185)$$

where τ_s is the surface stress. The surface stress thus acts as if it were a body force on the interior flow, and neither the Coriolis parameter nor the depth of the Ekman layer explicitly appear in this formula.

The Ekman layer is a very efficient way of communicating surface stresses to the interior. To see this, suppose that eddy mixing were the sole mechanism of transferring stress from the surface to the fluid interior, and there were no Ekman layer. Then the time scale of spindown of the fluid would be given by using

$$\frac{d\zeta}{dt} = A \frac{\partial^2 \zeta}{\partial z^2}, \quad (3.186)$$

implying a turbulent spin-down time, T_{turb} , of

$$T_{turb} \sim \frac{H^2}{A}, \quad (3.187)$$

where H is the depth over which we require a spin-down. This is much longer than the spin-down of a fluid that has an Ekman layer, for we have

$$\frac{T_{turb}}{T_{Ek}} = \frac{(H^2/A)}{(2H/f_0d)} = \frac{H}{d} \gg 1, \quad (3.188)$$

using $d = \sqrt{2A/f_0}$. The effects of friction are enhanced because of the presence of a secondary circulation confined to the Ekman layers (as in Fig. 3.4) in which the vertical scales are much smaller than those in the fluid interior and so where viscous effects become significant; these frictional stresses are then communicated to the fluid interior via the induced vertical velocities at the edge of the Ekman layers.

CHAPTER 4

ROSSBY WAVES

WEEKS 8 TO 10

4.1 FUNDAMENTALS AND FORMALITIES

4.1.1 Definitions and kinematics

A wave may be considered to be a propagating disturbance that has a characteristic relationship between its frequency and size. A linear wave might be defined as a disturbance that satisfies a *dispersion relation*. In order to see what all this means, and what a dispersion relation is, suppose that a disturbance, $\psi(\mathbf{x}, t)$ (where ψ might be velocity, streamfunction, pressure, etc), satisfies some equation

$$L(\psi) = 0, \quad (4.1)$$

where L is a linear operator, typically a polynomial in time and space derivatives. If (4.1) has constant coefficients then harmonic solutions may often be found that are a superposition of *plane waves*, each of which satisfy

$$\psi = \text{Re } \tilde{\psi} e^{i\theta(\mathbf{x}, t)} = \text{Re } \tilde{\psi} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (4.2)$$

where $\tilde{\psi}$ is a complex constant, θ is the phase, ω is the wave frequency and $\mathbf{k}$ is the vector wavenumber (k, l, m) (also written as (k^x, k^y, k^z) or, in subscript notation, k_i). The prefix Re denotes the real part of the expression, but we will drop it if there is no ambiguity.

Substituting (4.2) into (4.1) results in a *dispersion relation* of the form.

$$\omega = \Omega(\mathbf{k}) \quad (4.3)$$

where $\Omega(\mathbf{k})$ [or $\Omega(k_i)$, and meaning $\Omega(k, l, m)$] is some function determined by the form of L in (4.1) and which thus depends on the particular type of wave — the function is different for sound waves, light waves and the Rossby waves and gravity waves we will encounter in this book (peak ahead to (4.28) and (5.5), and there is more discussion in section 4.1.3). Unless it is necessary to explicitly distinguish the function Ω from the frequency ω , we will often write $\omega = \omega(\mathbf{k})$.

4.1.2 Wave propagation and phase speed

Phase speed

Let us consider the propagation of monochromatic plane waves. Given (4.2) a wave will propagate in the direction of $\mathbf{k}$ (Fig. 4.1). At a given instant and location we can align our coordinate axis along this direction, and we write $\mathbf{k} \cdot \mathbf{x} = Kx^*$, where x^* increases in the direction of $\mathbf{k}$ and $K^2 = |\mathbf{k}|^2$ is the magnitude of the wavenumber. With this, we can write (4.2) as

$$\psi = \text{Re } \tilde{\psi} e^{i(Kx^* - \omega t)} = \text{Re } \tilde{\psi} e^{iK(x^* - ct)}, \quad (4.4)$$

where $c = \omega/K$. From this equation it is evident that the phase of the wave propagates at the speed c in the direction of $\mathbf{k}$, and we define the *phase speed* by

$$c_p \equiv \frac{\omega}{K}. \quad (4.5)$$

The wavelength of the wave, λ , is the distance between two wavecrests — that is, the distance between two locations along the line of travel whose phase differs by 2π — and evidently this is given by

$$\lambda = \frac{2\pi}{K}. \quad (4.6)$$

In (for simplicity) a two-dimensional wave, and referring to Fig. 4.1, the wavelength and wave vectors in the x - and y -directions are given by,

$$\lambda^x = \frac{\lambda}{\cos \phi}, \quad \lambda^y = \frac{\lambda}{\sin \phi}, \quad k^x = K \cos \phi, \quad k^y = K \sin \phi. \quad (4.7)$$

In general, lines of constant phase intersect both the coordinate axes and propagate along them. The speed of propagation along these axes is given by

$$c_p^x = c_p \frac{l^x}{l} = \frac{c_p}{\cos \phi} = c_p \frac{K}{k^x} = \frac{\omega}{k^x}, \quad c_p^y = c_p \frac{l^y}{l} = \frac{c_p}{\sin \phi} = c_p \frac{K}{k^y} = \frac{\omega}{k^y}, \quad (4.8)$$

using (4.5) and (4.7), and again referring to Fig. 4.1 for notation. The speed of phase propagation along any one of the axis is in general *larger* than the phase speed in the primary direction of the wave. The phase speeds are clearly *not* components of a vector: for example, $c_p^x \neq c_p \cos \phi$. Analogously, the wavevector $\mathbf{k}$ is a true vector, whereas the wavelength λ is not.

To summarize, the phase speed and its components are given by

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k^x}, \quad c_p^y = \frac{\omega}{k^y}. \quad (4.9)$$

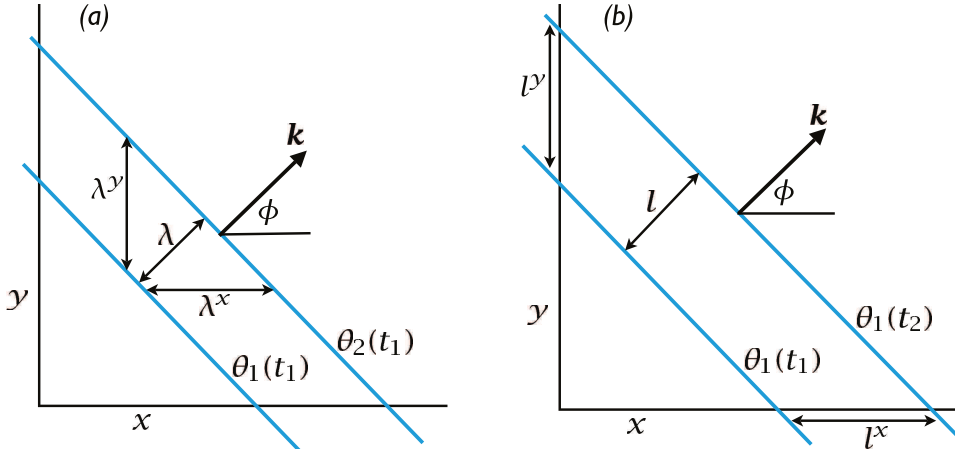


Figure 4.1 The propagation of a two-dimensional wave. (a) Two lines of constant phase (e.g., two wavecrests) at a time t_1 . The wave is propagating in the direction $\mathbf{k}$ with wavelength λ . (b) The same line of constant phase at two successive times. The phase speed is the speed of advancement of the wavecrest in the direction of travel, and so $c_p = l/(t_2 - t_1)$. The phase speed in the x -direction is the speed of propagation of the wavecrest along the x -axis, and $c_p^x = l^x/(t_2 - t_1) = c_p / \cos \phi$.

4.1.3 The dispersion relation

The above description is mostly kinematic and a little abstract, applying to almost any disturbance that has a wavevector and a frequency. The particular *dynamics* of a wave are determined by the relationship between the wavevector and the frequency; that is, by the *dispersion relation*. Once the dispersion relation is known a great many of the properties of the wave follow in a more-or-less straightforward manner, as we will see. Picking up from (4.3), the dispersion relation is a functional relationship between the frequency and the wavevector of the general form

$$\omega = \Omega(\mathbf{k}). \quad (4.10)$$

Perhaps the simplest example of a linear operator that gives rise to waves is the one-dimensional equation

$$\frac{\partial \psi}{\partial t} + c \frac{\partial \psi}{\partial x} = 0. \quad (4.11)$$

Substituting a trial solution of the form $\psi = \text{Re } A e^{i(kx - \omega t)}$, where Re denotes the real part, we obtain $(-i\omega + cik)A = 0$, giving the dispersion relation

$$\omega = ck. \quad (4.12)$$

The phase speed of this wave is $c_p = \omega/k = c$. A few other examples of governing equations, dispersion relations and phase speeds are:

$$\frac{\partial \psi}{\partial t} + \mathbf{c} \cdot \nabla \psi = 0, \quad \omega = \mathbf{c} \cdot \mathbf{k}, \quad c_p = |\mathbf{c}| \cos \theta, \quad c_p^x = \frac{\mathbf{c} \cdot \mathbf{k}}{k}, \quad c_p^y = \frac{\mathbf{c} \cdot \mathbf{k}}{l} \quad (4.13a)$$

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0, \quad \omega^2 = c^2 K^2, \quad c_p = \pm c, \quad c_p^x = \pm \frac{cK}{k}, \quad c_p^y = \pm \frac{cK}{l}, \quad (4.13b)$$

$$\frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0, \quad \omega = \frac{-\beta k}{K^2}, \quad c_p = \frac{\omega}{K}, \quad c_p^x = -\frac{\beta}{K^2}, \quad c_p^y = -\frac{\beta k/l}{K^2}. \quad (4.13c)$$

where $K^2 = k^2 + l^2$ and θ is the angle between $\mathbf{c}$ and $\mathbf{k}$, and the examples are all two-dimensional, with variation in x and y only.

A wave is said to be *nondispersive* or *dispersionless* if the phase speed is independent of the wavelength. This condition is clearly satisfied for the simple example (4.11) but is manifestly not satisfied for (4.13c), and these waves (Rossby waves, in fact) are *dispersive*. Waves of different wavelengths then travel at different speeds so that a group of waves will spread out — disperse — even if the medium is homogeneous. When a wave is dispersive there is another characteristic speed at which the waves propagate, known as the group velocity, and we come to this in the next section.

Most media are, of course, inhomogeneous, but if the medium varies sufficiently slowly in space and time — and in particular if the variations are slow compared to the wavelength and period — we may still have a *local* dispersion relation between frequency and wavevector,

$$\omega = \Omega(\mathbf{k}; \mathbf{x}, t). \quad (4.14)$$

Although Ω is a function of $\mathbf{k}$, $\mathbf{x}$ and t the semi-colon in (4.14) is used to suggest that $\mathbf{x}$ and t are slowly varying parameters of a somewhat different nature than $\mathbf{k}$. We'll resume our discussion of this topic in section ??, but before that we must introduce the group velocity.

4.2 GROUP VELOCITY

Information and energy clearly cannot travel at the phase speed, for as the direction of propagation of the phase line tends to a direction parallel to the y -axis, the phase speed in the x -direction tends to infinity! Rather, it turns out that most quantities of interest, including energy, propagate at the *group velocity*, a quantity of enormous importance in wave theory.¹¹ Roughly speaking, group velocity is the velocity at which a packet or a group of waves will travel, whereas the individual wave crests travel at the phase speed. To introduce the idea we will consider the superposition of plane waves, noting that a monochromatic plane wave already fills space uniformly so that there can be no propagation of energy from place to place. We will restrict attention to waves propagating in one direction, but the argument may be extended to two or three dimensions.

4.2.1 Superposition of two waves

Consider the linear superposition of two waves. Limiting attention to the one-dimensional case for simplicity, consider a disturbance represented by

$$\psi = \text{Re } \tilde{\psi}(e^{i(k_1 x - \omega_1 t)} + e^{i(k_2 x - \omega_2 t)}). \quad (4.15)$$

Wave Fundamentals

- A wave is a propagating disturbance that has a characteristic relationship between its frequency and size, known as the dispersion relation. Waves typically arise as solutions to a linear problem of the form

$$L(\psi) = 0, \quad (\text{WF.1})$$

where L is, commonly, a linear operator in space and time. Two examples are

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0 \quad \text{and} \quad \frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0. \quad (\text{WF.2})$$

The first example is so common in all areas of physics it is sometimes called ‘the’ wave equation. The second example gives rise to Rossby waves.

- Solutions to the governing equation are often sought in the form of plane waves that have the form

$$\psi = \text{Re } A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (\text{WF.3})$$

where A is the wave amplitude, $\mathbf{k} = (k, l, m)$ is the wavevector, and ω is the frequency.

- The dispersion relation connects the frequency and wavevector through an equation of the form $\omega = \Omega(\mathbf{k})$ where Ω is some function. The relation is normally derived by substituting a trial solution like (WF.3) into the governing equation (WF.1). For the examples of (WF.2) we obtain $\omega = c^2 K^2$ and $\omega = -\beta k / K^2$ where $K^2 = k^2 + l^2 + m^2$ or, in two dimensions, $K^2 = k^2 + l^2$.
- The phase speed is the speed at which the wave crests move. In the direction of propagation and in the x , y and z directions the phase speed is given by, respectively,

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k}, \quad c_p^y = \frac{\omega}{l}, \quad c_p^z = \frac{\omega}{m}. \quad (\text{WF.4})$$

where $K = 2\pi/\lambda$ where λ is the wavelength. The wave crests have both a speed (c_p) and a direction of propagation (the direction of $\mathbf{k}$), like a vector, but the components defined in (WF.4) are not the components of that vector.

- The group velocity is the velocity at which a wave packet or wave group moves. It is a vector and is given by

$$\mathbf{c}_g = \frac{\partial \omega}{\partial \mathbf{k}} \quad \text{with components} \quad c_g^x = \frac{\partial \omega}{\partial k}, \quad c_g^y = \frac{\partial \omega}{\partial l}, \quad c_g^z = \frac{\partial \omega}{\partial m}. \quad (\text{WF.5})$$

Most physical quantities of interest are transported at the group velocity.

- If the coefficients of the wave equation are not constant (for example if the medium is inhomogeneous) then, if the coefficients are only slowly varying, approximate solutions may sometimes be found in the form

$$\psi = \text{Re } A(\mathbf{x}, t) e^{i\theta(\mathbf{x}, t)}, \quad (\text{WF.6})$$

where the amplitude A is also slowly varying and the local wavenumber and frequency are related to the phase, θ , by $\mathbf{k} = \nabla \theta$ and $\omega = -\partial \theta / \partial t$. The dispersion relation is then a *local* one of the form $\omega = \Omega(\mathbf{k}; \mathbf{x}, t)$.

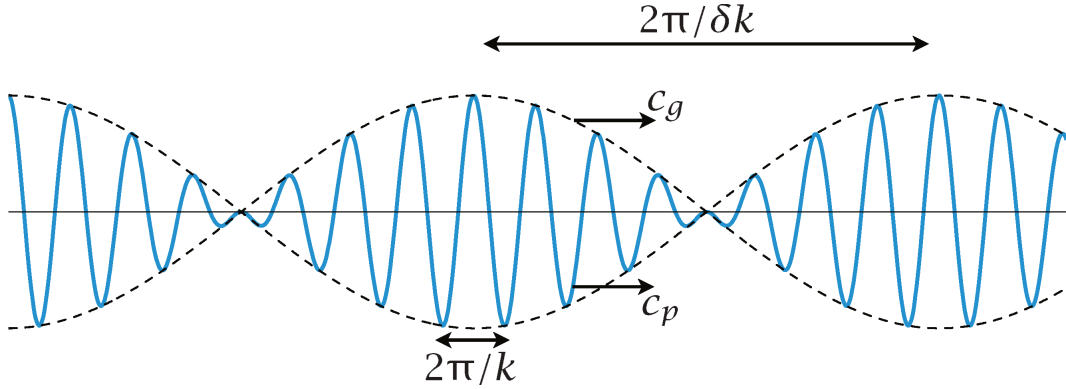


Figure 4.2 Superposition of two sinusoidal waves with wavenumbers k and $k + \delta k$, producing a wave (solid line) that is modulated by a slowly varying wave envelope or wave packet (dashed line). The envelope moves at the group velocity, $c_g = \partial\omega/\partial k$ and the phase of the wave moves at the phase speed $c_p = \omega/k$.

Let us further suppose that the two waves have similar wavenumbers and frequency, and, in particular, that $k_1 = k + \Delta k$ and $k_2 = k - \Delta k$, and $\omega_1 = \omega + \Delta\omega$ and $\omega_2 = \omega - \Delta\omega$. With this, (4.15) becomes

$$\begin{aligned}\psi &= \text{Re } \tilde{\psi} e^{i(kx - \omega t)} [e^{i(\Delta k x - \Delta\omega t)} + e^{-i(\Delta k x - \Delta\omega t)}] \\ &= 2 \text{Re } \tilde{\psi} e^{i(kx - \omega t)} \cos(\Delta k x - \Delta\omega t).\end{aligned}\tag{4.16}$$

The resulting disturbance, illustrated in Fig. 4.2 has two aspects: a rapidly varying component, with wavenumber k and frequency ω , and a more slowly varying envelope, with wavenumber Δk and frequency $\Delta\omega$. The envelope modulates the fast oscillation, and moves with velocity $\Delta\omega/\Delta k$; in the limit $\Delta k \rightarrow 0$ and $\Delta\omega \rightarrow 0$ this is the *group velocity*, $c_g = \partial\omega/\partial k$. Group velocity is equal to the phase speed, ω/k , only when the frequency is a linear function of wavenumber. The energy in the disturbance must move at the group velocity — note that the node of the envelope moves at the speed of the envelope and no energy can cross the node. These concepts generalize to more than one dimension, and if the wavenumber is the three-dimensional vector $\mathbf{k} = (k, l, m)$ then the three-dimensional envelope propagates at the group velocity given by

$$\mathbf{c}_g = \frac{\partial\omega}{\partial\mathbf{k}} \equiv \left(\frac{\partial\omega}{\partial k}, \frac{\partial\omega}{\partial l}, \frac{\partial\omega}{\partial m} \right).\tag{4.17}$$

The group velocity is also written as $\mathbf{c}_g = \nabla_{\mathbf{k}}\omega$ or, in subscript notation, $c_{gi} = \partial\omega/\partial k_i$, with the subscript i denoting the component of a vector.

4.3 ROSSBY WAVES

4.3.1 The linear equation of motion

For most of the rest of this chapter we will be concerned with the quasi-geostrophic equations of motion for which (as discussed in chapter 3) the inviscid, adiabatic potential vorticity equation is

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0, \quad (4.18)$$

where $q(x, y, z, t)$ is the potential vorticity and $\mathbf{u}(x, y, z, t)$ is the horizontal velocity. The velocity is related to a streamfunction by $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ and the potential vorticity is some function of the streamfunction, which might differ from system to system. Two examples, one applying to a continuously stratified system and the second to a single layer system, are

$$q = f + \zeta + \frac{\partial}{\partial z} \left(S(z) \frac{\partial \psi}{\partial z} \right), \quad q = \zeta + f - k_d^2 \psi. \quad (4.19a,b)$$

where $S(z) = f_0^2/N^2$, $\zeta = \nabla^2\psi$ is the relative vorticity and $k_d = 1/L_d$ is the inverse radius of deformation for a shallow water system. (Note that definitions of k_d and L_d can vary, typically by factors of 2, π , etc.) Boundary conditions may be needed to form a complete system.

We now *linearize* (4.18); that is, we suppose that the flow consists of a time-independent component (the ‘basic state’) plus a perturbation, with the perturbation being small compared with the mean flow. The basic state must satisfy the time-independent equation of motion, and it is common and useful to linearize about a zonal flow, $\bar{u}(y, z)$. The basic state is then purely a function of y and so we write

$$q = \bar{q}(y, z) + q'(x, y, t), \quad \psi = \bar{\psi}(y, z) + \psi'(x, y, z, t) \quad (4.20)$$

with a similar notation for the other variables. Note that $\bar{u} = -\partial\bar{\psi}/\partial y$ and $\bar{v} = 0$. Substituting into (4.18) gives, without approximation,

$$\frac{\partial q'}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{q} + \bar{\mathbf{u}} \cdot \nabla q' + \mathbf{u}' \cdot \nabla \bar{q} + \mathbf{u}' \cdot \nabla q' = 0. \quad (4.21)$$

The primed quantities are presumptively small so we neglect terms involving their products. Further, we are assuming that we are linearizing about a state that is a solution of the equations of motion, so that $\bar{\mathbf{u}} \cdot \nabla \bar{q} = 0$. Finally, since $\bar{v} = 0$ and $\partial\bar{q}/\partial x = 0$ we obtain

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0. \quad (4.22)$$

This equation or one very similar appears very commonly in studies of Rossby waves. To proceed, let us consider the simple example of waves in a single layer.

4.3.2 Waves in a single layer

Consider a system obeying (4.18) and (4.19b). The equation could be written in spherical coordinates with $f = 2\Omega \sin \vartheta$, but the dynamics are more easily illustrated on Cartesian β -plane for which $f = f_0 + \beta y$, and since f_0 is a constant it does not appear in our subsequent derivations.

Infinite deformation radius

If the scale of motion is much less than the deformation scale then we make the approximation that $k_d = 0$ and the equation of motion may be written as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0 \quad (4.23)$$

We linearize about a constant zonal flow, $\bar{u}$, by writing

$$\psi = \bar{\psi}(y) + \psi'(x, y, t), \quad (4.24)$$

where $\bar{\psi} = -\bar{u}y$. Substituting (4.24) into (4.23) and neglecting the nonlinear terms involving products of ψ' to give

$$\frac{\partial}{\partial t} \nabla^2 \psi' + \bar{u} \frac{\partial \nabla^2 \psi'}{\partial x} + \beta \frac{\partial \psi'}{\partial x} = 0. \quad (4.25)$$

This equation is just a single-layer version of (4.22), with $\partial \bar{q} / \partial y = \beta$, $q' = \nabla^2 \psi'$ and $v' = \partial \psi' / \partial x$.

The coefficients in (4.25) are not functions of y or z ; this is not a requirement for wave motion to exist but it does enable solutions to be found more easily. Let us seek solutions in the form of a plane wave, namely

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly - \omega t)}, \quad (4.26)$$

where $\tilde{\psi}$ is a complex constant and Re indicates the real part of the function (a notation sometimes omitted if no ambiguity is so-caused). Solutions of this form are valid in a domain with doubly-periodic boundary conditions; solutions in a channel can be obtained using a meridional variation of $\sin ly$, with no essential changes to the dynamics. The amplitude of the oscillation is given by $\tilde{\psi}$ and the phase by $kx + ly - \omega t$, where k and l are the x - and y -wavenumbers and ω is the frequency of the oscillation.

Substituting (4.26) into (4.25) yields

$$[(-\omega + Uk)(-K^2) + \beta k] \tilde{\psi} = 0, \quad (4.27)$$

where $K^2 = k^2 + l^2$. For non-trivial solutions this implies

$$\omega = Uk - \frac{\beta k}{K^2}. \quad (4.28)$$

This is the *dispersion relation* for barotropic Rossby waves, and evidently the velocity U Doppler shifts the frequency. The components of the phase speed and group velocity are given by, respectively,

$$c_p^x \equiv \frac{\omega}{k} = \bar{u} - \frac{\beta}{K^2}, \quad c_p^y \equiv \frac{\omega}{l} = \bar{u} \frac{k}{l} - \frac{\beta k}{K^2 l}, \quad (4.29a,b)$$

and

$$c_g^x \equiv \frac{\partial \omega}{\partial k} = \bar{u} + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \quad c_g^y \equiv \frac{\partial \omega}{\partial l} = \frac{2\beta k l}{(k^2 + l^2)^2}. \quad (4.30a,b)$$

The phase speed in the absence of a mean flow is *westwards*, with waves of longer wavelengths travelling more quickly, and the eastward current speed required to hold the waves of a particular wavenumber stationary (i.e., $c_p^x = 0$) is $U = \beta/K^2$. The background flow $\bar{u}$ evidently just provides a uniform shift to the phase speed, and could be transformed away by a change of coordinate.

Finite deformation radius

For a finite deformation radius the basic state $\Psi = -Uy$ is still a solution of the original equations of motion, but the potential vorticity corresponding to this state is $q = Uyk_d^2 + \beta y$ and its gradient is $\nabla q = (\beta + Uk_d^2)\mathbf{j}$. The linearized equation of motion is thus

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) (\nabla^2 \psi' - \psi' k_d^2) + (\beta + \bar{u} k_d^2) \frac{\partial \psi'}{\partial x} = 0. \quad (4.31)$$

Substituting $\psi' = \tilde{\psi} e^{i(kx + ly - \omega t)}$ we obtain the dispersion relation,

$$\omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2} = Uk - k \frac{\beta + Uk_d^2}{K^2 + k_d^2}. \quad (4.32)$$

The corresponding components of phase speed and group velocity are

$$c_p^x = \bar{u} - \frac{\beta + \bar{u}k_d^2}{K^2 + k_d^2} = \frac{\bar{u}K^2 - \beta}{K^2 + k_d^2}, \quad c_p^y = \bar{u} \frac{k}{l} - \frac{k}{l} \left(\frac{\bar{u}K^2 - \beta}{K^2 + k_d^2} \right) \quad (4.33a,b)$$

and

$$c_g^x = \bar{u} + \frac{(\beta + \bar{u}k_d^2)(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \quad c_g^y = \frac{2kl(\beta + \bar{u}k_d^2)}{(k^2 + l^2 + k_d^2)^2}. \quad (4.34a,b)$$

The uniform velocity field now no longer provides just a simple Doppler shift of the frequency, nor a uniform addition to the phase speed. From (4.33a) the waves are stationary when $K^2 = \beta/\bar{u} \equiv K_s^2$; that is, the current speed required to hold waves of a particular wavenumber stationary is $\bar{u} = \beta/K_s^2$. However, this is *not* simply the magnitude of the phase speed of waves of that wavenumber in the absence of a current — this is given by

$$c_p^x = \frac{-\beta}{K_s^2 + k_d^2} = \frac{-\bar{u}}{1 + k_d^2/K_s^2} \neq -\bar{u}. \quad (4.35)$$

Why is there a difference? It is because the current does not just provide a uniform translation, but, if k_d is non-zero, it also modifies the basic potential vorticity gradient. The basic state height field η_0 is sloping; that is $\eta_0 = -(f_0/g)\bar{u}y$, and the ambient potential vorticity field increases with y and $q = (\beta + Uk_d^2)y$. Thus, the basic state defines a preferred frame of reference, and the problem is not Galilean invariant.¹²

We also note that, from (4.33b), the group velocity is negative (westward) if the x -wavenumber is sufficiently small, compared to the y -wavenumber or the deformation wavenumber. That is, said a little loosely, *long waves move information westward and short waves move information eastward*, and this is a common property of Rossby waves. The x -component of the phase speed, on the other hand, is always westward relative to the mean flow.

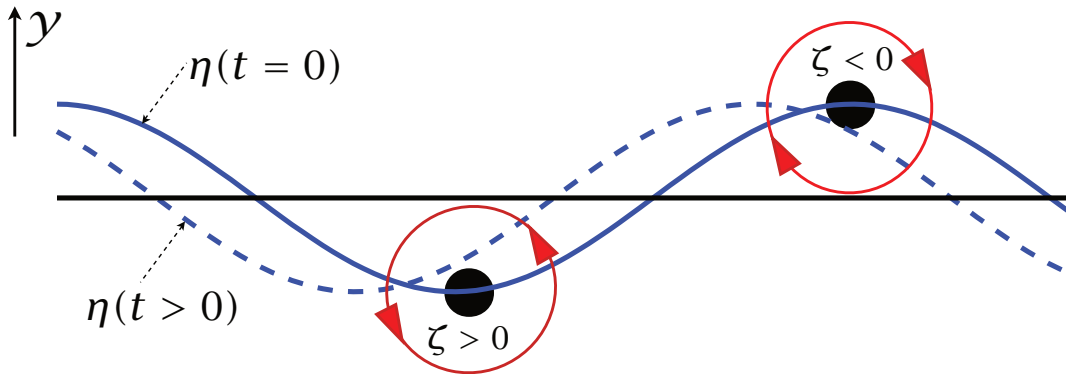


Figure 4.3 The mechanism of a two-dimensional (x - y) Rossby wave. An initial disturbance displaces a material line at constant latitude (the straight horizontal line) to the solid line marked $\eta(t = 0)$. Conservation of potential vorticity, $\beta y + \zeta$, leads to the production of relative vorticity, as shown for two parcels. The associated velocity field (arrows on the circles) then advects the fluid parcels, and the material line evolves into the dashed line. The phase of the wave has propagated westwards.

4.3.3 The mechanism of Rossby waves

The fundamental mechanism underlying Rossby waves is easily understood. Consider a material line of stationary fluid parcels along a line of constant latitude, and suppose that some disturbance causes their displacement to the line marked $\eta(t = 0)$ in Fig. 4.3. In the displacement, the potential vorticity of the fluid parcels is conserved, and in the simplest case of barotropic flow on the β -plane the potential vorticity is the absolute vorticity, $\beta y + \zeta$. Thus, in either hemisphere, a northward displacement leads to the production of negative relative vorticity and a southward displacement leads to the production of positive relative vorticity. The relative vorticity gives rise to a velocity field which, in turn, advects the parcels in material line in the manner shown, and the wave propagates westwards.

In more complicated situations, such as flow in two layers, considered below, or in a continuously stratified fluid, the mechanism is essentially the same. A displaced fluid parcel carries with it its potential vorticity and, in the presence of a potential vorticity gradient in the basic state, a potential vorticity anomaly is produced. The potential vorticity anomaly produces a velocity field (an example of potential vorticity inversion) which further displaces the fluid parcels, leading to the formation of a Rossby wave. The vital ingredient is a basic state potential vorticity gradient, such as that provided by the change of the Coriolis parameter with latitude.

4.4 ROSSBY WAVES IN STRATIFIED QUASI-GEOSTROPHIC FLOW

4.4.1 Setting up the problem

Let us now consider the dynamics of linear waves in stratified quasi-geostrophic flow on a β -plane, with a resting basic state.

The interior flow is governed by the potential vorticity equation, (3.112), and linearizing this about a state of rest gives

$$\frac{\partial}{\partial t} \left[\nabla^2 \psi' + \frac{1}{\bar{\rho}(z)} \frac{\partial}{\partial z} \left(\bar{\rho}(z) F(z) \frac{\partial \psi'}{\partial z} \right) \right] + \beta \frac{\partial \psi'}{\partial x} = 0, \quad (4.36)$$

where $\bar{\rho}$ is the density profile of the basic state and $F(z) = f_0^2/N^2$. (F is the square of the inverse Prandtl ratio, N/f_0 .) In the Boussinesq approximation $\bar{\rho} = \rho_0$, i.e., a constant. The vertical boundary conditions are determined by the thermodynamic equation, (3.114). If the boundaries are flat, rigid, slippery surfaces then $w = 0$ at the boundaries and if there is no surface buoyancy gradient the linearized thermodynamic equation is

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) = 0. \quad (4.37)$$

We apply this at the ground and, with somewhat less justification, at the tropopause: we assume the higher static stability of the stratosphere inhibits vertical motion. If the ground is not flat or if friction provides a vertical velocity by way of an Ekman layer, the boundary condition must be correspondingly modified, but we will stay with the simplest case and apply (4.37) at $z = 0$ and $z = H$.

4.4.2 Wave motion

As in the single-layer case, we seek solutions of the form

$$\psi' = \text{Re } \tilde{\psi}(z) e^{i(kx+ly-\omega t)}, \quad (4.38)$$

where $\tilde{\psi}(z)$ will determine the vertical structure of the waves. The case of a sphere is more complicated but introduces no truly new physical phenomena. If $\bar{\rho}$ and $F(z)$ are constants then the problem further simplifies and we can seek solutions of the form

$$\psi' = \text{Re } \tilde{\psi}(z) e^{i(kx+ly+mz-\omega t)}, \quad (4.39)$$

but we treat the more general case as it is not much harder.

Substituting (4.38) into (4.36) gives

$$\omega \left[-K^2 \tilde{\psi}(z) + \frac{1}{\bar{\rho}} \frac{d}{dz} \left(\bar{\rho} F(z) \frac{d\tilde{\psi}}{dz} \right) \right] - \beta k \tilde{\psi}(z) = 0. \quad (4.40)$$

Now, if $\tilde{\psi}$ satisfies

$$\frac{1}{\bar{\rho}} \frac{d}{dz} \left(\bar{\rho} F(z) \frac{d\tilde{\psi}}{dz} \right) = -\Gamma \tilde{\psi}, \quad (4.41)$$

where Γ is a constant, then the equation of motion becomes

$$-\omega [K^2 + \Gamma] \tilde{\psi} - \beta k \tilde{\psi} = 0, \quad (4.42)$$

and the dispersion relation follows, namely

$$\omega = -\frac{\beta k}{K^2 + \Gamma}. \quad (4.43)$$

Equation (4.41) constitutes an eigenvalue problem for the vertical structure; the boundary conditions, derived from (4.37), are $\partial\tilde{\psi}/\partial z = 0$ at $z = 0$ and $z = H$. The resulting eigenvalues, Γ are proportional to the inverse of the squares of the deformation radii for the problem and the eigenfunctions are the vertical structure functions. If $F(z)$ and $\tilde{\rho}$ were constants then we would easily find that $\Gamma = Fm^2$, with m defined in (4.39).

A simple example

Consider the case in which $F(z)$ and $\tilde{\rho}$ are constant, and in which the domain is confined between two rigid surfaces at $z = 0$ and $z = H$. Then the eigenvalue problem for the vertical structure is

$$F \frac{d^2 \tilde{\psi}}{dz^2} = -\Gamma \tilde{\psi} \quad (4.44a)$$

with boundary conditions of

$$\frac{d\tilde{\psi}}{dz} = 0, \quad \text{at } z = 0, H. \quad (4.44b)$$

There is a sequence of solutions to this, namely

$$\tilde{\psi}_n(z) = \cos(n\pi z/H), \quad n = 1, 2, \dots \quad (4.45)$$

with corresponding eigenvalues

$$\Gamma_n = n^2 \frac{F\pi^2}{H^2} = (n\pi)^2 \left(\frac{f_0}{NH} \right)^2, \quad n = 1, 2, \dots \quad (4.46)$$

Equation (4.46) may be used to define the deformation radii for this problem, namely

$$L_n \equiv \frac{1}{\sqrt{\Gamma_n}} = \frac{NH}{n\pi f_0}. \quad (4.47)$$

The first deformation radius is the same as the expression obtained by dimensional analysis, namely NH/f , except for a factor of π . (Definitions of the deformation radii both with and without the factor of π are common in the literature, and neither is obviously more correct. In the latter case, the first deformation radius in a problem with uniform stratification is given by NH/f , equal to $\pi/\sqrt{\Gamma_1}$.) In addition to these baroclinic modes, the case with $n = 0$, that is with $\tilde{\psi} = 1$, is also a solution of (4.44) for any $F(z)$.

Using (4.43) and (4.46) the dispersion relation becomes

$$\omega = -\frac{\beta k}{K^2 + (n\pi)^2 (f_0/NH)^2}, \quad n = 0, 1, 2, \dots \quad (4.48)$$

and, of course, the horizontal wavenumbers k and l are also quantized in a finite domain. The dynamics of the barotropic mode are independent of height and independent of the stratification of the basic state, and so these Rossby waves are *identical* with the Rossby waves in a homogeneous fluid contained between two flat rigid surfaces.

4.5 ♦ ENERGY PROPAGATION OF ROSSBY WAVES

We now consider how energy is fluxed in Rossby waves. To keep matters reasonably simple from an algebraic point of view we will consider waves in a single layer and without a mean flow, but we will allow for a finite radius of deformation. To remind ourselves, the dynamics are governed by the evolution of potential vorticity and the linearized evolution equation is

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \psi + \beta \frac{\partial \psi}{\partial x} = 0. \quad (4.49)$$

The dispersion relation follows in the usual way and is

$$\omega = \frac{-k\beta}{K^2 + k_d^2}, \quad (4.50)$$

which is a simplification of (4.32), and the group velocities are

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(K^2 + k_d^2)^2}, \quad c_g^y = \frac{2\beta kl}{(K^2 + k_d^2)^2}, \quad (4.51a,b)$$

which are simplifications of (4.34), and as usual $K^2 = k^2 + l^2$.

To obtain an energy equation multiply (4.49) by $-\psi$ to obtain, after a couple of lines of algebra,

$$\frac{1}{2} \frac{\partial}{\partial t} ((\nabla \psi)^2 + k_d^2 \psi^2) - \nabla \cdot \left(\psi \nabla \frac{\partial \psi}{\partial t} + \mathbf{i} \frac{\beta}{2} \psi^2 \right) = 0, \quad (4.52)$$

where $\mathbf{i}$ is the unit vector in the x direction. The first group of terms are the energy itself, or more strictly the energy density. (An energy density is an energy per unit mass or per unit volume, depending on the context.) The term $(\nabla \psi)^2/2 = (u^2 + v^2)/2$ is the kinetic energy and $k_d^2 \psi^2/2$ is the potential energy, proportional to the displacement of the free surface, squared. The second term is the energy flux, so that we may write

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{F} = 0. \quad (4.53)$$

where $E = (\nabla \psi)^2/2 + k_d^2 \psi^2/2$ and $\mathbf{F} = -(\psi \nabla \partial \psi / \partial t + \mathbf{i} \beta \psi^2)$. We haven't yet used the fact that the disturbance has a dispersion relation, and if we do so we may expect, following the derivations of section 4.2, that the energy moves at the group velocity. Let us now demonstrate this explicitly.

We assume a solution of the form

$$\psi = A(x) \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) = A(x) \cos(kx + ly - \omega t) \quad (4.54)$$

Essentials of Rossby Waves

- Rossby waves owe their existence to a gradient of potential vorticity in the fluid. If a fluid parcel is displaced, it conserves its potential vorticity and so its relative vorticity will in general change. The relative vorticity creates a velocity field that displaces neighbouring parcels, whose relative vorticity changes and so on.
- A common source of a potential vorticity gradient is differential rotation, or the β -effect, and *planetary waves* is the name given to this type of Rossby wave. In the presence of non-zero β the ambient potential vorticity increases northward and the phase of the Rossby waves propagates westward. In general, Rossby waves propagate pseudo-westwards, meaning to the left of the direction of the potential vorticity gradient.
- A common equation of motion for Rossby waves is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (\text{RW.1})$$

with an overbar denoting the basic state and a prime a perturbation. In the case of a single layer of fluid with no mean flow this equation becomes

$$\frac{\partial}{\partial t} (\nabla^2 + k_d^2) \psi' + \beta \frac{\partial \psi'}{\partial x} = 0 \quad (\text{RW.2})$$

with dispersion relation

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \quad (\text{RW.3})$$

- The phase speed in the zonal direction ($c_p^x = \omega/k$) is always negative, or westward, and is larger for large waves. For (RW.2) components of the group velocity are given by

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \quad c_g^y = \frac{2\beta k l}{(k^2 + l^2 + k_d^2)^2}. \quad (\text{RW.4})$$

The group velocity is westward if the zonal wavenumber is sufficiently small, and eastward if the zonal wavenumber is sufficiently large.

- Rossby waves exist in stratified fluids, and have a similar dispersion relation to (RW.3) with an appropriate vertical wavenumber appearing in place of the inverse deformation radius, k_d .
- The reflection of such Rossby waves at a wall is specular, meaning that the group velocity of the reflected wave makes the same angle with the wall as the group velocity of the incident wave. The energy flux of the reflected wave is equal and opposite to that of the incoming wave in the direction normal to the wall.

where $A(x)$ is assumed to vary slowly compared to the nearly plane wave. (Note that $\mathbf{k}$ is the wave vector, to be distinguished from $\mathbf{k}$, the unit vector in the z -direction.) The kinetic energy in a wave is given by

$$KE = \frac{A^2}{2} (\psi_x^2 + \psi_y^2) \quad (4.55)$$

so that, averaged over a wave period,

$$\overline{KE} = \frac{A^2}{2} (k^2 + l^2) \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \sin^2(\mathbf{k} \cdot \mathbf{x} - \omega t) dt. \quad (4.56)$$

The time-averaging produces a factor of one half, and applying a similar procedure to the potential energy we obtain

$$\overline{KE} = \frac{A^2}{4} (k^2 + l^2), \quad \overline{PE} = \frac{A^2}{4} k_d^2, \quad (4.57)$$

so that the average total energy is

$$\overline{E} = \frac{A^2}{4} (K^2 + k_d^2), \quad (4.58)$$

where $K^2 = k^2 + l^2$.

The flux, F , is given by

$$F = - \left(\psi \nabla \frac{\partial \psi}{\partial t} + \mathbf{i} \frac{\beta}{2} \psi^2 \right) = -A^2 \cos^2(\mathbf{k} \cdot \mathbf{x} - \omega t) \left(\mathbf{k}\omega + \mathbf{i} \frac{\beta}{2} \right), \quad (4.59)$$

so that evidently the energy flux has a component in the direction of the wavevector, $\mathbf{k}$, and a component in the x -direction. Averaging over a wave period straightforwardly gives us additional factors of one half:

$$\overline{F} = -\frac{A^2}{2} \left(\mathbf{k}\omega + \mathbf{i} \frac{\beta}{2} \right). \quad (4.60)$$

We now use the dispersion relation $\omega = -\beta k / (K^2 + k_d^2)$ to eliminate the frequency, giving

$$\overline{F} = \frac{A^2 \beta}{2} \left(\mathbf{k} \frac{k}{K^2 + k_d^2} - \mathbf{i} \frac{1}{2} \right), \quad (4.61)$$

and writing this in component form we obtain

$$\overline{F} = \frac{A^2 \beta}{4} \left[\mathbf{i} \left(\frac{k^2 - l^2 - k_d^2}{K^2 + k_d^2} \right) + \mathbf{j} \left(\frac{2kl}{K^2 + k_d^2} \right) \right] \quad (4.62)$$

Comparison of (4.62) with (4.51) and (4.58) reveals that

$$\overline{F} = \mathbf{c}_g \overline{E} \quad (4.63)$$

so that the energy propagation equation (4.53), when averaged over a wave, becomes

$$\frac{\partial \bar{E}}{\partial t} + \nabla \cdot \mathbf{c}_g \bar{E} = 0 . \quad (4.64)$$

It is interesting that the variation of A plays no role in the above manipulations, so that the derivation appears to go through if the amplitude $A(\mathbf{x}, t)$ is in fact a constant and the wave is a single plane wave. This seems hard to reconcile with our previous discussion, in which we noted that the group velocity was the velocity of a wave *packet* involving a superposition of plane waves. Indeed, the derivative of the frequency with respect to wavenumber means little if there is only one wavenumber. In fact there is nothing wrong with the above derivation if A is a constant and only a single plane wave is present. The resolution of the paradox arises by noting that a plane wave fills all of space and time; in this case there is no convergence of the energy flux and the energy propagation equation is trivially true.

4.6 VERTICAL PROPAGATION OF ROSSBY WAVES

The vertical propagation of Rossby waves is important not just because it must be taken into account to obtain an accurate picture of the tropospheric response to topographic and thermal forcing, but because it can excite motion in the stratosphere. We will continue to use the stratified quasi-geostrophic equations, but we now allow the model to be compressible and semi-infinite, extending from $z = 0$ to $z = \infty$.

Let us now consider how Rossby waves propagate in an inhomogeneous, stratified medium. It is simplest to first consider the problem slightly generally, without regard to boundary conditions, for this reveals some of the essential conditions under which Rossby waves propagate. In section ?? we will be a little more definite and consider the lower boundary conditions and the requirements for waves propagate vertically, possibly into the stratosphere. Our governing equation is the quasi-geostrophic potential vorticity equation and with applications to the stratosphere in mind we will use log-pressure co-ordinates so that the equation of motion is

$$\frac{\partial q}{\partial t} + J(\psi, q) = 0, \quad q = \nabla^2 \psi + \beta y + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi}{\partial z} \right), \quad (4.65)$$

where $\rho_R = \rho_0 e^{-z/H}$ with H being a specified density scale height, typically $RT(0)/g$.

4.6.1 Conditions for wave propagation

Let us linearize (4.65) about a zonal wind that depends only on z ; that is, we let

$$\psi = -\bar{u}(z)y + \psi', \quad (4.66)$$

and obtain

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad \frac{\partial \bar{q}}{\partial y} = \beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \bar{u}}{\partial z} \right), \quad (4.67)$$

or equivalently, in terms of streamfunction,

$$\begin{aligned} \left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] \\ + \frac{\partial \psi'}{\partial x} \left[\beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \bar{u}}{\partial z} \right) \right] = 0. \end{aligned} \quad (4.68)$$

The first term in square brackets is the perturbation potential vorticity, q' and the second term equals $\partial \bar{q}/\partial y$.

We may seek solutions to (4.68) of the form

$$\psi' = \text{Re } \tilde{\psi}(z) e^{i(kx+ly-kct)}. \quad (4.69)$$

Solutions of (4.68) then satisfy

$$\left[\frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \tilde{\psi}}{\partial z} \right) \right] = \tilde{\psi} \left(K^2 - \frac{\partial \bar{q}/\partial y}{\bar{u} - c} \right) \quad (4.70)$$

Let us simplify by assuming that both $\bar{u}$ and N^2 are constants, so that $\partial \bar{q}/\partial y = \beta$. Eq. (4.70) further simplifies if we define

$$\Phi(z) = \tilde{\psi}(z) \left(\frac{\rho_R}{\rho_R(0)} \right)^{1/2} = \tilde{\psi}(z) e^{-z/2H} \quad (4.71)$$

whence we obtain

$$\frac{d^2 \Phi}{dz^2} + m^2 \Phi = 0, \quad \text{where} \quad (4.72a,b)$$

where

$$m^2 = \frac{N^2}{f_0^2} \left(\frac{\beta}{\bar{u} - c} - K^2 - \gamma^2 \right), \quad (4.73)$$

where $\gamma^2 = f_0^2/(4N^2H^2) = 1/(2L_d)^2$, where L_d is the deformation radius as sometimes defined (i.e., $L_d = NH/f_0$). If the parameters on the right-hand side of (4.73) are constant then so is m and (4.72) has solutions of the form $\Phi(z) = \Phi_0 e^{imz}$ so that the streamfunction itself varies as

$$\psi' = \text{Re } \Phi_0 e^{i(kx+ly+mz-kct)+z/2H}. \quad (4.74)$$

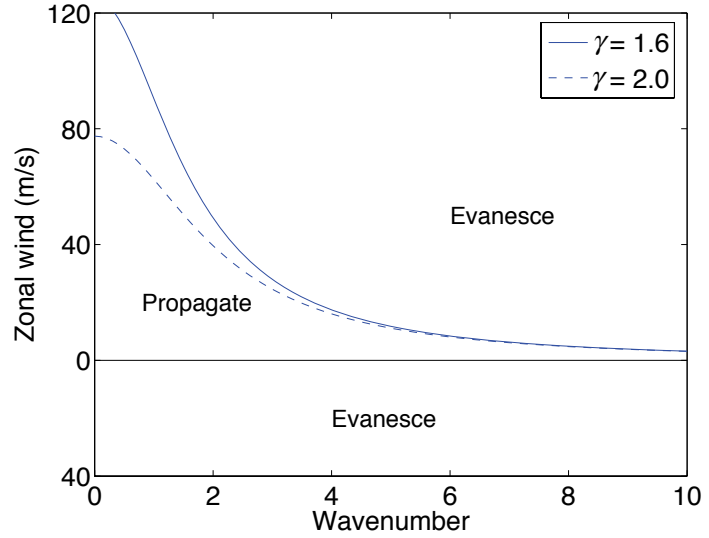
For waves to propagate upwards we require that $m^2 > 0$ and, from (4.73), that

$$0 < \bar{u} - c < \frac{\beta}{K^2 + \gamma^2}, \quad (4.75)$$

where $u_c = \beta/(K^2 + \gamma^2)$ is the Rossby critical velocity. For waves of some given frequency ($\omega = kc$) the above expression provides a condition on $\bar{u}$ for the vertical propagation of planetary waves. For stationary waves $c = 0$ and the criterion is

$$0 < \bar{u} < \frac{\beta}{K^2 + \gamma^2}, \quad (4.76)$$

Figure 4.4 The boundary between propagating waves and evanescent waves as a function of zonal wind and wavenumber, using (4.76), for a couple of values of γ . With $N = 2 \times 10^{-2} \text{ s}^{-1}$, $\gamma = 1.6$ ($\gamma = 2$) corresponds to a scale height of 7.0 km (5.5 km) and a deformation radius NH/f of 1400 km (1100 km).



and this is illustrated in Fig. 4.4. That is to say, the vertical propagation of stationary Rossby waves occurs only in westerly winds, and winds that are weaker than some critical value, $u_c = \beta/(K^2 + \gamma^2)$ that depends on the scale of the wave.

Stationary, vertically oscillatory modes can exist only for zonal flows that are eastwards and that are less than the critical velocity $U_c = \beta/(K^2 + \gamma^2)$. One way to interpret this condition is note that in a resting medium the Rossby wave frequency has a minimum value (and maximum absolute value), when $m = 0$, of

$$\omega = -\frac{\beta k}{K^2 + \gamma^2}. \quad (4.77)$$

Note too that in a frame moving with speed $\bar{u}$ our Rossby waves (stationary in the Earth's frame) have frequency $-\bar{u}k$, and this is the forcing frequency arising from the now-moving bottom topography. Thus, (4.76) is equivalent to saying that for oscillatory waves to exist *the forcing frequency must lie within the frequency range of vertically propagating Rossby waves*.

For westward flow, or for sufficiently strong eastward flow, the waves decay exponentially as $\Phi = \Phi_0 \exp(-\alpha z)$ where

$$\alpha = \frac{N}{f_0} \left(K^2 + \gamma^2 - \frac{\beta}{\bar{u}} \right)^{1/2}. \quad (4.78)$$

Note that the critical velocity $u_c = (\beta/K^2 + \gamma^2)$ is a function of wavenumber, and that it increases with horizontal wavelength. Thus, for a given eastward flow long waves may penetrate vertically when short waves are trapped, an effect sometimes referred to as 'Charney–Drazin filtering'.¹³ One important consequence of this is that the stratospheric motion is typically of larger scales than that of the troposphere, because Rossby waves tend to be excited first in the troposphere (by baroclinic instability and by flow over topography, among other things), but the shorter waves are trapped and only the longer ones reach the stratosphere. In the summer, the stratospheric

winds are often westwards (because the pole is warmer than the equator) and all waves are trapped in the troposphere; the eastward stratospheric winds that favour vertical penetration occur in the other three seasons, although very strong eastward winds can suppress penetration in mid-winter.

4.6.2 Dispersion relation and group velocity

Noting that $\omega = ck$ and rearranging (4.73) we obtain the dispersion relation for three-dimensional Rossby waves

$$\omega = \bar{u}k - \frac{\beta k}{K^2 + \gamma^2 + m^2 f_0^2 / N^2}. \quad (4.79)$$

The three components of the group velocity for these waves are then:

$$c_g^x = \bar{u} + \frac{\beta[k^2 - (l^2 + m^2 f_0^2 / N^2 + \gamma^2)]}{(K^2 + m^2 f_0^2 / N^2 + \gamma^2)^2}, \quad (4.80a)$$

$$c_g^y = \frac{2\beta kl}{(K^2 + m^2 f_0^2 / N^2 + \gamma^2)^2}, \quad c_g^z = \frac{2\beta km f_0^2 / N^2}{(K^2 + m^2 f_0^2 / N^2 + \gamma^2)^2}. \quad (4.80b,c)$$

The propagation in the horizontal is analogous to the propagation in a shallow water model [c.f. (4.33b)]; note also that higher baroclinic modes (bigger m) will have a more westward group velocity. The vertical group velocity is proportional to m , and for waves that propagate signals upward we must choose m to have the same sign as k so that c_g^z is positive. If there is no mean flow then the zonal wavenumber k is negative (in order that frequency is positive) and m must then also be negative. Energy then propagates upward but the phase propagates downward.

4.7 THE MECHANISM OF JET PRODUCTION

4.7.1 I. The vorticity budget

Suppose that the absolute vorticity normal to the surface (i.e., $\zeta + 2\Omega \sin \vartheta$) increases monotonically polewards. (A sufficient condition for this is that the fluid is at rest.) By Stokes' theorem, the circulation around a line of latitude circumscribing the polar cap, I , is equal to the integral of the absolute vorticity over the cap. That is,

$$I_i = \int_{\text{cap}} \boldsymbol{\omega}_{ia} \cdot d\mathbf{A} = \oint_C u_{ia} dl = \oint_C (u_i + \Omega a \cos \vartheta) dl, \quad (4.81)$$

where $\boldsymbol{\omega}_{ia}$ and u_{ia} are the initial absolute vorticity and velocity, respectively, u_i is the initial zonal velocity in the Earth's frame of reference, and the line integrals are around the line of latitude. For simplicity let us take $u_i = 0$ and suppose there is a disturbance equatorwards of the polar cap, and that this results in a distortion of the material line around the latitude circle C (Fig. 4.6). Since we are supposing the source of the disturbance to be distant from the latitude of interest, then if

Figure 4.5 A schematic east-west section of an upwardly propagating Rossby wave. The slanting lines are lines of constant phase and ‘high’ and ‘low’ refer to the pressure or stream-function values. Both k and m are negative so the phase lines are oriented up and to the west. The phase propagates westward and downward, but the group velocity is upward.

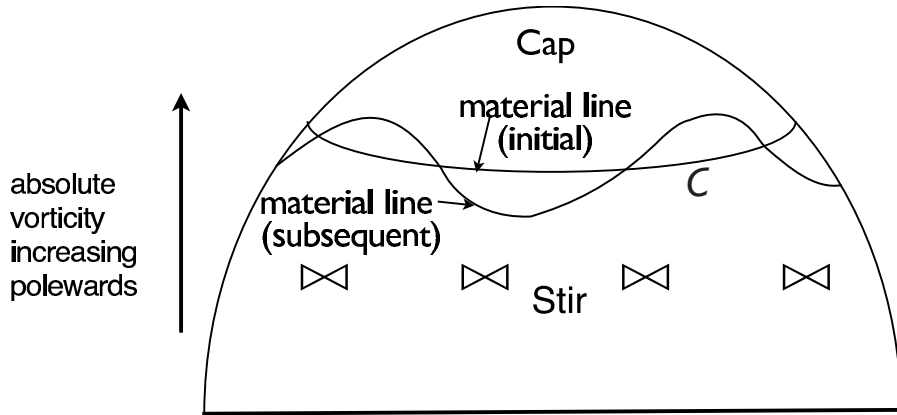
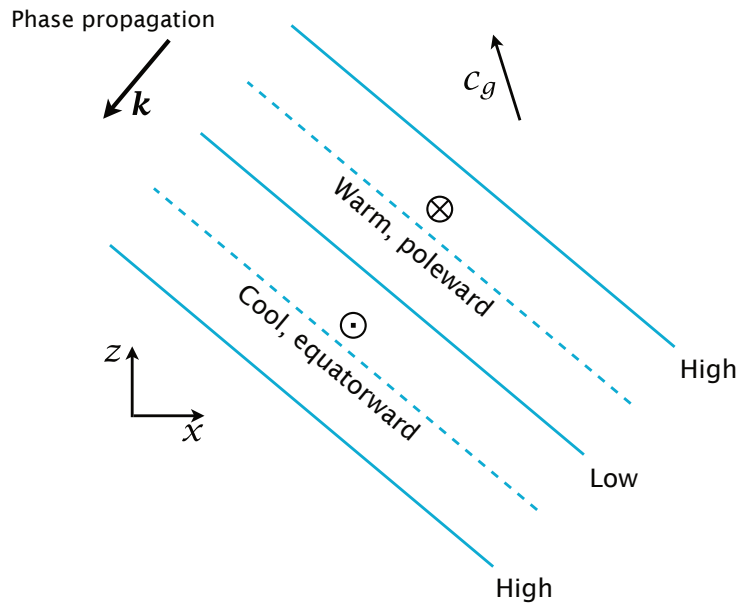


Figure 4.6 The effects of a mid-latitude disturbance on the circulation around the latitude line C . If initially the absolute vorticity increases monotonically polewards, then the disturbance will bring fluid with lower absolute vorticity into the cap region. Then, using Stokes' theorem, the velocity around the latitude line C will become more westwards.

we neglect viscosity the circulation along the material line is conserved, by Kelvin's circulation theorem. Thus, vorticity with a lower value is brought into the region of the polar cap — that is, the region polewards of the latitude line C . Using Stokes' theorem again the circulation around the latitude circle C must therefore fall; that is, denoting values after the disturbance with a subscript f ,

$$I_f = \int_{\text{cap}} \omega_{fa} \cdot d\mathbf{A} < I_i \quad (4.82)$$

so that

$$\oint_C (u_f + \Omega a \cos \vartheta) dl < \oint_C (u_i + \Omega a \cos \vartheta) dl \quad (4.83)$$

and

$$\bar{u}_f < \bar{u}_i \quad (4.84)$$

with the overbar indicating a zonal average. Thus, there is a tendency to produce *westward* flow polewards of the disturbance. By a similar argument westward flow is also produced equatorwards of the disturbance — to see this one might apply Kelvin's theorem over all of the globe south of the source of the disturbance (taking care to take the dot-product correctly between the direction of the vorticity vector and the direction normal to the surface). Finally, note that the overall situation is the same in the Southern Hemisphere. Thus, on the surface of a rotating sphere, external stirring will produce westward flow *away* from the region of the stirring.

Now suppose, furthermore, that the disturbance imparts no net angular momentum to the fluid. Then the integral of $ua \cos \vartheta$ over the entire hemisphere must be constant. But the fluid is accelerating westwards away from the disturbance. Therefore, the fluid in the region of the disturbance must accelerate *eastwards*; that is, angular momentum must converge into the stirred region, producing an eastward flow. This simple mechanism is the essence of the production of eastward eddy-driven jets in the atmosphere, and of the eastward surface winds in mid-latitudes. The stirring that here we have externally imposed comes, in reality, from baroclinic instability.

If the stirring subsides then the flow may reversibly go back to its initial condition, with a concomitant reversal of the momentum convergence that caused the zonal flow. Thus, we must have some form of dissipation and irreversibility in order to produce permanent changes, and in particular we need to irreversibly mix vorticity. If the fluid is continuously mixed, then of course we also need a source that restores the absolute vorticity gradient, otherwise we will completely homogenize the vorticity over the hemisphere, so let us now set up a simple model that shows how a permanent jet structure can be maintained.

4.7.2 II. Rossby waves and momentum flux

We saw above that a mean gradient of vorticity is an essential ingredient in the mechanism whereby a mean flow is generated by stirring. Given such, we expect Rossby waves to be excited, and we now show how Rossby waves are intimately related to the momentum flux maintaining the mean flow.

If a stirring is present in mid-latitudes then we expect that Rossby waves will be generated there, propagate away and break and dissipate. To the extent that the waves are quasi-linear and do not interact, then just away from the source region each wave has the form

$$\psi = \text{Re } C e^{i(kx+ly-\omega t)} = \text{Re } C e^{i(kx+ly-kct)}, \quad (4.85)$$

where C is a constant, with dispersion relation

$$\omega = ck = \bar{u}k - \frac{\beta k}{k^2 + l^2} \equiv \omega_R, \quad (4.86)$$

provided that there is no meridional shear in the zonal flow. The meridional component of the group velocity is given by

$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{2\beta kl}{(k^2 + l^2)^2}. \quad (4.87)$$

Now, the direction of the group velocity must be *away* from the source region; this is a radiation condition (discussed more in the next subsection), demanded by the requirement that Rossby waves transport energy *away* from the disturbance. Thus, northwards of the source kl is positive and southwards of the source kl is negative. That the product kl can be positive or negative arises because for each k there are two possible values of l that satisfy the dispersion relation (4.86), namely

$$l = \pm \left(\frac{\beta}{\bar{u} - c} - k^2 \right)^{1/2}, \quad (4.88)$$

assuming that the quantity in parentheses is positive.

The velocity variations associated with the Rossby waves are

$$u' = -\text{Re } C i l e^{i(kx+ly-\omega t)}, \quad v' = \text{Re } C i k e^{i(kx+ly-\omega t)}, \quad (4.89a,b)$$

and the associated momentum flux is

$$\overline{u'v'} = -\frac{1}{2} C^2 kl. \quad (4.90)$$

Thus, given that the sign of kl is determined by the group velocity, northwards of the source the momentum flux associated with the Rossby waves is southward (i.e., $\overline{u'v'}$ is negative), and southwards of the source the momentum flux is northward (i.e., $\overline{u'v'}$ is positive). That is, the momentum flux associated with the Rossby waves is *toward* the source region. Momentum converges in the region of the stirring, producing net eastward flow there and westward flow to either side (Fig. 4.7).

Another way of describing the same effect is to note that if kl is positive then lines of constant phase ($kx + ly = \text{constant}$) are tilted north-west/south-east, and the momentum flux associated with such a disturbance is negative ($\overline{u'v'} < 0$). Similarly, if kl is negative then the constant-phase lines are tilted north-east/south-west and the associated momentum flux is positive ($\overline{u'v'} > 0$). The net result is a convergence of momentum flux into the source region. In physical space this is reflected by having eddies that are ‘bow-shaped’, as in Fig. 4.8.

4.7.3 III. The pseudomomentum budget

The kinematic relation between vorticity flux and momentum flux for non-divergent two-dimensional flow is

$$v\zeta = \frac{1}{2} \frac{\partial}{\partial x} (v^2 - u^2) - \frac{\partial}{\partial y} (uv). \quad (4.91)$$

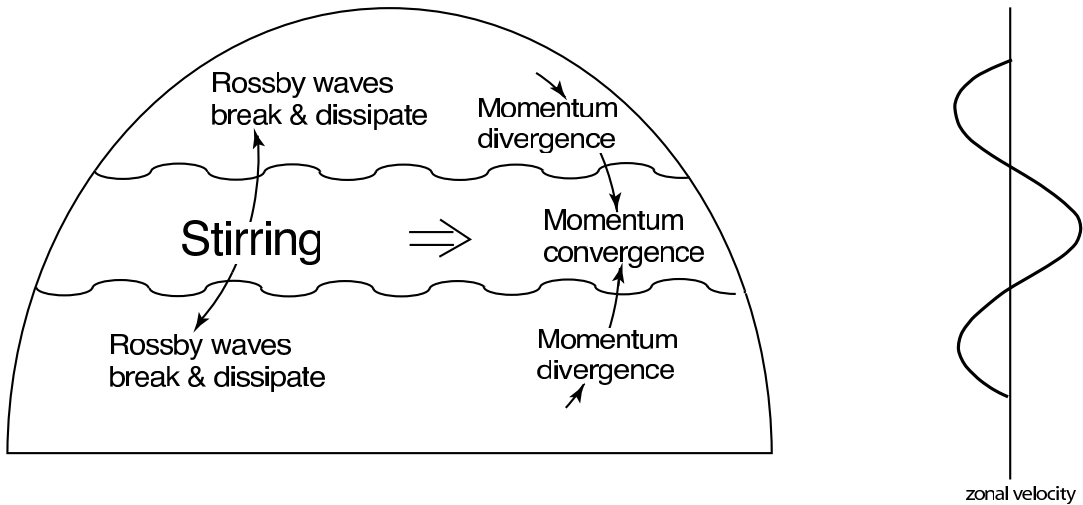


Figure 4.7 Generation of zonal flow on a β -plane or on a rotating sphere. Stirring in mid-latitudes (by baroclinic eddies) generates Rossby waves that propagate away from the disturbance. Momentum converges in the region of stirring, producing eastward flow there and weaker westward flow on its flanks.

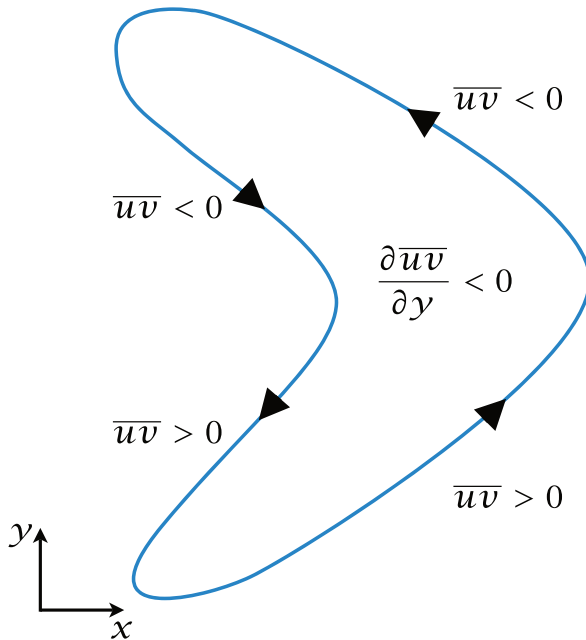


Figure 4.8 The momentum transport in physical space, caused by the propagation of Rossby waves away from a source in mid-latitudes. The ensuing bow-shaped eddies are responsible for a convergence of momentum, as indicated in the idealization pictured.

After zonal averaging this gives

$$\overline{v'\zeta'} = -\frac{\partial \overline{u'v'}}{\partial y}, \quad (4.92)$$

noting that $\bar{v} = 0$ for two-dimensional incompressible (or geostrophic) flow. For reference, in spherical coordinates this expression becomes

$$\overline{v'\zeta'} \cos \vartheta = -\frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}). \quad (4.93)$$

If (4.92) [or (4.93)] is integrated with respect to y between two quiescent latitudes then the right-hand side vanishes. That is the zonally averaged meridional vorticity flux vanishes when integrated over latitude.

Now, the barotropic zonal momentum equation is (for horizontally non-divergent flow)

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} - f v = -\frac{\partial \phi}{\partial x} + F_u - D_u, \quad (4.94)$$

where F_u and D_u represent the effects of any forcing and dissipation. Zonal averaging, with $\bar{v} = 0$, gives

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \bar{uv}}{\partial y} + \bar{F}_u - \bar{D}_u, \quad (4.95)$$

or, using (4.92),

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'\zeta'} + \bar{F}_u - \bar{D}_u. \quad (4.96)$$

Thus, the zonally averaged wind is maintained by the zonally averaged vorticity flux. On average there is little if any direct forcing of horizontal momentum and we may set $\bar{F}_u = 0$, and if the dissipation is parameterized by a linear drag (4.96) becomes

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'\zeta'} - r \bar{u}, \quad (4.97)$$

where the constant r is an inverse frictional time scale.

Now consider the maintenance of this vorticity flux. The barotropic vorticity equation is

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + v \beta = F_\zeta - D_\zeta, \quad (4.98)$$

where F_ζ and D_ζ are forcing and dissipation of vorticity. Linearize about a mean zonal flow to give

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + \gamma v' = F'_\zeta - D'_\zeta, \quad (4.99)$$

where

$$\gamma = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} \quad (4.100)$$

is the meridional gradient of absolute vorticity. Multiply (4.99) by ζ'/γ and zonally average, assuming that $\bar{u}_{yy}$ is small compared to β or varies only slowly, to form the pseudomomentum equation,

$$\frac{\partial \mathcal{A}}{\partial t} + \overline{v'\zeta'} = \frac{1}{\gamma}(\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta}), \quad (4.101)$$

where

$$\mathcal{A} = \frac{1}{2\gamma}\overline{\zeta'^2} \quad (4.102)$$

is a wave activity density, equal to the (negative of) the pseudomomentum for this problem. The parameter γ is positive if the average absolute vorticity increases monotonically northwards, and this is usually the case in both Northern and Southern Hemispheres.

In the absence of forcing and dissipation, (4.97) and (4.101) imply an important relationship between the change of the mean flow and the pseudomomentum, namely

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial \mathcal{A}}{\partial t} = 0. \quad (4.103)$$

Now if for some reason $\mathcal{A}$ increases, perhaps because a wave enters an initially quiescent region because of stirring elsewhere, then mean flow must decrease. However, because the vorticity flux integrates to zero, the zonal flow cannot decrease everywhere. Thus, if the zonal flow decreases in regions away from the stirring, it must *increase* in the region of the stirring. In the presence of forcing and dissipation this mechanism can lead to the production of a statistically steady jet in the region of the forcing, since (4.97) and (4.101) combine to give

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial \mathcal{A}}{\partial t} = -r\bar{u} + \frac{1}{\gamma}(\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta}), \quad (4.104)$$

and in a statistically steady state

$$r\bar{u} = \frac{1}{\gamma} \left(\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta} \right). \quad (4.105)$$

The terms on the right-hand side represent the stirring and dissipation of vorticity, and integrated over latitude their sum will vanish, or otherwise the pseudomomentum budget cannot be in a steady state. However, let us suppose that forcing is confined to mid-latitudes. In the forcing region, the first term on the right-hand side of (4.105) will be larger than the second, and an eastward mean flow will be generated. Away from the direct influence of the forcing, the dissipation term will dominate and westward mean flows will be generated, as sketched in Fig. 4.9. Thus, *on a β -plane or on the surface of a rotating sphere an eastward mean zonal flow can be maintained by a vorticity stirring that imparts no net momentum to the fluid.* In general, stirring in the presence of a vorticity gradient will give rise to a mean flow, and on a spherical planet the vorticity gradient is provided by differential rotation.

Figure 4.9 Mean flow generation by a meridionally confined stirring. Because of Rossby wave propagation away from the source region, the distribution of pseudomomentum dissipation is broader than that of pseudomomentum forcing, and the sum of the two leads to the zonal wind distribution shown, with positive (eastward) values in the region of the stirring.

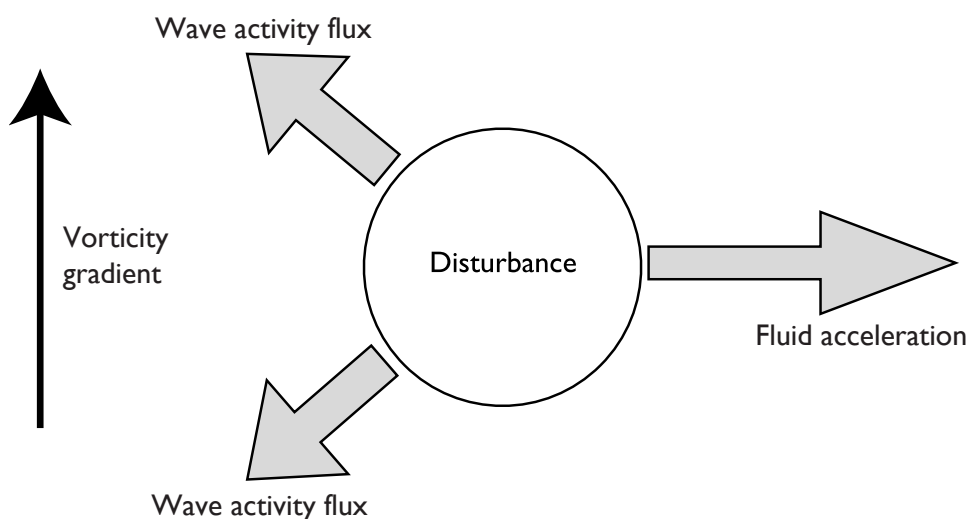
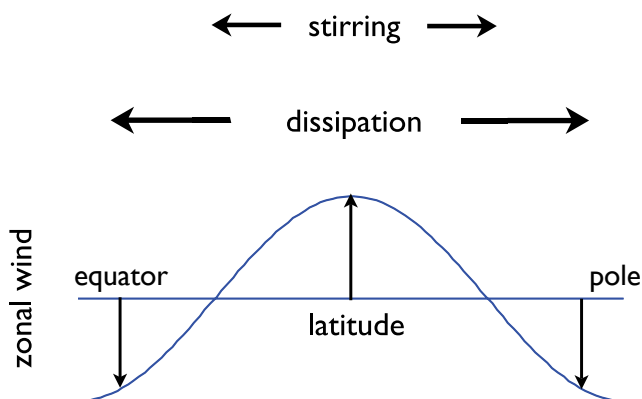


Figure 4.10 If a region of fluid on the β -plane or on a rotating sphere is stirred, then Rossby waves propagate away from the disturbance, and this is the direction of the wave activity density flux vector. Thus, there is divergence of wave activity in the stirred region, and using (4.110) this produces an eastward acceleration.

It is crucial to the generation of a mean flow that the dissipation has a broader latitudinal distribution than the forcing: if all the dissipation occurred in the region of the forcing then from (4.105) no mean flow would be generated. However, Rossby waves are generated in the forcing region, and these propagate meridionally before dissipating thus broadening the dissipation distribution and allowing the generation of a mean flow.

IV. The Eliassen–Palm flux

The Eliassen–Palm (EP) flux provides a convenient framework for determining how waves affect the mean flow, and the barotropic case is a particularly simple and instructive example. In the

unforced case, the zonally averaged momentum equation may be written as

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla_x \cdot \mathcal{F}, \quad (4.106)$$

where $\bar{v}^*$ is the residual meridional velocity and $\mathcal{F}$ is the Eliassen–Palm flux, and $\nabla_x \cdot$ is the divergence in the meridional plane. In the barotropic case $\bar{v}^* = 0$ and

$$\mathcal{F} = -\mathbf{j} \overline{u'v'}, \quad \nabla_x \cdot \mathcal{F} = -\frac{\partial}{\partial y} \overline{u'v'} = \overline{v'\zeta'} \quad (4.107)$$

If the momentum flux is primarily the result of interacting nearly monochromatic waves, then the EP flux obeys the group velocity property, namely that the flux of wave activity density is equal to the group velocity times the wave activity density. Thus,

$$\mathcal{F}^y \equiv \mathbf{j} \cdot \mathcal{F} \approx c_g^y \mathcal{A}, \quad (4.108)$$

where $\mathcal{A}$ is the wave activity density, or pseudomomentum,

$$\mathcal{A} = \frac{\overline{\zeta'^2}}{2\bar{q}_y} = \frac{\overline{\zeta'^2}}{2\gamma}, \quad (4.109)$$

and, if $\gamma > 0$, $\mathcal{A}$ is a positive-definite quantity. The zonal momentum equation and the Eliassen–Palm relation become, respectively,

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial y} (c_g^y \mathcal{A}), \quad \frac{\partial \mathcal{A}}{\partial t} + \frac{\partial}{\partial y} (c_g^y \mathcal{A}) = 0, \quad (4.110a,b)$$

and so

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \mathcal{A}}{\partial t}, \quad (4.111)$$

as in (4.103).

Now suppose that we initiate a disturbance at some latitude, and then let the fluid evolve freely. The disturbance generates Rossby waves whose group velocity will be directed away from the region of disturbance, and the wave activity density $\mathcal{A}$ will diminish in the region of the disturbance (and increase elsewhere). However, from (4.110a) the zonal velocity will *increase* in the region of the disturbance, and an eastward flow will be generated. That is, momentum converges in the region of the disturbance and an eastward jet is generated, as sketched in Fig. 4.10.

This EP flux argument, the pseudomomentum argument and the Rossby wave argument are, in essence, just different expressions of the same physical process, different explications but the same explanation. Indeed, the result of (4.90) can be regarded as illustrating the group velocity property of the EP flux for barotropic Rossby waves. The vorticity budget argument is arguably more general than any of these, because it does not depend on linearization or small amplitude disturbances. At the same time, none of these arguments is truly turbulent in nature, and although they are all fundamentally nonlinear, it is the presence of β -effect — a linear term in the vorticity equation — that is crucial in the development of a mean flow.

CHAPTER 5

GRAVITY WAVES

WEEK 10-11

This is a short chapter on internal gravity waves.

5.1 INTERNAL WAVES IN A CONTINUOUSLY STRATIFIED FLUID

We now turn our attention to *internal gravity waves*, namely waves that are internal to a given fluid and that owe their existence to the restoring force of gravity. Let us consider a continuously stratified Boussinesq fluid, initially at rest, in which the background buoyancy varies only with height and so the buoyancy frequency, N , is a function only of z . Linearizing the equations of motion about this basic state gives the linear momentum equations,

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi' \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (5.1a,b)$$

the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (5.2a,b)$$

Our notation is such that $\mathbf{u} \equiv u\mathbf{i} + v\mathbf{j}$, $\mathbf{v} \equiv u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$, and the gradient operator is two-dimensional unless noted. Thus, $\nabla \equiv \mathbf{i}\partial_x + \mathbf{j}\partial_y$ and $\nabla_3 \equiv \mathbf{i}\partial_x + \mathbf{j}\partial_y + \mathbf{k}\partial_z$.

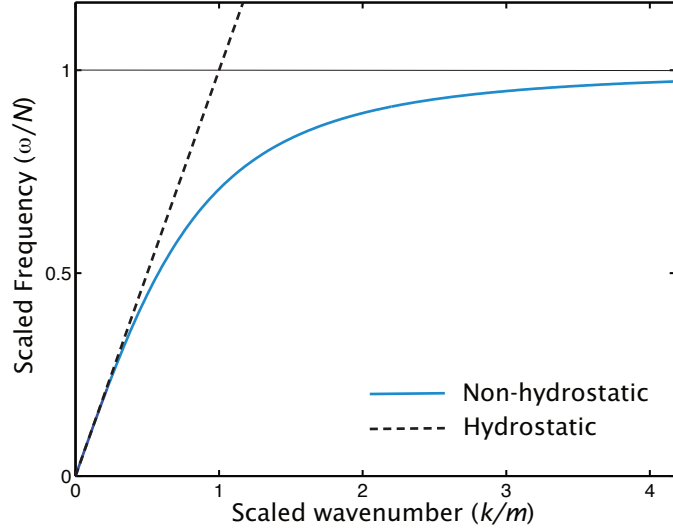
A little algebra gives a single equation for w' ,

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + N^2 \nabla^2 \right] w' = 0. \quad (5.3)$$

This equation is evidently *not* isotropic. If N^2 is a constant — that is, if the background buoyancy varies linearly with z — then the coefficients of each term are constant, and we may then seek solutions of the form

$$w' = \text{Re } \tilde{w} e^{i(kx + ly + mz - \omega t)}, \quad (5.4)$$

Figure 5.1 Scaled frequency, ω/N , plotted as a function of scaled horizontal wavenumber, k/m , using the full dispersion relation of (5.5) with $l = 0$ (solid line, asymptoting to unit value for large k/m), and with the hydrostatic dispersion relation (5.11) (dashed line, tending to ∞ for large k/m).



where Re denotes the real part, a denotation that will frequently be dropped unless ambiguity arises, and other variables oscillate in a similar fashion. Using (5.4) in (5.3) yields the dispersion relation:

$$\omega^2 = \frac{(k^2 + l^2)N^2}{k^2 + l^2 + m^2} = \frac{K^2 N^2}{K_3^2}, \quad (5.5)$$

where $K^2 = k^2 + l^2$ and $K_3^2 = k^2 + l^2 + m^2$. The frequency (see Fig. 5.1) is thus always less than N , approaching N for small horizontal scales, $K^2 \gg m^2$. If we neglect pressure perturbations, as in the parcel argument, then the two equations,

$$\frac{\partial \omega'}{\partial t} = b', \quad \frac{\partial b'}{\partial t} + \omega' N^2 = 0, \quad (5.6)$$

form a closed set, and give $\omega^2 = N^2$.

If the basic state density increases with height then $N^2 < 0$ and we expect this state to be unstable. Indeed, the disturbance grows exponentially according to $\exp(\sigma t)$ where

$$\sigma = i\omega = \frac{\pm K \tilde{N}}{K_3}, \quad (5.7)$$

where $\tilde{N}^2 \equiv -N^2$. Most convective activity in the ocean and atmosphere is, ultimately, related to an instability of this form, although of course there are many complicating issues — water vapour in the atmosphere, salt in the ocean, the effects of rotation and so forth.

Another form of the dispersion relation

We can write the dispersion relation, (5.5), as

$$\omega = \pm N \cos \vartheta, \quad (5.8)$$

where $\cos^2 \vartheta = K^2/(K^2 + m^2)$ so that ϑ is the angle between the three-dimensional wave-vector, $\mathbf{k} = k\mathbf{i} + l\mathbf{j} + m\mathbf{k}$, and the horizontal. The frequency is evidently a function only of N and ϑ , and, if this is given, the frequency is not a function of wavelength. This has some interesting consequences for wave reflection, as we see below.

We can also write the dispersion relation, (5.5), as

$$\frac{\omega^2}{N^2 - \omega^2} = \frac{K^2}{m^2}. \quad (5.9)$$

Thus, and consistently with our first point, given the wave frequency the ratio of the vertical to the horizontal wavenumber is fixed.

5.1.1 Hydrostatic internal waves

Let us now suppose that the fluid satisfies the hydrostatic Boussinesq equations, and for simplicity assume that $l = 0$. The linearized two-dimensional equations of motion become

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi', \quad 0 = -\frac{\partial \phi'}{\partial z} + b', \quad (5.10a)$$

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (5.10b)$$

where these are the horizontal and vertical momentum equations, the mass continuity equation and the thermodynamic equation respectively. A little algebra gives the dispersion relation,

$$\omega^2 = \frac{K^2 N^2}{m^2}. \quad (5.11)$$

The frequency and, if N^2 is negative, the growth rate, is unbounded as $K^2/m^2 \rightarrow \infty$, and the hydrostatic approximation thus has quite unphysical behaviour for small horizontal scales (see also problem 2.11). Many numerical models of the large-scale circulation in the atmosphere and ocean do make the hydrostatic approximation. In these models convection must be *parameterized*; otherwise, it would simply occur at the smallest scale available, namely the size of the numerical grid, and this type of unphysical behaviour should be avoided. Of course in non-hydrostatic models convection must also be parameterized if the horizontal resolution of the model is too coarse to properly resolve the convective scales.

5.2 INTERNAL WAVES IN A ROTATING FRAME OF REFERENCE

In the presence of both a Coriolis force and stratification a displaced fluid will feel two restoring forces — that due to gravity and that due to rotation. The first effect gives rise to gravity waves, as we have discussed, and the second to inertial waves. When the two forces are together simultaneously the resulting waves are called *inertia-gravity waves*.

5.2.1 Equations of motion

In a rotating frame of reference, specifically on an f -plane, the linearized equations of motion are the momentum equations

$$\frac{\partial \mathbf{u}'}{\partial t} + \mathbf{f}_0 \times \mathbf{u}' = -\nabla \phi' \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (5.12a,b)$$

and the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (5.12c,d)$$

These are similar to (5.1) and (5.2), with the addition of a Coriolis term in the horizontal momentum equations.

To obtain a single equation for w' we take the horizontal divergence of (5.12a) and use the continuity equation to give

$$\frac{\partial}{\partial t} \left(\frac{\partial w'}{\partial z} \right) + f_0 \zeta' = \nabla^2 \phi' \quad (5.13)$$

where $\zeta' \equiv (\partial v'/\partial x - \partial u'/\partial y)$ is the vertical component of the vorticity. We may obtain an evolution equation for that vorticity by taking the curl of (5.12a), giving

$$\frac{\partial \zeta'}{\partial t} = f_0 \frac{\partial w'}{\partial z}. \quad (5.14)$$

Eliminating vorticity between these equations gives

$$\left(\frac{\partial^2}{\partial t^2} + f_0^2 \right) \frac{\partial w'}{\partial z} = \frac{\partial}{\partial t} \nabla^2 \phi'. \quad (5.15)$$

We may obtain another equation linking pressure and vertical velocity by eliminating the buoyancy between (5.12b) and (5.12d), so giving

$$\frac{\partial^2 w'}{\partial t^2} + N^2 w' = -\frac{\partial}{\partial t} \frac{\partial \phi'}{\partial z}. \quad (5.16)$$

Eliminating ϕ' between (5.15) and (5.16) gives a single equation for w' analogous to (5.3), namely

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] w' = 0. \quad (5.17)$$

If we assume a time dependence of the form $w' = \widehat{w} e^{-i\omega t}$, this equation may be written in the sometimes useful form,

$$\frac{\partial^2 \widehat{w}}{\partial z^2} = \left(\frac{N^2 - \omega^2}{\omega^2 - f_0^2} \right) \nabla^2 \widehat{w}. \quad (5.18)$$

5.2.2 Dispersion Relation

Assuming wave solutions to (5.17) of the form

$$w' = \tilde{w} e^{i(kx+ly+mz-\omega t)}, \quad (5.19)$$

we readily obtain the dispersion relation

$$\omega^2 = \frac{f_0^2 m^2 + (k^2 + l^2) N^2}{k^2 + l^2 + m^2}. \quad (5.20)$$

It is clear from this expression that ω lies between N and f .

We can also write the dispersion relation as

$$\omega^2 = f_0^2 \sin^2 \vartheta + N^2 \cos^2 \vartheta, \quad (5.21)$$

or

$$\omega^2 = f_0^2 + (N^2 - f_0^2) \cos^2 \vartheta, \quad \text{or} \quad \omega^2 = N^2 - (N^2 - f_0^2) \sin^2 \vartheta, \quad (5.22)$$

where ϑ is the angle of the wavevector with the horizontal. The frequency therefore lies between N and f_0 . The waves satisfying (5.20) are sometimes called inertia-gravity and are analogous to surface gravity waves in a rotating frame — that is, Poincaré waves — discussed in section 2.6.2.

In many atmospheric and oceanic situations $f_0 \ll N$ (in fact typically $N/f_0 \sim 100$, the main exception being weakly stratified near-surface mixed layers in the ocean) and $f < \omega < N$. From (5.21) the frequency is dependent only on the angle the wavevector makes with the horizontal, and the surfaces of constant frequency again form cones in wavenumber space, although depending on the values of f and ω the frequency does not necessarily decrease monotonically with ϑ as in the non-rotating case.

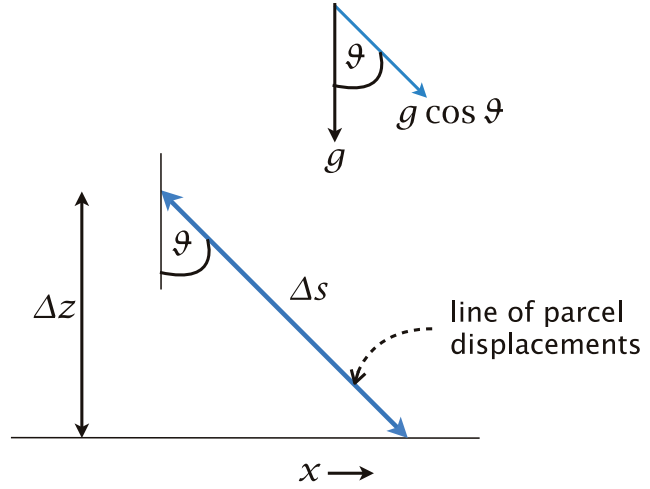
5.3 ♦ PARCEL ARGUMENTS FOR THE DISPERSION RELATION

5.3.1 Non-rotating case

Let us consider first the dispersion relation itself and try to derive it more physically, or at least heuristically. Let us suppose there is a wave propagating in the (x, z) plane at some angle ϑ to the horizontal, with fluid parcels moving parallel to the troughs and crests, as in Fig. 5.4. In general the restoring force on a parcel is due to both the pressure gradient and gravity, but along the crests there is no pressure gradient. Referring to Fig. 5.2, for a total displacement Δs the restoring force in the direction of the particle displacement, F_{re} , is

$$\begin{aligned} F_{re} &= g \cos \vartheta \times \Delta \rho = g \cos \vartheta \times \frac{\partial \rho}{\partial z} \Delta z \\ &= g \cos \vartheta \times \frac{\partial \rho}{\partial z} \Delta s \cos \vartheta = \rho_0 \frac{\partial b}{\partial z} \cos^2 \vartheta \Delta s, \end{aligned} \quad (5.23)$$

Figure 5.2 Parcel displacements and associated forces in a internal gravity wave in which the parcel displacements are occurring at an angle ϑ to the vertical, as in Fig. 5.4.



noting that $\Delta z = \cos \vartheta \Delta s$. The equation of motion of a parcel moving along a trough or crest is therefore

$$\rho_0 \frac{d^2 \Delta s}{dt^2} = -\rho_0 N^2 \cos^2 \vartheta \Delta s, \quad (5.24)$$

which implies a frequency $\omega = N \cos \vartheta$, as in (5.8). One of the $\cos \vartheta$ factors in (5.24) comes from the fact that the parcel displacement is at an angle to the direction of gravity, and the other comes from the fact that the restoring force that a parcel experiences is proportional to $N \cos \vartheta$. (The reader may also wish to refer ahead to Fig. 5.3 and section 5.3 for a similar argument.)

Now consider the wave illustrated in Fig. 5.4. For this wave both k and m are positive, and the frequency is assumed positive by convention to avoid duplicative solutions. The slanting solid and dashed lines are lines of constant phase, and from (5.31b) the buoyancy and pressure are $1/4$ of a wavelength out of phase. When k and m are both positive the extrema in the buoyancy field lag the extrema in the extrema in the vertical velocity by $\pi/2$, as illustrated. The perturbation velocities are zero along the lines of extreme buoyancy. This follows because the velocities are in phase with the pressure, which as we noted is out of phase with the buoyancy.

5.3.2 Rotating Case

Consider a parcel that is displaced along a slantwise path in the x - z plane, as shown in Fig. 5.3, with a horizontal displacement of Δx and a vertical displacement of Δz . Let us suppose that the fluid is Boussinesq and that there is a stable and uniform stratification given by $N^2 = -g\rho_0^{-1}\partial\rho_0/\partial z = \partial b/\partial z$. Referring to (5.23) as needed, the component of the restoring buoyancy force, F_b say, in the direction of the parcel oscillation is given by (5.23)

$$F_b = -N^2 \cos \vartheta \Delta z = -N^2 \cos^2 \vartheta \Delta s. \quad (5.25)$$

The parcel will also experience a restoring Coriolis force, F_C , and the component of this in

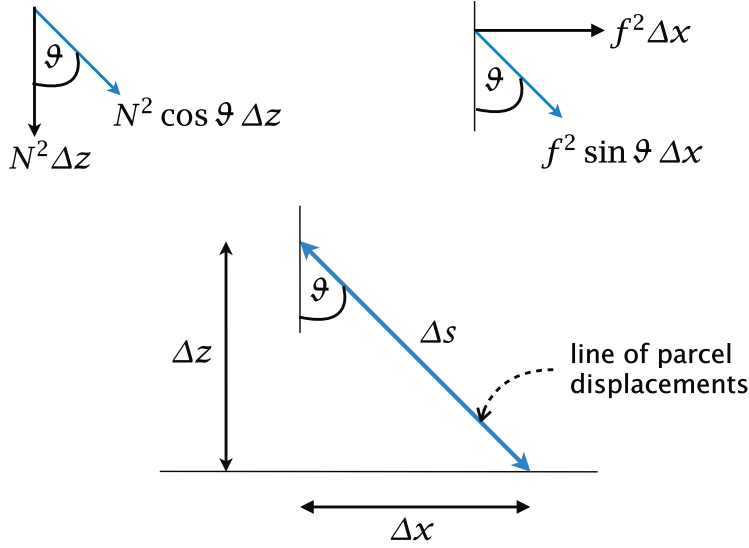


Figure 5.3 Parcel displacements and associated forces in an inertia-gravity wave in which the parcel displacements are occurring at an angle ϑ to the vertical. Both Coriolis and buoyancy forces are present, and $\Delta s = \Delta z / \cos \vartheta = \Delta x / \sin \vartheta$.

the direction of the parcel displacement is

$$F_C = -f^2 \sin \vartheta \Delta x = -f^2 \sin^2 \vartheta \Delta s. \quad (5.26)$$

Using (5.25) and (5.26) the (Lagrangian) equation of motion for a displaced parcel is

$$\frac{d^2 \Delta s}{dt^2} = -(N^2 \cos^2 \vartheta + f^2 \sin^2 \vartheta) \Delta s, \quad (5.27)$$

and hence the frequency is given by

$$\omega^2 = N^2 \cos^2 \vartheta + f^2 \sin^2 \vartheta. \quad (5.28)$$

Now, nearly everywhere in both atmosphere and ocean, $N^2 > f^2$. From (5.28) we then see that the frequency lies in the interval $N^2 > \omega^2 > f^2$. (To see this, put $N = f$ or $f = N$ in (5.28), and use $\sin^2 \vartheta + \cos^2 \vartheta = 1$.) If the parcel displacements approach the vertical then the Coriolis force diminishes and $\omega \rightarrow N$, and similarly $\omega \rightarrow f$ as the displacements become horizontal. The ensuing waves are then pure inertial waves.

We can write (5.28) in terms of wavenumbers if we note that, in the x - z plane,

$$\cos^2 \vartheta = \frac{k^2}{k^2 + m^2}, \quad \sin^2 \vartheta = \frac{m^2}{k^2 + m^2} \quad (5.29)$$

where k and m are the horizontal and vertical wavenumbers. The dispersion relation becomes

$$\omega^2 = \frac{N^2 k^2 + f^2 m^2}{k^2 + m^2}. \quad (5.30)$$

5.4 ♦ MORE PROPERTIES OF INTERNAL WAVES

Internal waves have a number of interesting and counter-intuitive properties so let's point a few of them out.

Polarization relations

If the pressure field is oscillating like $\phi' = \tilde{\phi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] = \tilde{\phi} \exp[i(kx + ly + mz - \omega t)]$ then, using (5.1a), the horizontal velocity components have the phases

$$(\tilde{u}, \tilde{v}) = (k, l)\omega^{-1}\tilde{\phi} \quad (5.31a)$$

As the frequency is real, the velocities are in phase with the pressure. A little algebra also reveals that the buoyancy perturbation is related to the pressure perturbation by

$$\tilde{b} = \frac{imN^2}{N^2 - \omega^2}\tilde{\phi} = \frac{iN^2K^2}{m\omega^2}\tilde{\phi} = \frac{iK_3^2}{m}\tilde{\phi}, \quad (5.31b)$$

using the dispersion relation, so that the buoyancy and pressure perturbations are $\pi/2$ out of phase.

The vertical velocity is related to the pressure perturbation by

$$\tilde{w} = \frac{-\omega m}{N^2 - \omega^2}\tilde{\phi} = \frac{-K^2}{m\omega}\tilde{\phi}, \quad (5.31c)$$

where the second expression uses (5.9). The vertical velocity is in phase with the pressure perturbation, and for regions of positive m (and so with upward phase propagation) regions of high relative pressure are associated with downward fluid motion.

The pressure, buoyancy and velocity fields are all real fields and we can write the above phase relationships in terms of sines and cosines as follows.

$$\phi = \Phi_0 \cos(kx + ly + mz - \omega t), \quad (5.32a)$$

$$(u, v) = (k, l)\frac{\Phi_0}{\omega} \cos(kx + ly + mz - \omega t), \quad (5.32b)$$

$$w = \left(\frac{-\omega m}{N^2 - \omega^2} = \frac{-K^2}{m\omega} \right) \Phi_0 \cos(kx + ly + mz - \omega t) \quad (5.32c)$$

$$b = \left(\frac{mN^2}{N^2 - \omega^2} = \frac{N^2K^2}{m\omega^2} \right) \Phi_0 \sin(kx + ly + mz - \omega t), \quad (5.32d)$$

where Φ_0 is a constant. We might equally well have chosen ϕ to have a sine dependence in (5.32a), in which case (5.32b,c,d) should be changed appropriately. The relations between pressure, buoyancy and velocity in (5.31) and (5.32) are known as *polarization relations*.

Relation between wave vector and velocity

$$\mathbf{k} \cdot \tilde{\mathbf{v}} = 0. \quad (5.33)$$

This means that, at any instant, the wave vector is perpendicular to the velocity vector, and the velocity is therefore aligned *along* the direction of the troughs and crests, along which there is no pressure gradient. If the wave vector is purely horizontal (i.e., $m = 0$), then the motion is purely vertical and $\omega = N$.

The vertical and horizontal velocities are related to the wave wavenumbers. Suppose for simplicity (and with little loss of generality) that the motion is all in the x - z plane, with $l = 0$ and $v = 0$. Then

$$\frac{\tilde{u}}{\tilde{w}} = -\frac{m}{k}. \quad (5.34)$$

Furthermore, from (5.4) with $l = 0$, at any given instant all of the perturbation quantities in the wave are constant along the lines $kx + mz = \text{constant}$. Thus, *all fluid parcel motions are parallel to the wave fronts*. Now, since the wave frequency is related to the background buoyancy frequency by $\omega = \pm N \cos \vartheta$, it follows that the fluid parcels oscillate along lines that are at an angle $\vartheta = \cos^{-1}(\omega/N)$ to the vertical.

The polarization relations and the group and phase velocities are illustrated in Fig. 5.4. Let us now discuss them, and the figure, in a little more detail.

5.4.1 Group velocity and phase speed

As we noted above, the frequency of internal waves is given by $\omega = N \cos \vartheta$, where ϑ is the angle the wave vector makes with the horizontal. This means that the surfaces of constant frequency are *cones*, as illustrated in Fig. 5.5.

To evaluate phase and group velocities in a useful way it is convenient to use spherical polar coordinates, as in Fig. 5.6, in which

$$k = K_3 \cos \vartheta \cos \lambda, \quad l = K_3 \cos \vartheta \sin \lambda, \quad m = K_3 \sin \vartheta, \quad (5.35)$$

so that $\mathbf{k} = K_3(\cos \vartheta \cos \lambda, \cos \vartheta \sin \lambda, \sin \vartheta)$. The angles are ϑ , the angle of the wave vector with the horizontal and λ , which determines the orientation in the horizontal plane. (The notation is similar to the spherical coordinates of chapter 2 — see Fig. 2.3 — although here ϑ is the angle with the horizontal, not the angle with the equatorial plane.) We also note that

$$\sin^2 \vartheta = \frac{m^2}{k^2 + l^2 + m^2}, \quad \cos^2 \vartheta = \frac{K_3^2}{K_3^2} = \frac{k^2 + l^2}{k^2 + l^2 + m^2}, \quad \tan \lambda = \frac{l}{k}. \quad (5.36)$$

In many problems we can align the direction of the wave propagation with the x -axis and take $l = 0$ and $\tan \lambda = 0$.

The phase speed of the internal waves in the direction of the wave vector (sometimes referred to as the phase velocity) is given by

$$c_p = \frac{\omega}{K_3} = \frac{N}{K_3} \cos \vartheta = \frac{NK}{K_3^2}. \quad (5.37)$$

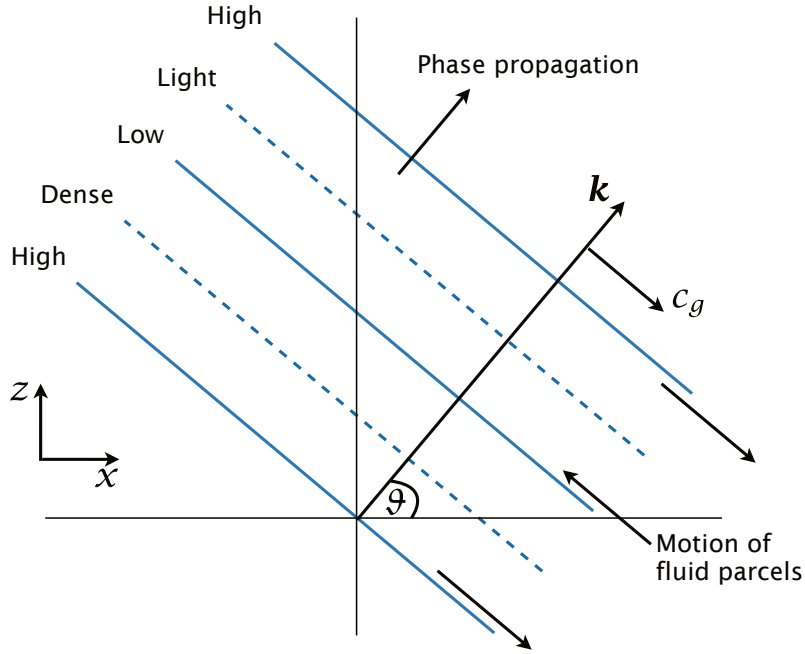


Figure 5.4 An internal wave propagating in the direction $\mathbf{k}$. If the motion is the the x - z plane then both k and m are positive for the wave shown. The solid lines show crests and troughs of constant pressure, and the dashed lines the corresponding crests and troughs of buoyancy (or density). The motion of the fluid parcels in along the lines of constant phase, as shown, and is parallel to the group velocity and perpendicular to the phase speed.

The phase speeds (as conventionally-defined) in the x, y and z directions are

$$c_p^x \equiv \frac{\omega}{k} = \frac{N}{k} \cos \vartheta, \quad c_p^y \equiv \frac{\omega}{l} = \frac{N}{l} \cos \vartheta, \quad c_p^z \equiv \frac{\omega}{m} = \frac{N}{m} \cos \vartheta. \quad (5.38a,b,c)$$

As noted in section 4.1.2 these quantities are the speed of propagation of the wave crests in the respective directions. In general, each speed is *larger* than the phase speed in the direction perpendicular to the wave crests (that is, in the direction of the wave vector), but no information is transmitted at these speeds.

The group velocity is given by

$$\mathbf{c}_g = \left(\frac{\partial \omega}{\partial k}, \frac{\partial \omega}{\partial l}, \frac{\partial \omega}{\partial m} \right). \quad (5.39)$$

Using (5.5) we find

$$c_g^x = \frac{\partial \omega}{\partial k} = \frac{Nm}{K_3^2} \frac{km}{KK_3} = \left(\frac{N}{K_3} \sin \vartheta \right) \cos \lambda \sin \vartheta, \quad (5.40a)$$

$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{Nm}{K_3^2} \frac{lm}{KK_3} = \left(\frac{N}{K_3} \sin \vartheta \right) \sin \lambda \sin \vartheta, \quad (5.40b)$$

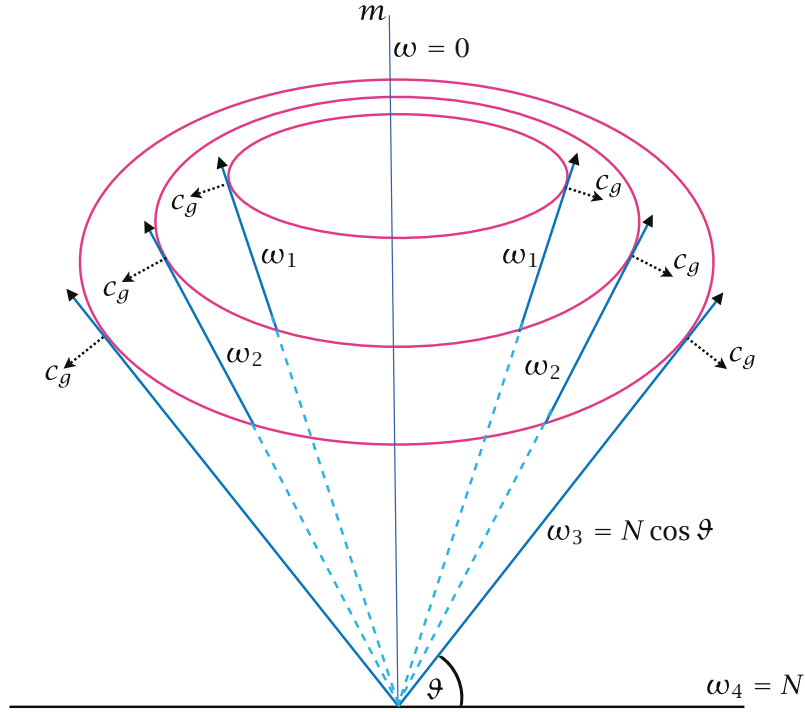


Figure 5.5 Internal wave cones. The surfaces of constant frequency are cones, defined by the surface that has a constant angle to the horizontal. The wave vector, and so the phase velocity, points along the cone away from the origin, and the frequency of any wave with a wave vector in the cone is $N \cos \vartheta$. The group velocity is at right angles to the cone and pointed in the direction of increasing frequency, as indicated by the arrows on the dotted lines. In the vertical direction the phase speed and group velocity have opposite signs.

$$c_g^z = \frac{\partial \omega}{\partial m} = -\frac{Nm}{K_3^2} \frac{K}{K_3} = -\left(\frac{N}{K_3} \sin \vartheta\right) \cos \vartheta. \quad (5.40c)$$

The magnitude of the group velocity is evidently

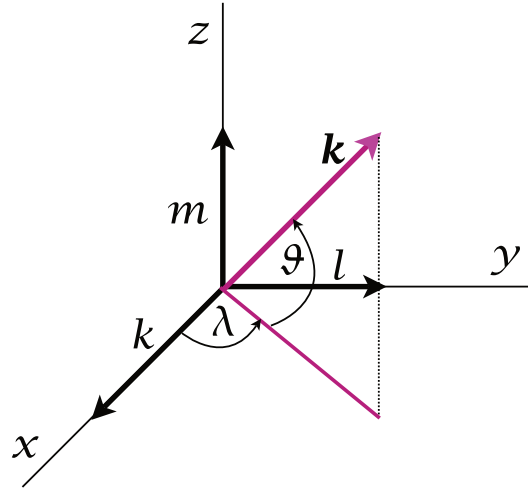
$$|c_g| = \frac{N}{K_3} \sin \vartheta, \quad (5.41)$$

and the group velocity vector is directed at an angle ϑ to the vertical, as in Fig. 5.5. This angle is perpendicular to the cone itself; that is, the group velocity is perpendicular to the wave vector, as may be verified by taking the dot product of (5.35) and (5.40) which gives

$$\mathbf{k} \cdot \mathbf{c}_g = 0. \quad (5.42)$$

The group velocity is therefore parallel to the motion of the fluid parcels, as illustrated in Fig. 5.4. Furthermore, because energy propagates with the group velocity, and the latter is *parallel* to

Figure 5.6 The spherical coordinates used to describe internal waves, as in (5.35). The angle ϑ is the angle of the wave vector with the horizontal, and λ determines the orientation in the horizontal plane. The wave vector $\mathbf{k}$ is given by $\mathbf{k} = (k, l, m)$ in the direction of increasing (x, y, z) .



lines of constant phase, energy propagates perpendicular to the direction of phase propagation — very different from the case of acoustic waves or even shallow water waves. In the vertical direction we see from (5.38c) and (5.40c) that

$$\frac{\omega}{m} \frac{\partial \omega}{\partial m} = -\frac{N^2}{K^2} \cos^2 \vartheta < 0. \quad (5.43)$$

That is, the phase speed and the group velocity have opposite signs, meaning that if the wave crests move downward the group moves upward!

Effect of a mean flow

Suppose that there is a mean flow, U , in the x -direction, as is common in both atmosphere and ocean. The dispersion relation, (5.5), simply becomes

$$(\omega - Uk)^2 = \frac{K^2 N^2}{K^2 + m^2}. \quad (5.44)$$

The frequency is Doppler shifted, as expected, but the upward propagation of waves is affected in an interesting way. From (5.44) we find that the vertical component of the group velocity may be written as

$$\frac{\partial \omega}{\partial m} = \frac{-m(\omega - Uk)}{K^2 + m^2} = \frac{-mk(c - U)}{K^2 + m^2}. \quad (5.45)$$

where $c = \omega/k$ is the phase speed in the x -direction. If U is not constant but is varying slowly with z then (5.45) still holds, although m itself will also vary slowly with z . The point to note is that the group velocity goes to zero at the location where $U = c$, that is at a critical layer and the wave stalls. Of course m itself may become large near a critical layer (as we consider in more detail in section ??). In this case — which is essentially the hydrostatic one, with $m^2 \gg K^2$ — we

obtain

$$\frac{\partial \omega}{\partial m} = \frac{-k(c - U)}{m} = \frac{-k^2(c - U)^2}{KN}. \quad (5.46)$$

The physical consequence of group velocity going to zero as the wave approaches a critical layer is that any dissipation that may be present has more time to act. That is, we can expect a wave to be preferentially dissipated near a critical layer, giving up its momentum to the mean flow and its energy to create mixing — the former being important in the atmosphere (for this is the mechanism producing the quasi-biennial oscillation) and the latter in the ocean.

5.4.2 ♦ Energetics of internal waves

In this section we explore the energetics of internal waves, and we first show that the linearized equations conserve a sensible form of energy. Linearized equations do not, of course, automatically conserve energy even if the original nonlinear equations from which they derive do: an unstable wave will draw energy from the background state and grow in amplitude, as we saw in chapter 6 on baroclinic instability.

Energy Conservation

To obtain an energy equation we proceed much as in the nonlinear case described in section A2.4.3. From (5.1) we obtain an equation for the evolution of kinetic energy, namely

$$\frac{\partial}{\partial t} \left(\frac{\mathbf{v}'^2}{2} \right) = b' w' - \nabla_3 \cdot (\phi' \mathbf{v}'), \quad (5.47)$$

where $\mathbf{v}'^2 = u'^2 + v'^2 + w'^2$, and from (5.2) we obtain

$$\frac{1}{N^2} \frac{\partial b'^2}{\partial t} + w' b' = 0. \quad (5.48)$$

Adding the above two equations gives

$$\frac{\partial}{\partial t} \frac{1}{2} \left(\mathbf{v}'^2 + \frac{b'^2}{N^2} \right) + \nabla_3 \cdot (\phi' \mathbf{v}') = 0 \quad (5.49)$$

This is the linear version of (A2.112). Two differences are apparent: (i) The transport of energy is only by way of the pressure term and the advective transport is absent, as expected in a linear model; (ii) the potential energy term bz of the linear model is replaced by b'^2/N^2 . It is less obvious why this should be so. However, the quantity

$$A = \frac{1}{2} \int \frac{\overline{b'^2}}{\partial \bar{b} / \partial z} dz dA = \frac{1}{2} \int \frac{\overline{b'^2}}{N^2} dz dA \quad (5.50)$$

is just the *available potential energy* of a Boussinesq fluid in which the isopycnal surfaces vary only slightly from a stable, purely horizontal, resting state (see section A3.10.1).

If we integrate (5.49) over a volume such that the normal component of the velocity vanishes at the boundaries (for example, we integrate over a volume enclosed by rigid walls) then the divergence term vanishes and we obtain the integral conservation statement:

$$\hat{E} = \frac{1}{2} \int \left(\mathbf{v}'^2 + \frac{b'^2}{N^2} \right) dV, \quad \frac{d\hat{E}}{dt} = 0. \quad (5.51)$$

The quantity $\hat{E}$ is an example of a *wave activity*: a conserved quantity that is quadratic in wave amplitude. This conservation statement (5.49) is true whether or not the basic state is stably stratified; that is, whether or not N^2 is positive. However, (5.51) only provides a bound on growing perturbations if N^2 is positive, in which case all the terms that constitute $\hat{E}$ are positive definite. If $N^2 < 0$ then both $\mathbf{v}'^2$ and b'^2 can grow without bound even as $\hat{E}$ itself remains constant.

Consider now the energy in a *wave*, and we will denote by $\bar{E}$ the energy density, meaning the mean perturbation energy per unit volume, averaged over a wavelength. Thus

$$2\bar{E} = \overline{\mathbf{v}'^2} + \frac{\overline{b'^2}}{N^2}. \quad (5.52)$$

If we use the polarization relations of section 5.4 then the kinetic and potential energy densities may be written in terms of the pressure amplitude as

$$2\overline{KE} = \left(\frac{k^2}{\omega^2} + \frac{l^2}{\omega^2} + \frac{(k^2 + l^2)^2}{m^2\omega^2} \right) |\tilde{\phi}|^2 = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2, \quad (5.53a)$$

$$2\overline{PE} = \frac{N^2 K^4}{m^2\omega^4} = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2, \quad (5.53b)$$

using also the dispersion relation, $\omega^2 K_3^2 = K^2 N^2$. Thus, there is *equipartition* between the kinetic and potential energies, a common feature of waves in non-rotating systems (although not a universal feature of waves). The total energy density is thus

$$\bar{E} = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2 = \frac{K_3^2}{K^2} |\tilde{\omega}|^2 = \frac{|\tilde{\omega}|^2}{\cos^2 \vartheta}. \quad (5.54)$$

where $\tilde{\omega}$ is the amplitude of the vertical component of the velocity perturbation.

♦ *Energy propagation and the group velocity property*

If a wave activity, A , and its flux, $\mathbf{F}$ obeyed a conservation law of the form $\partial A / \partial t + \nabla \cdot \mathbf{F} = 0$, and if the wave activity and its flux were both quadratic functions of the wave amplitude, then the flux is related to the wave activity by $\mathbf{F} = c_g A$. The internal wave energy density and its flux do have these properties — see (5.49) — so we should expect the group velocity property to hold, and we now demonstrate that explicitly, albeit briefly.

The energy flux vector for internal waves is

$$\mathbf{F} = \overline{\phi' \mathbf{v}'} \quad (5.55)$$

and using (5.31a) and (5.31c) this is

$$\mathbf{F} = \left(\frac{k}{\omega}, \frac{l}{\omega}, -\frac{K^2}{m\omega} \right) |\tilde{\phi}|^2. \quad (5.56)$$

Using (5.40) and (5.54) the group velocity times the energy density is

$$c_g^x \times \bar{E} = \left[\frac{Nm^2}{K_3^3} \frac{k}{K} \right] \times \left[\frac{K^2 K_3^2}{m^2 \omega^2} |\tilde{\phi}|^2 \right] = \frac{k}{\omega} |\tilde{\phi}|^2, \quad (5.57a)$$

$$c_g^y \times \bar{E} = \left[\frac{Nm^2}{K_3^3} \frac{l}{K} \right] \times \left[\frac{K^2 K_3^2}{m^2 \omega^2} |\tilde{\phi}|^2 \right] = \frac{l}{\omega} |\tilde{\phi}|^2, \quad (5.57b)$$

$$c_g^z \times \bar{E} = \left[\frac{NmK}{K_3^3} \right] \times \left[\frac{K^2 K_3^2}{m^2 \omega^2} |\tilde{\phi}|^2 \right] = -\frac{K^2}{m\omega} |\tilde{\phi}|^2, \quad (5.57c)$$

which evidently is the same as (5.56), completing our demonstration.

